RECTOR: Rule-Enforced Constrained Trajectory Orchestrator for Compliance-Aware Autonomous Driving

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***Abstract*—Modern multi-modal motion predictors can propose diverse futures, but the final trajectory is still often chosen based on confidence rather than on explicit traffic-rule priorities. We present RECTOR (Rule-Enforced Constrained Trajectory ORchestrator), a rule-aware selection layer that preserves a generator's candidate set and reranks trajectories using the strict priority order Safety $\succ$ Legal $\succ$ Road $\succ$ Comfort. RECTOR combines three elements: (1) rule-applicability estimation over a 28-rule catalog; (2) differentiable proxy evaluators for 24 rules that assign continuous violation severities; and (3) a deterministic lexicographic selector whose decisions are traceable at the tier level. The current implementation includes a learned applicability head, but the broader framework also supports conservative non-learned applicability policies for audit and ablation.**

On WOMD `validation_interactive`, the canonical evaluation over 43,219 augmented scenarios reports strong candidate-set accuracy (minADE = 0.687 m, minFDE = 1.274 m, miss rate = 18.46%). Confidence-only selection already leaves 75.94% of selected trajectories with no reported Protocol A violation, so the benchmark is better read as a *selection* problem over a mostly already-plausible candidate set than as a broad correction of pervasive rule breaking. Under the proxy-24, learned-applicability setting, lexicographic selection raises selected-mode compliance to 84.82% (equivalently, total violations fall from 24.06% to 15.18%, about 3.8k fewer violating selections) and reduces Safety+Legal violations from 22.09% to 13.39%, while also improving selected-trajectory accuracy from 2.199 m to 2.138 m selADE. The strongest signal is concentrated in the Safety tier and in tail-risk cases where confidence ranking would otherwise choose a higher-risk mode. A weighted-sum baseline matches the same compliance rates on this benchmark and attains lower selected ADE (2.048 m), so the contribution supported by the present evidence is not a blanket benchmark win over scalar baselines, but a selector with explicit priority guarantees, infeasibility signaling, and auditable decision logic.

Verification supports the practical reliability of the stack: semantic invariants pass at 99.2–100%, post-mitigation NaN/Inf propagation is 0.0%, and proxy–full-evaluator pairwise candidate-ranking concordance reaches 0.948. Mean inference latency is 7.3 ms per sample, and a 55-scenario Waymax log-replay study spanning 3,338.5 s records zero overlap steps, confirming end-to-end integration of the selection pipeline. Taken together, the results suggest that explicit rule-prioritized selection is a practical way to make multi-modal prediction outputs more accountable, interpretable, and compliance-oriented without retraining the underlying generator. Limitations, scope boundaries, and deployment qualifications are consolidated in Sections IX and X.

***Index Terms*—autonomous driving, trajectory selection, rule compliance, lexicographic optimization, traffic rule enforcement, multi-modal prediction, Waymo Open Motion Dataset, safety guarantees, differentiable proxies, rule applicability prediction**

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I. Introduction

Trajectory prediction and planning for autonomous vehicles must produce behavior that is not only geometrically accurate but also safe, legal, and socially compliant. Modern deep multi-modal predictors—including anchor-based methods, latent-variable generators, and diffusion-based samplers—can generate diverse candidate futures. However, the selection of the final executed trajectory remains an underspecified stage in most systems. When several plausible modes exist, the system must choose one that satisfies explicit, priority-ordered traffic rules.

This paper addresses that problem by introducing RECTOR (**R**ule-**E**nforced **C**onstrained **T**rajectory **OR**chestrator), a rule-aware selection framework that ranks candidate ego trajectories using a tiered rulebook and a tolerance-based lexicographic ordering. The framework is built around three components: (i) a neural head that predicts *which rules apply* in a given scene, (ii) differentiable smooth proxies that evaluate *how severely* each applicable rule is violated, and (iii) a deterministic lexicographic selector that enforces the non-negotiable priority hierarchy Safety $\succ$ Legal $\succ$ Road $\succ$ Comfort. Because RECTOR operates entirely at selection time, it is architecturally compatible with upstream multi-modal trajectory generators without modifying the predictor itself, and its empirical value should be interpreted as reranking the same candidate set rather than repairing the generator.

The primary empirical result is that, even though confidence-only selection already yields no reported Protocol A violation in 76.14% of scenarios, rule-aware selection—whether lexicographic or weighted-sum—still changes which candidate is executed in the minority of hard scenes: selected-mode compliance rises to 84.97%, total violations fall by 8.83 pp, and the gain is concentrated in Safety-tier tail-risk filtering rather than broad correction of benchmark-wide behavior. On the current benchmark, the lexicographic selector produces identical binary compliance outcomes to weighted-sum across all 250 tested weight configurations (including fully flat weights $w_S{=}w_C{=}1$, which diverge by exactly 1 scenario out of 12,800); adversarial confidence injection likewise confirms that both rule-aware strategies reject safety-violating candidates at the same rate (3.6–5.7%) while confidence-only

selection fails at 100%. These experiments establish that the empirical lex–WS equivalence is a structural property of the WOMD validation distribution (Safety proxy scores are large, median ≈ 1.0; cross-tier Comfort advantages are small, effective misspecification boundary at $w_S/w_C \approx 1$) rather than an artifact of the tested weight range. RECTOR's lexicographic formulation contributes (i) a formal guarantee that higher-priority violations cannot be traded for lower-priority gains independent of score scale, weight specification, or distribution of compliance margins—a guarantee that weighted-sum satisfies only when weights are correctly calibrated—and (ii) complete elimination of weight tuning from the selector, which matters when calibration data are scarce, the score distribution shifts, or auditability of the decision is required. The following subsections elaborate on this motivation, outline the concrete contributions, and define the scope of the current work.

A. Motivation: The Compliance Gap

To illustrate the selection problem, consider standard metrics such as Minimum Average Displacement Error (minADE). While these metrics measure how well predictions cover the ground truth, they do not assess the safety or legality of the executed mode. In complex scenes, two candidates may be geometrically similar yet critically different: one might violate a red light or fail to yield even with a low geometric error. This discrepancy is especially pronounced in "long-tail" interactive scenarios, where error distributions are heavy-tailed (Fig. 2). In these regimes, selection is not a minor post-processing step but a primary mechanism for ensuring plan acceptability.

B. Key Technical Observations

RECTOR's design is motivated by four technical observations. First, traffic rules follow a strict priority structure: safety dominates legality, which in turn dominates road-level and comfort considerations. Unlike scalar penalties, lexicographic ordering ensures that higher-priority tiers are never traded off for lower-priority gains. Second, rule applicability is context-dependent: a stop-sign rule is irrelevant without a stop sign. Rather than relying on brittle hand-coded logic, RECTOR uses a learned model to predict the "active rule set" directly from scene features. Third, many rules admit smooth approximations; core constraints such as collision margins can be expressed as continuous proxy functions for numerically stable scoring.

Fourth, *when no candidate satisfies high-priority constraints, the system must degrade gracefully*. A finite candidate set may not always contain a fully compliant option. In such cases, the selector should choose the *least-violating* trajectory and signal infeasibility to downstream components, rather than falling back to confidence-only selection. We formalize this as a minimum-violation fallback with an explicit infeasibility flag (Section III-I).

C. Approach Overview

Taken together, these observations suggest a two-stage architecture. In the reference instantiation used in this paper, a Transformer–CVAE generator inspired by the M2I interactive prediction family generates $K = 6$ candidate ego trajectories over a fixed 5-second horizon ($T = 50$ timesteps at 10 Hz). Each mode is a sequence of ego states with a confidence score [1]. RECTOR's rule-aware selection layer is decoupled from the generator and can be attached to the candidate set produced upstream.

In the second stage, each candidate is evaluated against a catalog of R=28 traffic rules organized into four tiers that reflect the normative priority structure motivated in Section I-B. The four tiers are:

- **Safety (Tier 0):** collision avoidance and minimum distance constraints (5 rules),
- **Legal (Tier 1):** traffic-signal and stop-sign compliance, speed limits, and right-of-way rules (7 rules),[1]
- **Road (Tier 2):** lane keeping and road boundary adherence (2 rules),
- **Comfort (Tier 3):** acceleration, jerk, and steering bounds and smooth motion preferences (14 rules).

Candidates are then ranked by tolerance-based lexicographic ordering. The selector minimizes Safety-tier violations first. Only near-ties, within a small tolerance, are resolved using the Legal tier, then the Road tier, and finally the Comfort tier. This procedure makes priorities explicit and removes weight tuning from the selector itself. In particular, a small improvement in comfort can never displace a safer or more lawful candidate. The result is a deterministic, auditable decision based only on tier scores and tolerance parameters, not on tuned scalar weights.

D. Contributions

Against this background, the specific contributions of this paper are as follows.

C1 Negative Result on Applicability Learning with Immediate Deployment Implications. We systematically compare four applicability policies—learned head, hybrid conservative (always-on for Safety/Legal, learned for Road/Comfort), fully always-on, and oracle—under cross-evaluation on 43,219 scenarios (Table XXIII). The central finding is negative but practically important: a *parameter-free always-on policy strictly dominates* the 3.33 M-parameter learned head on *both* Safety-tier compliance (-2.64 pp: $23.06\% \rightarrow 20.43\%$) *and* selected-trajectory accuracy (-0.089 m selADE: $2.138 \rightarrow 2.049$ m). The accuracy improvement occurs because activating more Safety rules steers the selector away from collision-prone, geometrically extreme modes. We therefore recommend always-on activation for Safety and Legal tiers as the default deployment policy and treat the learned head as residual support for Road and Comfort tiers, where false-positive activation incurs measurable accuracy cost. The learned head's failure on Safety-tier rules is a class-imbalance problem: per-rule F1 is almost entirely determined by oracle

[1]Tier labels are used as research abstractions within the WOMD-based rule catalog; deployment and jurisdictional qualifications are discussed in Sections IX and X.

support (ρ=0.994); rules with >5,000 positive instances achieve mean F_1=0.75 while rules below 1,000 achieve F_1=0.00. This finding has direct practical implications: for safety-critical tiers, a parameter-free conservative policy should be preferred over learned prediction until recall on rare Safety events is substantially improved.

C2 Practical Rule-Aware Selection Integration Beyond Prior Rulebooks. Building on the lexicographic rulebook concept of Censi et al., we integrate priority-ordered selection into a learned multi-modal trajectory pipeline. The concrete additions beyond prior rulebook formulations are the learned applicability head, differentiable proxies for 24 of 28 rules, and a batched selector compatible with GPU inference while preserving the intended hierarchy. The measured benefit is a reranking effect on the same candidate set, with most of the headroom coming from avoiding Safety-tier tail-risk selections rather than from broad changes in an already mostly compliant benchmark.

The primary empirical result is that *rule-aware selection of any kind* raises selected-mode compliance from 76.14% to 84.97% and reduces Safety+Legal violations from 21.87% to 13.15% on 12,800 scenarios. On this benchmark, weighted-sum and RECTOR produce near-identical binary compliance outcomes: extending the weight grid to 250 configurations (including fully flat weights w_S=w_C=1) shows that WS diverges from lex on at most 1 scenario out of 12,800 (+0.0078 pp) and matches lex exactly for $w_S \geq 5$; the 0.100 m selADE gap in favour of weighted-sum is a tie-breaking artefact explained in Section VII-B3. Adversarial confidence injection (Section VII-B2) confirms that both rule-aware strategies reject safety-violating candidates at the same rate (3.6–5.7%) while confidence-only selection picks the injected violator at 100%: the separation is rule-aware vs. confidence-only, not lexicographic vs. weighted-sum. Because the benchmark already contains many compliant or near-compliant modes, the relevant gain is the ability to filter the unsafe tail of the candidate set, not to convert a broadly unlawful predictor into a lawful one.

The contribution of lexicographic ordering is therefore not a measurable compliance gain on this distribution, but a *structural guarantee that holds independent of score scale and weight values*: lexicographic selection provably cannot trade a higher-priority violation for a lower-priority gain regardless of how the proxy scores are scaled or distributed. Weighted-sum satisfies this property only when weights are correctly calibrated (here, $w_S \geq 5w_C$); below that threshold it can fail. Lexicographic selection requires no weight specification and cannot be miscalibrated, which matters in deployment contexts where calibration data are scarce, scores shift under distribution shift, or the decision must be fully auditable without reference to tuned parameters.

C3 Correctness Properties of the Selector. We verify three properties of RECTOR's deterministic selector: order invariance (selection is independent of candidate input order), infeasibility fallback (minimum-violation selection when no fully compliant candidate exists), and scalarization correctness (a sufficiently large exponential base recovers lexicographic ranking from a scalar score; this condition is automatically satisfied by the tier-score normalization contract for any base $B > 3$). These are straightforward consequences of the selector's definition and of known results in lexicographic optimization [2], but we state and verify them explicitly to support reproducibility. The infeasibility fallback is proven formally and the implementation now emits an explicit `infeasible` flag when the selected candidate has non-zero Safety-tier score, enabling downstream fail-safe logic. Empirically, tolerance sweeps from 10^{-3} to 10^{-2} leave Safety, Legal, and total violations unchanged, confirming robustness to implementation parameters.

C4 Differentiable Proxy Catalog for Selection-Time Enforcement. We implement 24 differentiable proxy functions spanning Safety, Legal, Road, and Comfort constraints, enabling continuous rule severity scoring for batched selection-time enforcement while keeping 4 semantically difficult rules reserved for post-hoc audit. Proxy–full-evaluator alignment shows mean pairwise candidate-ranking concordance of 0.948 across all 24 proxied rules (0.875 for the 10 rules with measurable variance in both evaluators; 1,000 scenarios), indicating that the proxy reliably preserves tier-aggregate candidate ordering despite moderate per-rule severity correlations (mean Spearman ρ=0.25) (Section V).

C5 Verification Pipeline for Rule-Aware Selection. We complement the proxy layer with a verification pipeline covering semantic invariants (non-negativity, correct applicability gating, monotonicity, temporal consistency), numerical stability (0.0% post-mitigation NaN/Inf rate), and proxy–auditor alignment. A 55-scenario Waymax log-replay validation over 3,338.5 s reports zero overlap steps and confirms end-to-end simulator-bridge execution. Robustness experiments show that injected perception and map perturbations increase violation rates in a controlled, bounded manner (up to ∼6.1 pp for perception noise, ∼12.1 pp for worst-case map errors), demonstrating that the pipeline's checks meaningfully bound failure modes (Section VIII).

C6 Empirical Validation. On 12,800 WOMD `validation_interactive` scenarios, RECTOR shows that nontrivial selection headroom remains even when the candidate set is already mostly compliant: reported selected-mode compliance rises from 76.14% to 84.97% (23.86% → 15.03% total violations, about 1.1k fewer violating executed trajectories). The Safety tier provides the dominant improvement (21.87% → 13.15%, −8.72 pp) under the learned applicability head, which is the strongest evidence that the remaining headroom lies in tail-risk filtering rather than in correcting pervasive rule breaking; a uniformly weighted-sum baseline achieves numerically identical outcomes on this evaluation set. Scope boundaries,

deployment qualifications, and failure modes are consolidated in Sections IX and X.

E. System Architecture

RECTOR augments standard multi-modal generators with two rule-aware modules (Fig. 1). First, an **applicability mechanism** identifies active constraints from the scene embedding—for example, triggering stop-sign rules only at intersections and crosswalk-yield rules when pedestrians are present. In the reference model this mechanism is learned, but the results later show that always-on activation is the preferable default for Safety and Legal tiers on the current benchmark. Second, a **tiered lexicographic scorer** aggregates these violations and selects the candidate with the minimal tier-score vector. This architecture is intentionally modular: while the generator focuses on diversity and geometric fidelity, the scorer ensures that the final decision contains no learned parameters in the selector itself—the per-rule normalization constants and tolerance values are fixed engineering constants, not gradient-updated weights—and remains strictly compliant with the priority hierarchy. This separation allows RECTOR to improve compliance without requiring retraining of the upstream model. Full architecture details and parameter breakdowns are given in Section IV (Table IV).

F. Interpretability and Explainability

RECTOR's *selection step* is auditable by construction. For every selection decision, the system identifies the *first-differing tier*—the specific priority level that separated the winning trajectory from the runners-up. Combined with per-rule applicability predictions and proxy violation severities, this yields a natural-language-ready decision trace—for example: "Trajectory A rejected due to Legal-tier violation; Trajectory B selected despite lower Comfort score." Unlike post-hoc attribution methods [3], RECTOR's decision trace is a deterministic consequence of the tier scores and the lexicographic comparator, making every selection fully reproducible and storable as a lightweight audit artifact. An important qualification: while the selector itself is fully transparent, the *input* to the selector—the applicability mask produced by a learned neural network—is not inherently interpretable. The auditability claim therefore covers the selection logic and its trace, not the upstream applicability predictions that condition it.

G. Operating Setup

RECTOR performs open-loop ego trajectory selection over a 5-second horizon (T=50 timesteps at 10 Hz, K=6 modes), ensuring tactical safety and legality via 28 cataloged rules across 4 priority tiers.

Compliance therefore has both tactical and strategic dimensions in this paper. RECTOR addresses tactical compliance within the fixed 5-second prediction horizon; strategic compliance requires route-level planning beyond the present formulation.

H. Paper Organization

With these boundaries established, the remainder of this paper is organized as follows. Section II reviews related work along five thematic axes that map directly to RECTOR's design decisions. Section III formalizes the lexicographic selection objective, defines all notation and tier-score aggregation, and proves the three formal properties introduced in C3. Section IV then describes the RECTOR architecture in detail: the scene encoder, Transformer–CVAE decoder, rule applicability head, and tiered scorer. Section V specifies the 28-rule catalog, differentiable proxy implementations, and the data pipeline that underpins both training and auditing. Sections VI and VII present, respectively, the experimental protocol (dataset, metrics, baselines) and the full results—trajectory accuracy, selection strategy comparison, per-tier violation analysis, ablation studies, and computational efficiency. Section VIII summarizes the verification pipeline that ensures the reported compliance gains are not artifacts of evaluator instability. Finally, Section IX outlines the deployment roadmap and future directions, and Section X concludes.

II. Related Work

This section contextualizes RECTOR across five thematic axes that align with its core design decisions: multi-modal trajectory generation, constraint-enforcement strategies, rule representation and priority encoding, selection and reranking mechanisms, and system-level explainability. Although modern predictors are increasingly accurate and diverse, the *final* selected trajectory often still lacks an explicit guarantee of safety or rule compliance. RECTOR addresses this gap by treating selection as a first-class decision stage governed by a deterministic, tiered rulebook.

A. Candidate Trajectory Generation

The first axis concerns how candidate trajectories are produced, because the quality and diversity of these candidates determine the upper bound on what any downstream selector—including RECTOR—can achieve.

Most motion prediction models produce a set of candidate trajectories and then score or select them using likelihood or confidence. A large family uses structured anchor sets, goal proposals, or lane-graph reasoning to generate multimodal futures efficiently. Representative methods include LaneGCN [4], TNT and DenseTNT [5], [6], Transformer-based set predictors such as MTR [7] and Wayformer [8], and query-centric architectures such as QCNet [9]. Game-theoretic interaction modeling (e.g., GameFormer [10]) further captures multi-agent coupling, but it still selects the final mode based on confidence. These methods achieve strong best-of-K accuracy, yet their training objectives are primarily geometric and probabilistic and therefore do not enforce traffic-rule acceptability for the executed mode.

A second family relies on latent-variable generative modeling to capture long-tail behavior and interactive uncertainty. DESIRE [11] and PRECOG [12] are representative early approaches, while Trajectron++ [13] models multi-agent interactions with structured latent variables. More recent work uses

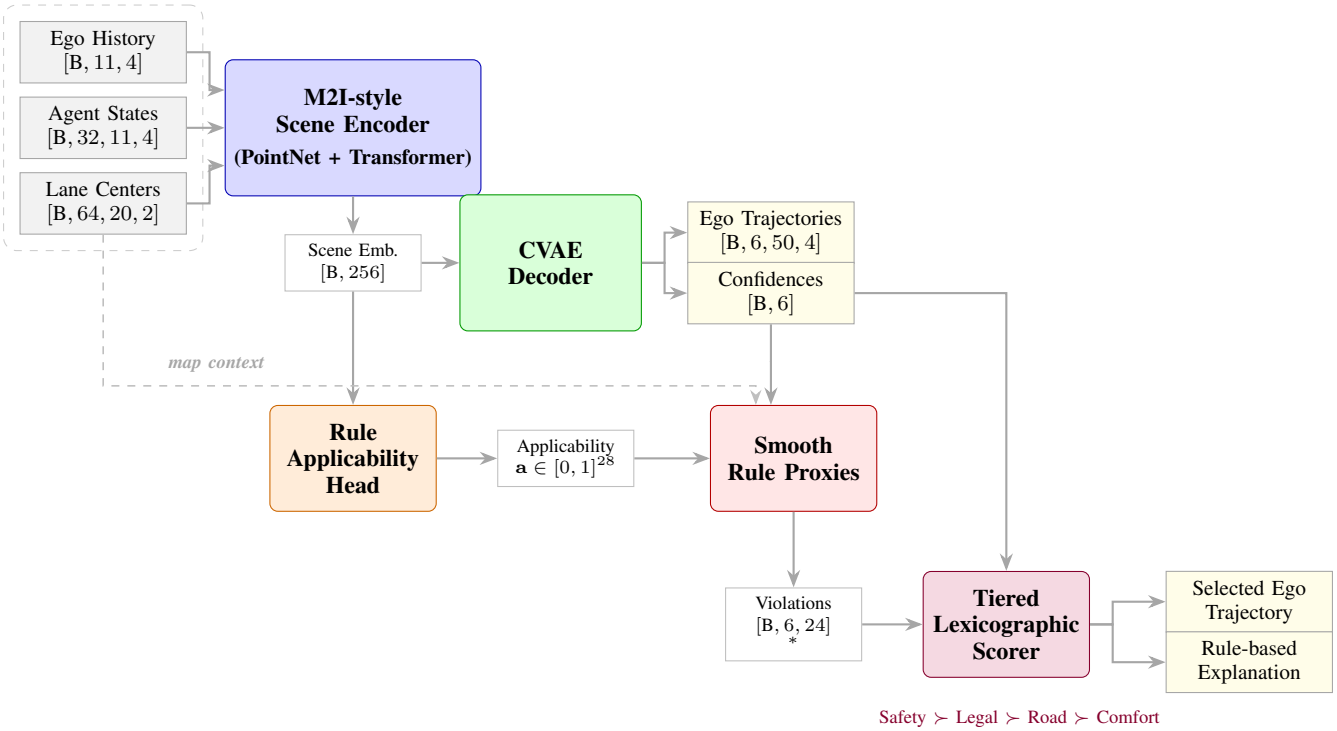


Fig. 1: **RECTOR system overview.** Scene inputs (ego history, agent states, lane geometry) flow through an M2I-style encoder (trained jointly with the decoder) to produce a dense scene embedding $\mathbf{e}_{\text{scene}} \in \mathbb{R}^{256}$. The Transformer–CVAE decoder generates six diverse ego-trajectory hypotheses, each with an associated confidence score. In parallel, the *rule applicability head* (the core learned component; 3.33 M of 8.83 M total parameters) predicts which of 28 traffic rules are relevant to the current scenario. Smooth proxies compute violation severities for each trajectory–rule pair (*24/28 rules have differentiable proxies; the remaining 4 are evaluated post-selection via the full-catalog auditor and contribute zero to proxy-based scores), and the tiered lexicographic scorer selects the ego trajectory with minimal violations respecting strict tier priority (Safety $\succ$ Legal $\succ$ Road $\succ$ Comfort). The selected trajectory includes a rule-based explanation of the selection decision.

large-capacity sequence models and large-scale pretraining—for example, MotionLM [14]—and diffusion-based generation for flexible sampling, such as MotionDiffuser [15]. A related line of work applies constraint guidance during diffusion-based generation: CTG [16] and CTG++ [17] guide trajectory sampling toward rule-satisfying regions. RECTOR's post-generation strategy is intentionally orthogonal to these constraint-guided approaches. By decoupling selection from generation, RECTOR avoids both the loss of candidate diversity and the *auditability gap* inherent in systems where compliance is embedded in the generation process rather than expressed as an explicit, rule-level decision trace.

Both families—anchor-based and generative—increase candidate diversity (or, in the constraint-guided case, targeted compliance), but they also amplify a practical issue: as the number of candidates grows, the decision of *which* candidate to execute becomes more consequential, and confidence-only selection can be unreliable in rare scenes where the model is under-calibrated. RECTOR is intentionally orthogonal to the generator. It can operate on top of any multi-modal model—anchor-based, latent-variable, Transformer, or diffusion—and focuses solely on the selection step: among diverse candidates, it deterministically prioritizes those that satisfy safety and legal constraints rather than relying only on probability mass.

B. When Are Rules Enforced? Constraint Entry Taxonomy

The previous subsection described *what* candidates are generated; this subsection asks *when* rules and safety constraints enter the pipeline, because the entry point determines both the guarantees a method can offer and its characteristic failure modes. Three entry points are relevant.

Planning-time enforcement integrates constraints directly into a control or optimization loop. Model predictive control (MPC) [18] and control barrier functions (CBFs) [19] are common representatives, and reachability analysis provides an additional formal tool [20]. These methods can offer strong guarantees under their modeling assumptions, but their use in large-scale data-driven prediction is limited by perceptual uncertainty, interaction complexity, and the need for real-time-constrained optimization at every step.

Policy-time enforcement introduces constraints during learning, for example via constrained reinforcement learning [21], [22] or shielding and safety filters such as SafetyNet [23]. These methods can empirically improve safety, but the learned constraint satisfaction may be brittle under distribution shifts, and the resulting behavior can be difficult to audit when constraints are not explicitly represented in the decision output.

Selection-time enforcement, by contrast, evaluates candidate trajectories against rules *after* generation and selects the best

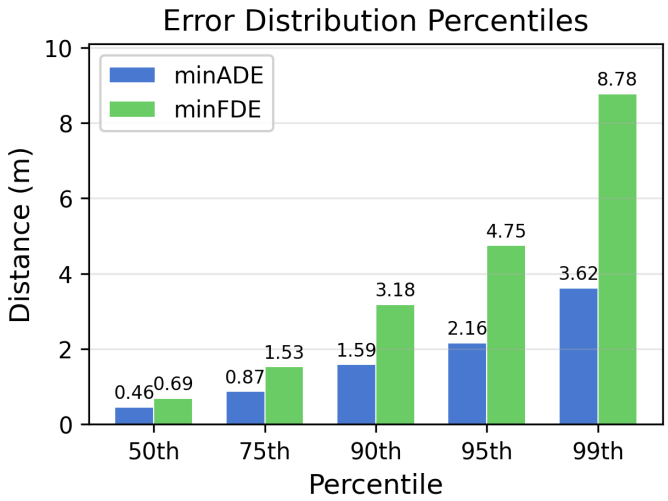


Fig. 2: Error distribution analysis showing heavy-tailed behavior on WOMD `validation_interactive`. Median minADE (0.46 m) is small, but the 95th percentile (2.16 m) and 99th percentile (3.62 m) are substantially larger, indicating that a small fraction of interactive scenes drive extreme errors. These long-tail scenarios are the primary contexts in which rule-aware selection provides the greatest benefit, because the selected trajectory must be executed without a guaranteed geometrically accurate option.

compliant option. This strategy remains underexplored, yet it is natural in multi-modal settings: if the generator already provides diversity, the selector should exploit it to avoid unsafe or illegal futures. RECTOR adopts this strategy, using smooth proxies and lexicographic tiering to prevent undesirable trade-offs while preserving full candidate diversity. Critically, selection-time enforcement yields an explicit, rule-level decision trace for post hoc auditing rather than embedding compliance in opaque generation internals.

C. Rule Representation and Priority Structures

Having established that RECTOR enforces rules at selection time, the next question is how to *represent* those rules and how to encode the priority relationships among them.

In practice, industry systems often rely on structured rule checkers and safety envelopes. RSS (Responsibility-Sensitive Safety) [24] is a well-known example that formalizes safe longitudinal and lateral distances together with responsibility constraints. Its operating point is fundamentally different from RECTOR's: RSS is a planning/control-time formal safety envelope with responsibility semantics, whereas RECTOR is a selection-time ranking layer over a learned candidate set. Censi et al. [25] introduced the concept of *liability-aware rulebooks* with lexicographic prioritization for autonomous driving, establishing that traffic rules should be organized into priority tiers in which higher-priority rules strictly dominate lower ones. This rulebook concept has since influenced multiple planning frameworks that use priority-ordered constraint satisfaction.

In learning-based prediction, however, rule constraints are often reduced to scalar penalties, which introduces a familiar issue from multi-objective optimization: weight tuning implicitly defines trade-offs, and small weight changes can reorder solutions in unintended ways (see, e.g., surveys in multi-objective methods [26]). This is particularly problematic for traffic rules, where priorities are not subjective preferences but normative requirements—safety and legality *should* dominate comfort, and that dominance should not depend on a particular weight vector.

RECTOR extends the rulebook concept of Censi et al. [25] by integrating lexicographic selection into a *learned* multi-modal prediction pipeline. The specific additions beyond the original rulebook framework are threefold: (i) a neural applicability head that predicts rule relevance from scene context, removing the need for hand-coded activation logic; (ii) smooth differentiable proxy functions that enable gradient-based training-time shaping; and (iii) a tolerance-based selector compatible with batched GPU inference, in which tolerances are used only to resolve near-ties so that the selector remains stable while preserving the intended hierarchy. To the best of our knowledge, prior AV prediction and selection pipelines do not treat rule applicability itself as a learned, scene-conditioned prediction target; relevance is typically hard-coded from map logic or left implicit in downstream rule checks.

D. Selection and Reranking over Candidate Sets

The preceding subsections described how candidates are generated, when constraints are applied, and how rules are prioritized. This subsection turns to the specific mechanism of *choosing* the final mode from the candidate set—the step that RECTOR targets.

Most multimodal predictors select the final mode either by taking the mode with maximum confidence (argmax over mode probabilities) or by sampling. Although this is computationally attractive, confidence is not a reliable proxy for acceptability, especially in rare interactive scenarios where the model is under-calibrated. An alternative is to rerank candidates using a learned scorer, a cost model, or a downstream-task objective. In adjacent areas, differentiable optimization layers (e.g., OptNet and related differentiable solvers) have been used to impose structured constraints during learning [27], [28]. Integrating full traffic-rule semantics into a differentiable solver remains difficult; moreover, strict constraints can collapse multimodality if they are applied too aggressively.

RECTOR addresses this gap with a non-parametric reranking mechanism that adds two components often missing from prior reranking approaches: (i) *applicability gating*, which activates only the rules relevant to the current scene and thereby avoids penalizing candidates for constraints that do not apply, and (ii) *tiered lexicographic ranking*, which prevents undesirable cross-tier trade-offs. Together, these features keep reranking deterministic and interpretable while maintaining real-time execution and compatibility with any upstream generator.

E. Explainability and Auditability

The selection mechanism just described not only picks a trajectory but also produces a structured record of *why* that trajectory was chosen—a property that connects naturally to the broader challenge of explainability in trajectory prediction.

Explainability in this domain is difficult because model decisions are distributed over time, across agents, and within latent representations. Post hoc explainers can provide local attributions, but they may not align with the true decision boundary and are often unstable across similar inputs [3], [29]. In safety-critical autonomy, a more actionable form of explainability is *auditability*: the system should report which constraints were considered, which were violated, and which priority level dominated the decision.

RECTOR satisfies this requirement by construction. For each candidate, it produces tier-wise scores and rule-level proxy violations; the final decision can be summarized by the first tier that differentiates among competing candidates. This output is closer to a safety-case artifact than to a generic feature-attribution method, and it supports debugging, dataset curation, and regression testing when rules change.

F. Benchmarks and Compliance Metrics

A related and complementary trend concerns the metrics and benchmarks used to evaluate trajectory predictors, because the choice of metric shapes research priorities.

Standard motion-forecasting benchmarks have historically emphasized geometric coverage of the ground truth. Datasets such as Argoverse [30] and nuScenes [31] popularized minADE and minFDE as primary metrics, which measure whether *some* predicted mode is close to the ground-truth future. This is appropriate for measuring multi-modal coverage, but it does not assess whether the *selected* mode is acceptable. More recent planning-aware benchmarks and closed-loop evaluations—including nuPlan [32] and scenario-centric evaluations in the Waymo ecosystem [33]—reinforce an important point: once a model is used to select an ego trajectory, the relevant metrics must include compliance and safety, not only geometric error.

RECTOR aligns with this direction by reporting selected-trajectory metrics (selADE/selFDE) alongside explicit rule-violation rates. Moreover, it distinguishes between proxy-based selection (for stable scoring and training compatibility) and full-catalog auditing (for semantic coverage), which matters when some rules depend on signals that are not available as smooth proxies.

G. Summary: Positioning of RECTOR

Taken together, the five axes reviewed above show that prior work provides strong multimodal candidate generators, mature constraint-enforcement methods for control and planning, several rule formalization frameworks, and an evolving set of compliance-aware benchmarks. What remains is a practical gap in *selection*: given a multimodal set of candidate ego futures, how should a system choose the trajectory to execute under explicitly prioritized traffic rules while remaining real-time and auditable?

RECTOR targets this gap as a selection layer that is architecturally independent of the upstream generator. It predicts rule applicability from scene context, evaluates tiered proxy severities, and applies tolerance-based lexicographic ordering to select a final trajectory while producing an auditable decision trace. In this paper, we instantiate RECTOR on top of a single CVAE generator to provide an end-to-end reference stack; the selection layer is designed to be composable with other upstream models, but this composability has not been empirically validated on additional generators. Table I summarizes this positioning across representative methods from each of the five axes.

III. Problem Formulation

We formalize trajectory selection as a *hierarchical* decision problem: given a set of diverse candidates from a multi-modal generator, the objective is to select the single most acceptable ego trajectory under a prioritized rulebook. We move beyond treating selection as a minor post-processing step, instead encoding the non-negotiable priority structure of traffic rules through tiered, tolerance-based lexicographic ordering. The resulting formulation provides a clean interface between (i) any multi-modal generator and (ii) an explicit, auditable rulebook evaluator.

The development proceeds in a deliberate sequence. We first fix notation and define the prediction setting (Sections III-A–III-B), then introduce the geometric metrics that characterize candidate-set quality and show why they are insufficient for acceptability (Sections III-C–III-D). With this motivation in place, we formalize the rule taxonomy, tier-score aggregation, and the lexicographic selection criterion (Sections III-E–III-G), establish its key properties (Section III-H), and verify three correctness properties (Section III-I). We close with a comparison to alternative strategies (Section III-J), the dual-protocol evaluation context (Section III-K), and the formal problem statement (Section III-L).

A. Notation and Conventions

Table II summarizes the notation used throughout the paper. We follow a scenario-centric viewpoint: each scenario s contains ego history, nearby agents, and map context, and the predictor produces a candidate set $\mathcal{T}$ along with confidence scores. Rules are organized into four tiers, and violations are aggregated per tier. All symbols introduced in this table are used consistently in the sections that follow; we reference them without redefining them.

B. Trajectory Prediction Setting

With the notation in place, we can define the prediction setting precisely. Let $s \sim \mathcal{D}$ denote a driving scenario drawn from a dataset distribution. A multi-modal predictor with parameters θ maps each scenario to (i) a finite set of candidate ego trajectories and (ii) a confidence score for each candidate:

$$(\mathcal{T}, \mathbf{p}) = f_\theta(s), \quad \mathcal{T} = \{\tau^{(1)}, \ldots, \tau^{(K)}\}, \quad \mathbf{p} = (p^{(1)}, \ldots, p^{(K)}),$$

where each candidate $\tau^{(k)}$ is a length-T sequence of ego states at sampling interval Δt. The goal is not only to generate

TABLE I: Positioning of RECTOR as a Generator-Agnostic Selection Layer Relative to Related Work

Method	Multi-Modal	Rule-Aware (Decision)	Tiered Priority	Decision Trace	Real-Time
Multi-Modal Prediction Methods					
TNT/DenseTNT [5], [6]	✓	—	—	—	✓
Trajectron++ [13]	✓	—	—	—	✓
MTR [7]	✓	—	—	—	✓
Wayformer [8]	✓	—	—	—	✓
MotionDiffuser [15]	✓	—	—	—	∼
QCNet [9]	✓	—	—	—	✓
GameFormer [10]	✓	—	—	—	✓
CTG/CTG++ [16], [17]	✓	✓	—	—	∼
Constraint Enforcement and Safety					
MPC-based [18]	—	✓	∼	✓	∼
CBF-based [19]	—	✓	∼	✓	∼
Constrained RL [22]	—	✓	—	—	✓
RSS [24]	—	✓	—	✓	∼
SafetyNet [23]	—	✓	—	—	✓
Rulebooks [25]	—	✓	✓	✓	—
RECTOR (This Work)					
RECTOR (Ours)	✓	✓	✓	✓	✓

Legend: ✓ = Full support. ∼ = Partial/conditional support. — = Not supported.

Note: Rows under *Multi-Modal Prediction Methods* are *candidate generators*. They do not natively include rule-aware, tiered decision logic; attaching RECTOR to any such generator yields a composite stack that is rule-aware, hierarchical, and interpretable *at selection time*. Characterisations in this table are based on published descriptions of each method as research baselines and may not reflect the current state of any commercial implementation.

diverse candidates, but also to select a single trajectory $\hat{\tau}$ that is preferable under a prioritized rulebook. The remainder of this section formalizes what "preferable means.

C. Standard Geometric Metrics

Before introducing the rule-based selection objective, we define the standard geometric metrics used in multi-modal trajectory prediction, because understanding their limitations motivates the formulation that follows. These metrics characterize the *coverage* quality of the candidate set $\mathcal{T}$, but—crucially—they do not determine whether the *selected* mode is acceptable under traffic rules.

Definition III.1 (Average Displacement Error)**.** *The Average Displacement Error (ADE) is the mean Euclidean distance over the prediction horizon:*

$$\mathit{ADE}(\tau, \tau^{\mathrm{gt}}) = \frac{1}{T}\sum_{t=1}^{T} \|\tau_t - \tau_t^{\mathrm{gt}}\|_2. \tag{1}$$

Definition III.2 (Final Displacement Error)**.** *The Final Displacement Error (FDE) is the Euclidean distance at the final timestep:*

$$\mathit{FDE}(\tau, \tau^{\mathrm{gt}}) = \|\tau_T - \tau_T^{\mathrm{gt}}\|_2. \tag{2}$$

For multi-modal prediction, we report the minimum error across candidates, which measures the best achievable match to the ground truth within $\mathcal{T}$:

$$\mathrm{minADE}(s) = \min_{k\in[K]} \mathrm{ADE}(\tau^{(k)}, \tau^{\mathrm{gt}}), \tag{3}$$

$$\mathrm{minFDE}(s) = \min_{k\in[K]} \mathrm{FDE}(\tau^{(k)}, \tau^{\mathrm{gt}}). \tag{4}$$

Definition III.3 (Miss Rate)**.** *The miss rate is the fraction of scenarios where the best candidate exceeds a threshold δ:*

$$\mathit{MissRate}_{\delta} = \mathbb{E}_{s\sim\mathcal{D}}\left[\nVdash\left[\mathit{minFDE}(s) > \delta\right]\right]. \tag{5}$$

We use $\delta = 2.0\,\mathrm{m}$ following the Waymo benchmark convention. MissRate does not enter the lexicographic selection objective; it is included here as a standard benchmark metric that characterizes candidate-set coverage. The actual motivation for rule-aware selection comes from the heavy-tailed error analysis in Section III-D, which uses percentile statistics rather than a single miss-rate threshold. In addition to these best-of-K metrics, our evaluation in later sections reports *selected-trajectory* metrics (selADE/selFDE), which quantify the quality of decisions at the selection stage rather than the coverage of the candidate set.

D. Error Distribution Analysis

The metrics above reveal a critical property of the prediction setting that directly motivates rule-aware selection: the distribution of prediction errors is *heavy-tailed* in interactive driving. This matters because selection mistakes in the tail can lead to safety- and law-critical failures, even when median performance is strong. Using scenario-level aggregation over N_{scen}=12,800 interactive WOMD scenarios (as distinct from the canonical sample-level count N_{eval}=25,448 defined in Table II and Section VI), the per-scenario minADE confirms this: the median is 0.46 m, but the 90th percentile reaches 1.59 m, the 95th reaches 2.16 m, and the 99th reaches 3.62 m—roughly 8× the median (Fig. 2).

In such cases, confidence-only selection is especially brittle, because mode probabilities can be poorly calibrated under rare interactions, while rule constraints (e.g., collision margins or

TABLE II: Key Notation and Symbol Definitions

Scenarios and Trajectories	
s	Driving scenario sampled from dataset distribution $\mathcal{D}$
$\mathbf{x}_{\text{ego}}$	Ego history (past states over T_h timesteps)
$\mathbf{X}_{\text{agents}}$	Neighbor-agent histories
$\mathcal{M}$	Map context (lanes, signals, crosswalks, boundaries, etc.)
τ	Ego future trajectory over T timesteps
τ_t	Ego state at timestep t (typically position and heading)
$\mathcal{T}$	Set of K candidate trajectories
τ^{gt}	Ground truth future trajectory
Δt	Sampling interval (0.1 s at 10 Hz)
Confidence and Modes	
$p^{(k)}$	Confidence score for candidate k
K	Number of candidate modes (typically 6)
T_h	History length (11 timesteps, 1.1 s)
T, T_f	Prediction horizon (50 timesteps, 5.0 s); T_f used in architecture descriptions
N_{scen}	Number of scenarios in a scenario-level aggregation (12,800 in WOMD `validation_interactive`)
N_{eval}	Number of evaluation samples (one per scenario in the current pipeline; $N_{\text{eval}} = N_{\text{scen}}$)
Rules and Compliance	
$\mathcal{R}$	Rule catalog; $R = \lvert\mathcal{R}\rvert = 28$
r	Individual traffic rule
$\ell(r)$	Tier assignment for rule r in $\{0, 1, 2, 3\}$
$a_r(s)$	Oracle applicability indicator (auditing): 1 if rule r applies in scenario s
$\hat{a}_r(s)$	Predicted applicability used by RECTOR for selection (thresholded from model output)
$\tilde{a}_r(s)$	Continuous applicability score before thresholding; $\tilde{a}_r = \sigma(\text{logit}_r)$
$V_r(\tau; s)$	Raw violation severity of τ under rule r in scenario s
$\bar{V}_r(\tau; s)$	Normalized violation severity (mapped and clamped to $[0, 1]$)
α_r	Per-rule normalization constant; $\bar{V}_r = \text{clamp}(V_r/\alpha_r,\ 0,\ 1)$
Tier Scores and Selection	
$\tilde{w}_r$	Intra-tier weight for rule r; $\sum_{r:\,\ell(r)=\ell} \tilde{w}_r = 1$ per tier (default: uniform)
$S_\ell(\tau; s)$	Aggregated tier-ℓ score
$\mathbf{S}(\tau; s)$	Tier score vector $[S_0, S_1, S_2, S_3]$
ϵ_ℓ	Tier tolerance used for robust lexicographic comparison
$\prec_{\text{lex},\epsilon}$	Shorthand for procedural lexicographic selection (Definition III.5); not a partial order
L	Number of tiers ($L = 4$)
Metrics	
minADE	Minimum Average Displacement Error over modes
minFDE	Minimum Final Displacement Error over modes
MissRate	Fraction of scenarios with minFDE $> \delta$

red-light compliance) remain decisive for acceptability. This motivates a selection mechanism grounded in traffic rules rather than likelihood alone. The formal machinery for such a mechanism is developed next.

E. Traffic Rule Taxonomy

To define rule-aware selection, we must first specify how traffic rules are represented. Let $\mathcal{R}$ denote a catalog of R=28 traffic rules. Each rule $r \in \mathcal{R}$ is assigned to one of four tiers via $\ell(r) \in \{0, 1, 2, 3\}$, reflecting the normative priority structure of traffic constraints:

$$\text{Safety} \succ \text{Legal} \succ \text{Road} \succ \text{Comfort}, \tag{6}$$

where higher tiers must dominate lower tiers in the selection decision (see Section I-B for the normative justification and the Legal-tier disclaimer).

Rules are not universally applicable—a stop-sign rule is irrelevant in a scene without a stop sign—and this context dependence is one of the central technical challenges addressed in this work. Two distinct applicability indicators are used throughout the paper, and the distinction between them is *operationally critical*:

- **Oracle applicability** $a_r(s) \in \{0, 1\}$: a ground-truth label derived from map and scenario metadata. It is used *exclusively* for post-hoc auditing (Protocol B in Section III-K) and *never* enters the selection objective.
- **Predicted applicability** $\hat{a}_r(s) \in \{0, 1\}$: the thresholded output of a learned neural head (Section IV) that predicts rule relevance from the scene embedding. This is the indicator used in the tier score aggregation (Definition III.4) and in the selection objective (Problem 1).

Oracle applicability is defined as:

$$a_r(s) = \begin{cases} 1 & \text{if rule } r \text{ applies in scenario } s, \\ 0 & \text{otherwise.} \end{cases} \tag{7}$$

The predicted counterpart $\hat{a}_r(s)$ is obtained by thresholding a continuous score $\tilde{a}_r(s) = \sigma(\text{logit}_r)$ produced by the applicability head. The gap between $a_r(s)$ and $\hat{a}_r(s)$ is the applicability prediction error, and it is one of the key sources

of approximation in the deployed system. This motivates the dual-protocol evaluation design (Section III-K).

Each applicable rule produces a non-negative violation severity $V_r(\tau; s)$ for a candidate trajectory τ in scenario s:

$$V_r(\tau; s) \geq 0. \tag{8}$$

Severity can be binary or continuous (e.g., margin-based), and it can depend on kinematics, interaction context, and map semantics. Because the 28 rules are heterogeneous—their raw severities carry different units (metres for collision margins, m/s for speed violations, dimensionless IoU for overlap rules)—a *normalization contract* is required before tier-level aggregation. We define the normalized violation as

$$\bar{V}_r(\tau; s) \;=\; \mathrm{clamp}\Big(\frac{V_r(\tau; s)}{\alpha_r},\; 0,\; 1\Big), \tag{9}$$

where $\alpha_r > 0$ is a per-rule normalization constant that maps the physically meaningful range of V_r to the unit interval. This ensures $\bar{V}_r(\tau; s) \in [0, 1]$ for every rule, and consequently each tier score $S_\ell(\tau; s) \in [0, 1]$ (Definition III.4). The specific values of α_r and the clamping details for each rule are given in Section V; only the contract $\bar{V}_r \in [0, 1]$ is needed for the selection formulation that follows. This separation is deliberate: the present section defines the *selection objective*, while Section V defines the *concrete evaluators* that instantiate it.

F. Tier Score Aggregation

Given the per-rule violations and applicability indicators defined above, the next step is to aggregate them into tier-level scores for comparison by the lexicographic selector. For each tier $\ell \in \{0, 1, 2, 3\}$, we aggregate violations across the rules assigned to that tier, using *predicted* applicability $\hat{a}_r(s)$ for gating during selection (not the oracle indicator, which is reserved for auditing).

Definition III.4 (Tier Score)**.** *The tier-ℓ score of trajectory τ in scenario s is*

$$S_\ell(\tau; s) = \sum_{r \in \mathcal{R}:\, \ell(r)=\ell} \tilde{w}_r\, \hat{a}_r(s)\, \bar{V}_r(\tau; s), \tag{10}$$

where $\tilde{w}_r \geq 0$ are per-rule weights satisfying $\sum_{r:\, \ell(r)=\ell} \tilde{w}_r = 1$ within each tier (defaulting to uniform: $\tilde{w}_r = 1/n_\ell$, where n_ℓ is the number of rules in tier ℓ). This normalization ensures $S_\ell(\tau; s) \in [0, 1]$ for all ℓ, regardless of the number of rules per tier—a property required by Proposition III.8.

The four tier scores are collected into a tier score vector:

$$\mathbf{S}(\tau; s) = \big[S_0(\tau; s),\, S_1(\tau; s),\, S_2(\tau; s),\, S_3(\tau; s)\big] \in \mathbb{R}^4_{\geq 0}. \tag{11}$$

This construction has four practical properties that make it suitable for hierarchical selection. First, it is non-negative and bounded in $[0, 1]$ per tier, which supports stable comparisons across candidates and guarantees scalarization correctness (Proposition III.8). Second, the intra-tier normalization ensures that a tier with 14 rules (Comfort) does not numerically dominate a tier with 2 rules (Road) at equal per-rule severity. Third, cross-tier priorities are represented *explicitly* in the tier structure rather than *implicitly* in a weight vector. Fourth, applicability gating prevents irrelevant rules from dominating the score when context is missing (e.g., stop-sign constraints in scenes without stop signs).

G. Lexicographic Selection Criterion

With tier scores defined, we now specify how they are compared across candidates to enforce the priority structure. The core mechanism is lexicographic ordering over the tier score vector. Because numeric scores can be close among candidates, we use a tolerance-based comparator to avoid instability due to floating-point noise and near-ties.

Definition III.5 (Tolerance-Based Lexicographic Selection (Procedural))**.** *Given candidates $\mathcal{T} = \{\tau^{(1)}, \ldots, \tau^{(K)}\}$, tier score vectors $\mathbf{S}^{(k)} = \mathbf{S}(\tau^{(k)}; s)$, and per-tier tolerances $(\varepsilon_0, \ldots, \varepsilon_3)$ with $\varepsilon_\ell \geq 0$, the selection proceeds by iterative pruning:*

1) *Initialise the surviving set $\mathcal{C}_0 = \{1, \ldots, K\}$.*
2) *For each tier $\ell = 0, 1, 2, 3$ in priority order:*
   a) *Compute $S^*_\ell = \min_{k \in \mathcal{C}_\ell} S^{(k)}_\ell$.*
   b) *Retain only near-optimal candidates: $\mathcal{C}_{\ell+1} = \{k \in \mathcal{C}_\ell : S^{(k)}_\ell \leq S^*_\ell + \varepsilon_\ell\}$.*
3) *Among the survivors $\mathcal{C}_4$, select $k^* = \arg\min_{k \in \mathcal{C}_4} \sum_{\ell=0}^{3} S^{(k)}_\ell$.*
4) *If an exact tie remains after this residual aggregate comparison, return the lowest candidate index in the tied set.*

Return $\hat{\tau} = \tau^{(k^*)}$.

The definition is deliberately *procedural* rather than order-theoretic. The tolerance relation $|S^{(i)}_\ell - S^{(j)}_\ell| \leq \varepsilon_\ell$ is *not* transitive, and consequently the pruning sets $\mathcal{C}_\ell$ depend on the global minimum within the current surviving pool, not on pairwise comparisons. This means the procedure does not induce a total order on $\mathbb{R}^4_{\geq 0}$ in the classical sense; it defines a deterministic selection *algorithm* rather than a classical preference relation. As shown in Proposition III.6, the output is nonetheless well-defined and invariant to candidate enumeration order.

For conciseness, we sometimes write $\mathbf{S}^{(i)} \prec_{\mathrm{lex},\epsilon} \mathbf{S}^{(j)}$ to mean that candidate i would survive and candidate j would not whenever both are present in the same candidate pool—but we emphasise that this notation is a shorthand for the procedural definition above, not a claim that $\prec_{\mathrm{lex},\epsilon}$ is a partial order.

Lexicographic selection requires at most $O(KL)$ tier comparisons (with L=4) and permits early termination: as soon as a candidate is pruned at a higher-priority tier, lower tiers cannot recover it. The residual aggregate tie-break is consulted only after tolerance pruning has exhausted the strict tier ordering, so it resolves near-ties within the surviving set rather than redefining the priority hierarchy.

H. Properties of Lexicographic Selection

Before proving formal guarantees, it is useful to state the qualitative properties that follow directly from the ordering

definition, because they clarify why lexicographic selection matches traffic-rule semantics better than scalarization.

- **Priority preservation.** If two candidates differ beyond tolerance at tier ℓ, then their ordering is decided at tier ℓ, independent of all lower tiers.
- **No weight-induced trade-offs.** Unlike weighted sums, the method does not require tuning cross-tier weights that implicitly allow comfort improvements to override safety or legality.
- **Determinism with controlled tie-handling.** The tolerance comparator reduces unstable ordering changes under small numerical perturbations; residual ties are resolved by aggregate residual score over the surviving set, with candidate index used only as a final exact-tie fallback.
- **Auditability.** The selected decision can be attributed to the first tier $\ell^\star$ that differentiates candidates, enabling concise explanations (e.g., "rejected due to Legal-tier red-light violation").

These properties support the central goal of this paper: selection should respect non-negotiable traffic priorities while remaining compatible with modern multi-modal generators. The following subsection strengthens these qualitative claims with three formal theorems.

I. Formal Guarantees

We now establish three formal properties of the filter-based lexicographic selector (Algorithm 1 in Section IV). *Order invariance*: the selection is a function of the candidate set, independent of enumeration order. *Infeasibility fallback*: when no fully compliant candidate exists, the system performs graceful degradation by selecting the least-violating option rather than reverting to confidence-only selection. *scalarization correctness*: the GPU-friendly scoring mechanism preserves the intended lexicographic ordering, ensuring lower-priority gains never override higher-priority requirements. Together, these results formalize the structural distinction introduced in Section I: RECTOR's lexicographic ordering *provably cannot* trade higher-priority violations for lower-priority gains, whereas a weighted-sum baseline merely *does not* under its current parameterization.

Proposition III.6 (Order Invariance)**.** *Algorithm 1 produces the same output regardless of the ordering of candidates in the input set $\mathcal{T}$, for any fixed tolerance vector $(\varepsilon_0, \ldots, \varepsilon_3)$ with $\varepsilon_\ell \geq 0$.*

Proof. At each tier ℓ, the algorithm computes $S^* = \min_{k \in \mathcal{C}} S_\ell^{(k)}$ and retains $\mathcal{C}' = \{k \in \mathcal{C} : S_\ell^{(k)} \leq S^* + \varepsilon_\ell\}$. Both the minimum operation and the threshold filter depend only on the *set of tier scores*, not on the order in which candidates are enumerated. The residual aggregate tie-break $\arg\min_{k \in \mathcal{C}} \sum_\ell S_\ell^{(k)}$ is also order-independent, and the final exact-tie fallback uses a fixed deterministic rule (lowest candidate index). Therefore the output is invariant to input permutation. □ □

The first proposition ensures that the selection is a well-defined function of the candidate *set*, not an artifact of enumeration order.

Remark (Scope of order invariance)**.** *Order invariance guarantees that Algorithm 1 produces the same selected index k^* regardless of the enumeration order of the K candidates. It does* not *assert that k^* is the unique global optimum under a transitive total order. Because the tolerance relation $S_\ell^{(i)} \approx_\varepsilon S_\ell^{(j)}$ (Section III-G) is* not *transitive, the output depends on which candidates survive each tier's pruning set $\mathcal{C}_\ell$; this set is determined by the minimum tier score in the current candidate pool, not by pairwise comparisons. In practice, with $\varepsilon_\ell \leq 10^{-3}$ and $K \leq 12$ (the main paper uses K=6), any non-transitive chain can involve only a small finite subset of candidates, and surviving candidates are typically numerically close enough that the residual aggregate tie-break resolves them deterministically.*

The second proposition addresses the practically important case in which the candidate set contains no fully compliant option.

Proposition III.7 (Infeasibility Fallback)**.** *When all K candidates violate Safety-tier rules (i.e., $S_0^{(k)} > 0$ for all $k \in [K]$), Algorithm 1 selects the candidate with minimum Safety-tier score within tolerance ε_0, breaking ties via lower tiers and then confidence. Formally, the algorithm returns the* least-violating *candidate under the full lexicographic ordering.*

Proof. The filter at tier 0 retains all candidates within ε_0 of $\min_k S_0^{(k)}$. Subsequent tiers further discriminate among these survivors. The procedure terminates with the candidate that minimizes the tier score vector lexicographically, which is, by construction, the least-violating trajectory. □ □

Remark (Infeasibility signalling). When the selected trajectory has $S_0 > 0$ (Safety-tier violation), the system detects infeasibility. The implementation now emits an explicit `infeasible` flag when the selected candidate has non-zero Safety-tier score, enabling downstream fail-safe logic such as emergency braking or conservative fallback.

The first two propositions address the selector's logical correctness. The third shows that the lexicographic ordering can be implemented efficiently on GPU hardware via a scalar surrogate, without sacrificing correctness.

Proposition III.8 (scalarization Correctness)**.** *The GPU-friendly scalarization $\mathrm{Score}^{(k)} = \sum_{\ell=0}^{3} B^{4-\ell} S_\ell^{(k)}$ with base B is order-preserving with respect to exact lexicographic ordering (i.e., $\varepsilon_\ell = 0$ for all ℓ) if and only if $\max_{k,\ell} S_\ell^{(k)} < B/(L-1)$.*

Proof. For exact lexicographic ordering, candidate i is preferred to j iff there exists ℓ^* such that $S_\ell^{(i)} = S_\ell^{(j)}$ for $\ell < \ell^*$ and $S_{\ell^*}^{(i)} < S_{\ell^*}^{(j)}$. Under the scalarization, the difference at tier ℓ^* contributes at least $B^{4-\ell^*}$ to the score gap, while all lower tiers together contribute at most $(L-\ell^*-1) \cdot \max_\ell S_\ell \cdot B^{3-\ell^*}$. This is dominated whenever $(L-1) \cdot \max_\ell S_\ell < B$, i.e. $\max_\ell S_\ell < B/(L-1)$.

In the RECTOR implementation, each tier score $S_\ell^{(k)}$ is a *normalized weighted average* of per-rule severities: $S_\ell^{(k)} = \sum_{r:\ell(r)=\ell} \tilde{w}_r \bar{V}_r^{(k)}$ where the intra-tier weights satisfy $\tilde{w}_r \geq 0$, $\sum_r \tilde{w}_r = 1$, and each normalized violation $\bar{V}_r^{(k)} \in [0,1]$ (see

Section V-E). Consequently, $\max_{k,\ell} S_\ell^{(k)} \leq 1$ regardless of the number of rules in any tier—in particular, regardless of the 14 rules in the Comfort tier. With $B = 1000$ and $L = 4$ the sufficient condition becomes $\max_{k,\ell} S_\ell^{(k)} \leq 1 < 1000/3 \approx 333$, which holds; any $B > L - 1 = 3$ would suffice. □ □

Corollary III.9. *The bound $\max_\ell S_\ell < B/(L-1)$ is independent of the number of rules per tier as long as tier scores are normalized to $[0, 1]$. A tier with 14 rules does not weaken the guarantee; only the normalization contract must be maintained by the proxy aggregator.*

Remark (Normalization contract and choice of B)**.** *Because tier scores are normalized to $[0, 1]$ by the intra-tier weight constraint $\sum_{r:\ell(r)=\ell} \tilde{w}_r = 1$ (Section III-F), the sufficient condition $\max_{k,\ell} S_\ell^{(k)} \leq 1 < B/(L-1)$ is automatically satisfied for* any *$B > L-1 = 3$. In other words,* ***Proposition III.8 is a straightforward consequence of the normalization contract****: given that tier scores lie in $[0, 1]$ by construction, the scalarization is order-preserving for any base $B > 3$, and no empirical verification or careful parameter choice is required. The reference implementation uses B=1000 for a single practical reason: it provides a $1000\times$ separation between adjacent tiers, giving numerical headroom for floating-point accumulation over $K \leq 6$ candidates in batched GPU execution. The maximum composite score ($B^4 = 10^{12}$) remains well within IEEE 754 single-precision range ($\approx 3.4 \times 10^{38}$). However, one subtlety deserves mention: with a Comfort-tier multiplier of 10^0=1 and Safety-tier contributions up to 10^{12}, the representable Comfort-tier resolution in the composite score is approximately 10^{-9} relative to the Safety contribution, which may cause correct Comfort-tier tie-breaking to be lost to floating-point cancellation when Safety-tier scores differ.* ***The scalarization is therefore used only as a GPU pre-filter****; final selection always uses the procedural lexicographic algorithm (Algorithm 2), which compares tier scores sequentially and is immune to floating-point cancellation. In practice, this is further mitigated by the tolerance-based pruning (Definition III.5), which resolves near-ties before the scalarization is consulted.*

Taken together, these three results establish that lexicographic selection is well-defined (Proposition III.6), gracefully degrading (Proposition III.7), and efficiently implementable (Proposition III.8). The "cannot versus does not" distinction follows directly: no re-parameterization of the lexicographic selector can cause it to trade a Safety-tier violation for a Comfort-tier gain, because the tier filtering is applied *before* lower tiers are consulted. A weighted-sum scalarization, by contrast, can always be reparameterized to produce such a trade-off, even if its current weights happen to avoid it.

J. Comparison with Alternative Selection Strategies

To make the formal properties above concrete, it is useful to compare lexicographic selection against two common alternatives that serve as baselines in our experiments.

Confidence-only selection chooses the most likely candidate:

$$\tau_{\text{conf}} = \arg\max_{\tau \in \mathcal{T}} p(\tau), \tag{12}$$

which is computationally simple but does not encode acceptability.

Weighted-sum selection aggregates rule penalties into a single scalar:

$$\tau_{\text{ws}} = \arg\min_{\tau \in \mathcal{T}} \sum_{r \in \mathcal{R}} w_r \, \hat{a}_r(s) \, \bar{V}_r(\tau; s). \tag{13}$$

All baselines use the same predicted applicability $\hat{a}_r(s)$ as RECTOR to ensure a fair comparison. While the weighted-sum approach is flexible, it introduces implicit trade-offs via w_r, and small changes in weights can reorder solutions in undesirable ways, especially across tiers with different semantics. Table III summarizes these strategies and highlights the structural role of tiered ordering in RECTOR.

K. Full-Catalog Evaluation Context

Before stating the formal problem, one further methodological distinction must be made because it affects how results are interpreted throughout the paper. In practice, not all rules have equally convenient smooth proxies, and some require semantic signals or discrete conditions. We therefore distinguish between two evaluation modes:

1) **Proxy-based selection** (Protocol A), where $\hat{a}_r(s)$ and smooth evaluators are used to rank candidates robustly across 24/28 rules, and
2) **Full-catalog auditing** (Protocol B), where the complete Waymo-style evaluator is applied to the selected trajectory using oracle applicability $a_r(s)$ across all 28 rules.

A key point is that human driving in the dataset is not violation-free under either protocol; therefore, the goal is not "zero violations," but a consistent reduction in high-priority (Safety/Legal) violations while preserving competitive accuracy. The proxy-based evaluation (24/28 rules) used for selection provides the appropriate like-for-like comparison across strategies. When the full 28-rule evaluator is applied, absolute violation rates increase for all methods—including ground-truth human trajectories—due to the stricter evaluator and the inclusion of rules without proxies. This methodological difference means that full-catalog absolute rates are not directly interpretable for safety comparisons between methods; see Section IX for further discussion.

L. Formal Problem Statement

With all definitions, properties, and evaluation context established, we can now state the rule-aware selection problem concisely as the objective that RECTOR optimizes.

Problem 1 (Rule-Aware Trajectory Selection)**.** *The rule-aware selection problem is specified by a five-component tuple $\langle f_\theta,\ \mathcal{R},\ \hat{a},\ \bar{V},\ \textsc{Select}\rangle$:*

(i) ***Generator*** *f_θ: maps scenario $s \sim \mathcal{D}$ to candidates and confidences $(\mathcal{T}, \mathbf{p})$;*
(ii) ***Rule catalog*** *$\mathcal{R}$ with tier mapping $\ell\colon \mathcal{R} \to \{0, 1, 2, 3\}$;*
(iii) ***Applicability predictor*** *$\hat{a}_r(s) \in \{0, 1\}$: a learned neural head that predicts which rules are active in scenario s (approximation source 1; see Section IV); smooth proxies*

TABLE III: Selection Strategy Comparison (Qualitative)

Strategy	Selection Criterion	Key Limitation	Interpretability
Confidence Only	$\arg\max p^{(k)}$	No safety or legality awareness	Low
Weighted Sum	$\arg\min \sum_r w_r a_r \bar{V}_r$	Weight tuning induces trade-offs	Medium
Lexicographic (RECTOR)	$\arg\min \mathbf{S}(\tau; s)$ under $\prec_{\text{lex},\epsilon}$	Requires tier structure and tolerances	High

that approximate full rule-violation semantics (approximation source 2; see Section V);

(iv) ***Selector*** SELECT*: tolerance-based lexicographic selection (Definition III.5) over tier scores* $\mathbf{S}(\tau; s)$ *aggregated from (ii)–(iv).*

The objective is to select $\hat{\tau} \in \mathcal{T}$ *that minimizes the tier score vector under the procedural lexicographic ordering:*

$$\hat{\tau} = \textsc{Select}\big(\mathcal{T},\ \mathbf{S}(\cdot\,; s),\ \mathbf{p},\ \boldsymbol{\varepsilon}\big). \tag{14}$$

The tuple formulation makes explicit that the deployed system involves two sources of approximation—applicability prediction (iii) and proxy evaluation (iv)—that can each introduce gaps relative to the idealised selection objective. Correctness of the selector (v) is established by Propositions III.6–III.8 under the assumption that tier scores are exact; the empirical impact of the approximation gaps is quantified via the dual-protocol evaluation (Section III-K) and the proxy-alignment analysis in Section V.

This formulation makes the separation between *generation* and *selection* explicit and enables the central claim of this work: when multi-modal diversity exists, rule-aware selection can convert that diversity into measurable compliance gains. In the present benchmark, the lexicographic selector contributes a formal priority guarantee and auditability rather than an empirical compliance separation from weighted-sum. Section IV describes the architecture that implements this objective, and Section VII presents the empirical evidence.

IV. RECTOR ARCHITECTURE

This section specifies the architecture that implements the rule-aware selection problem defined in Section III. RECTOR augments the multi-modal trajectory generator with two rule-aware components designed for auditability and real-time performance: an applicability mechanism and a deterministic tiered scorer. The reference implementation uses a learned applicability head, but the results section later shows that a parameter-free always-on policy is preferable for Safety and Legal tiers on the current benchmark. The separation between generator and selector is deliberate—it enables compatibility with any upstream trajectory model while keeping the compliance logic explicit, auditable, and stable under distribution shift.

The section is organized to follow the data through the system. We begin with a system overview and parameter budget (Section IV-A), then describe each architectural component in order of the inference pipeline: the scene encoder that produces a shared embedding (Section IV-B), the Transformer–CVAE decoder that generates candidate trajectories (Section IV-C), the applicability mechanism used in the reference model (Section IV-D), the tiered scorer that implements lexicographic selection (Section IV-E), and proxy validation against the full evaluator (Section IV-F). We then describe the training objective (Section IV-G), the inference procedure (Section IV-H), and close with a summary of how these components support the paper's central claim (Section IV-I).

A. System Overview

Given a scenario $s{=}(\mathbf{x}_{\text{ego}}, \mathbf{X}_{\text{agents}}, \mathcal{M})$, RECTOR follows a modular three-stage pipeline (Fig. 1):

1) **Scene Encoding & Candidate Generation:** An M2I-style encoder (PointNet + Transformer) converts heterogeneous history and map context into a dense $D{=}256$-dimensional scene embedding $\mathbf{e}_{\text{scene}}$. A Transformer–CVAE decoder then generates $K{=}6$ diverse ego-trajectory hypotheses $\{\boldsymbol{\tau}^{(k)}\}_{k=1}^{K}$, each with an associated confidence score $p^{(k)}$.
2) **Applicability Mechanism:** In parallel with trajectory generation, the reference model uses a 3.33 M-parameter neural head to predict which of the $R{=}28$ cataloged traffic rules are relevant to the current scenario, yielding the applicability mask $\hat{\mathbf{a}} \in \{0,1\}^R$. Later results show that this learned component is dominated by a parameter-free always-on policy for Safety and Legal tiers, so it should be understood as the reference implementation studied here rather than as the recommended deployment default.
3) **Tiered Lexicographic Scorer:** Each candidate is evaluated against the active rule set using smooth proxies; the scorer selects the trajectory with the minimal violation vector under the strict priority hierarchy Safety $\succ$ Legal $\succ$ Road $\succ$ Comfort (Section III-G).

We follow the standard supervised forecasting protocol: the generator conditions on a fixed history window of ego and neighbor-agent states over $T_h{=}11$ timesteps (1.1 s) and predicts $T_f{=}50$ future steps (5.0 s) at 10 Hz. Training minimizes a composite loss over trajectory generation and rule applicability (Section IV-G); rule signals are used primarily for *selection-time* evaluation, with proxy-based shaping treated as an optional weak regulariser.

Table IV summarizes the parameter distribution across components, highlighting an important design property: the applicability head adds substantial capacity (3.33 M parameters, 37.7%), whereas the tiered scorer itself contains no trainable parameters. This distinction matters because the results later show that the paper's robust empirical gains come from structured decision logic, while the learned applicability head is not the preferred policy for Safety and Legal deployment on this benchmark. The following subsections detail each component in turn.

Note: "Applicability Head + Scorer" includes the applicability MLP and the tiered scorer (fixed threshold/tolerance constants with zero

TABLE IV: Parameter Breakdown for the Reference Instantiation (CVAE Generator + RECTOR Selection Layer)

Component	Parameters	Fraction (%)
Scene Encoder	324,288	3.7
Trajectory Decoder	5,182,165	58.7
Applicability Head + Scorer	3,327,289	37.6
Total	**8,833,742**	**100.0**

trainable parameters). PriorNet and PosteriorNet are submodules of the CVAE trajectory decoder; their parameters are included in the decoder count. The 8.83 M total is from the canonical two-stage checkpoint (hash c90d4457e4996d22) used in all experiments. The scene encoder contains 324,288 parameters; the trajectory decoder 5,182,165; and the applicability head (MLP) together with the parameter-free tiered scorer and differentiable selector 3,327,289. The earlier single-stage configuration used a smaller PointNet encoder; both versions share the same decoder and applicability head architecture.

B. Scene Encoder

The first component in the pipeline is the scene encoder, which maps ego history, neighbor agent histories, and structured map context into a fixed-dimensional embedding $\mathbf{e}_{\text{scene}} \in \mathbb{R}^D$ (with $D{=}256$). We represent map entities (lanes, boundaries, crosswalks, stop lines, and signals) as polylines and treat each polyline as a set of points with semantic features; this vectorised representation supports scalable attention-based fusion across modalities.

Concretely, the encoder comprises three sub-encoders and a fusion stage (see Table IV for the parameter budget):

- **Agent encoder (PointNet-style):** encodes ego and neighbor agent histories into per-agent tokens via temporal encoding and pooling.
- **Map polyline encoder (PolylineNet):** encodes lane centrelines, boundaries, crosswalks, and other map features into per-polyline tokens via pointwise MLP and max-pooling.
- **Transformer fusion with cross-attention:** a multi-head Transformer encoder over the concatenated agent and map tokens to yield the global scene embedding $\mathbf{e}_{\text{scene}}$.

A compact formulation is:

$$\mathbf{h}_{\text{ego}} = \text{Conv1D}(\mathbf{x}_{\text{ego}}, \text{kernel} = 3) \in \mathbb{R}^{B\times 1\times D}, \tag{15}$$

$$\mathbf{h}_{\text{lane},\ell} = \max_{p}\ \text{MLP}\big(\mathcal{M}_{\text{lanes}}[\ell, p, :]\big) \in \mathbb{R}^{B\times 1\times D}, \tag{16}$$

$$\mathbf{X}_{\text{tok}} = \text{TransformerEncoder}\big(\mathbf{h}_{\text{ego}} \oplus \mathbf{H}_{\text{agents}} \oplus \mathbf{H}_{\text{lanes}}\big) \in \mathbb{R}^{B\times(1+N+L)\times D}, \tag{17}$$

$$\mathbf{e}_{\text{scene}} = \text{QueryAttn}(\mathbf{q}_{\text{ego}}, \mathbf{X}_{\text{tok}}, \mathbf{X}_{\text{tok}}) \in \mathbb{R}^{B\times D}, \tag{18}$$

where N is the number of agent tokens, L is the number of lane-polyline tokens, $\oplus$ concatenates along the token dimension, and $\mathbf{q}_{\text{ego}}$ is a learned ego query used to pool the fused token set into a single scene embedding. The encoder is intentionally modest in capacity (Table IV), because the compliance gains should not rely on inflating the backbone; they should arise from rule-aware selection. The scene embedding $\mathbf{e}_{\text{scene}}$ serves as the shared representation consumed by both the trajectory decoder and the applicability head—a branching point that is central to the architecture.

Reproducibility details. The Conv1D ego encoder uses 3 layers with kernel size 3, channels $[D_{\text{in}}, 128, D]$, ReLU activations, and batch normalization. The PolylineNet MLP processes each polyline point through a 2-layer MLP ($[D_{\text{in}}, 128, D]$) followed by channel-wise max-pooling. The Transformer fusion layer uses 2 encoder layers with 8 attention heads, $D{=}256$ model dimension, feed-forward dimension 512, and dropout 0.1. Agent history features comprise $(x, y, \cos\theta, \sin\theta, v_x, v_y, \text{width}, \text{length})$ per timestep ($D_{\text{in}}{=}8$); lane features comprise $(x, y, \text{type}, \text{speed_limit})$ per polyline point ($D_{\text{in}}{=}4$).

C. Trajectory Decoder

From the scene embedding, the trajectory decoder generates a candidate set of size $K{=}6$ over a horizon of $T{=}50$ timesteps at 10 Hz. RECTOR uses a Transformer–CVAE decoder that combines three mechanisms: (i) query-based goal conditioning for multi-modality, (ii) a latent variable $\mathbf{z} \in \mathbb{R}^{d_z}$ (with $d_z{=}64$) for residual uncertainty, and (iii) Transformer decoding for temporally coherent trajectories.

Goal conditioning and endpoint supervision. We generate K mode-specific goal embeddings using learnable queries $\mathbf{q}^{(k)} \in \mathbb{R}^D$:

$$\mathbf{g}_{\text{emb}}^{(k)} = \text{GoalNet}(\mathbf{e}_{\text{scene}}, \mathbf{q}^{(k)}), \quad k \in [K]. \tag{19}$$

A separate linear head predicts the 2D endpoint for supervision:

$$\mathbf{g}_{xy}^{(k)} = \text{Linear}_{D\to 2}\big(\mathbf{g}_{\text{emb}}^{(k)}\big) \in \mathbb{R}^2. \tag{20}$$

CVAE latent variable. To capture residual uncertainty, we model a shared Gaussian prior and posterior conditioned on the scene embedding alone:

$$p(\mathbf{z} \mid \mathbf{e}_{\text{scene}}) = \mathcal{N}(\boldsymbol{\mu}_{\text{prior}}, \text{diag}(\boldsymbol{\sigma}_{\text{prior}}^2)), \tag{21}$$

$$q(\mathbf{z} \mid \mathbf{e}_{\text{scene}}, \boldsymbol{\tau}^*) = \mathcal{N}(\boldsymbol{\mu}_{\text{post}}, \text{diag}(\boldsymbol{\sigma}_{\text{post}}^2)), \tag{22}$$

with parameters produced by lightweight MLPs (PriorNet and PosteriorNet). Mode-specific conditioning is provided by the distinct goal embeddings $\mathbf{g}_{\text{emb}}^{(k)}$ at the decoder stage rather than through separate per-mode latent distributions. During training, the posterior conditions on the ground-truth trajectory $\boldsymbol{\tau}^*$; at inference, independent samples are drawn from the shared prior for each mode.

Trajectory decoding and confidence. Each mode is decoded by conditioning on $(\mathbf{e}_{\text{scene}}, \mathbf{g}_{\text{emb}}^{(k)}, \mathbf{z}^{(k)})$. The Transformer decoder uses $L_{\text{dec}}{=}4$ layers with 8 attention heads, model dimension $D{=}256$, and feed-forward dimension 512. The GoalNet is a 2-layer MLP ($[D{+}D, 256, D]$) with ReLU. PriorNet and PosteriorNet are each 2-layer MLPs mapping from D (or $D{+}T{\times}2$) to $2d_z{=}128$ outputs (μ and $\log\sigma^2$). The decoder produces a full ego state sequence (x, y, θ, v) over the horizon, matching the $[T_f, 4]$ output shown in Fig. 3. Higher-order kinematic quantities required by some rules (e.g., acceleration, jerk, yaw rate) are then computed from this state sequence via discrete-time finite differences during proxy evaluation (Section V). A confidence head (linear $D{\to}1$) outputs logits, normalized into $p^{(k)}$ via softmax:

$$p^{(k)} = \frac{\exp(\text{logit}^{(k)})}{\sum_{j=1}^{K}\exp(\text{logit}^{(j)})}. \tag{23}$$

Fig. 3 details the decoder structure. At this point in the pipeline, the generator has produced $K{=}6$ diverse candidate trajectories and their associated confidences. The next two components—the applicability head and the tiered scorer—determine which of these candidates is selected for execution.

D. Applicability Head

The applicability head should be read as a *tested reference component*, not as the paper's strongest practical recommendation. In the reference model it predicts, from the scene embedding alone, which of the $R{=}28$ cataloged rules are relevant in the current scenario so that irrelevant rules do not influence the tier scores. As a secondary benefit, applicability gating can skip unnecessary rule evaluations when the catalog grows.

Formally, the head produces continuous applicability scores and thresholds them for inference-time gating:

$$\tilde{\mathbf{a}} = \sigma(\text{MLP}_{\text{app}}(\mathbf{e}_{\text{scene}})) \in [0, 1]^R, \tag{24}$$

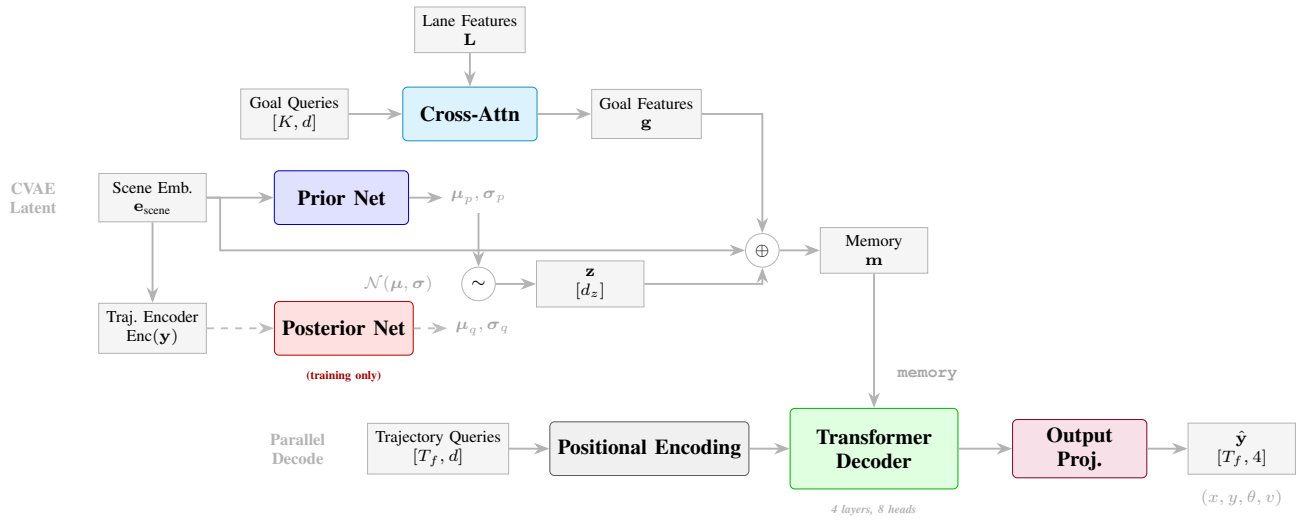


Fig. 3: **Transformer–CVAE Decoder.** Goal queries attend to lane features via cross-attention to produce goal features $\mathbf{g}$. The scene embedding feeds a prior network $p(\mathbf{z}|\mathcal{C})$; during training, a trajectory encoder and posterior network $q(\mathbf{z}|\mathcal{C}, \mathbf{y})$ provide the latent $\mathbf{z}$ (dashed path). The latent, goal features, and scene embedding are concatenated into a memory vector. Learnable trajectory queries with positional encoding attend to this memory through 4 transformer decoder layers, producing T_f=50 timesteps in parallel. The output projection yields ego trajectory states (x, y, θ, v).

with $\hat{\mathbf{a}} \in \{0,1\}^R$ obtained by thresholding the logits. During training, applicability is supervised using oracle labels $\mathbf{a}^* \in \{0,1\}^R$ derived from map and agent annotations:

$$\mathcal{L}_{\text{app}} = -\frac{1}{R}\sum_{r=1}^{R}\left[a_r^* \log(\tilde{a}_r) + (1-a_r^*)\log(1-\tilde{a}_r)\right]. \tag{25}$$

Architecturally, the implementation uses a tier-aware query mechanism over the shared scene representation, followed by a lightweight cross-tier refinement. Those details are retained for reproducibility, but the empirical takeaway is now clear from Section VII: the learned head is not the preferred policy for Safety and Legal tiers on this benchmark. Under oracle cross-evaluation, a parameter-free always-on policy strictly dominates the 3.33 M-parameter learned head on both compliance and selected-trajectory accuracy, so the learned module should be regarded as a negative result with residual value limited to Road/Comfort gating, auditable active-rule prediction, and future calibration work.

This result materially changes how the architecture should be interpreted. The compliance case for RECTOR does *not* rest on the applicability head; it rests on the deterministic tiered scorer and the proxy-evaluation stack. The learned head remains part of the reference implementation because it is how the experiments were run, but the deployment recommendation supported by the present evidence is conservative always-on activation for Safety and Legal tiers.

E. Tiered Scorer

With the applicability mask $\hat{\mathbf{a}}$ provided by the head (Section IV-D), the tiered scorer implements the prioritized ranking formalized in Section III-G. Tier scores are aggregated as:

$$S_\ell^{(k)}(s) = \sum_{r:\,\ell(r)=\ell} \tilde{w}_r\, \hat{a}_r(s) \cdot \bar{V}_{r,k}(s), \quad \ell \in \{0,1,2,3\}, \tag{26}$$

where $\tilde{w}_r \geq 0$ are per-rule weights normalized within each tier so that $\sum_{r:\,\ell(r)=\ell} \tilde{w}_r = 1$. This normalization ensures $S_\ell^{(k)}(s) \in [0,1]$ for all k and ℓ, regardless of the number of rules in the tier—a condition required by Proposition III.8. In the reference implementation (`tiered_scorer.py`), the weights $\tilde{w}_r$ default to uniform ($1/n_\ell$ where n_ℓ is the number of rules in tier ℓ) and are normalized via `weights / weights.sum()`.

Selection. The selector applies tolerance-based lexicographic ranking (Algorithm 1), which preserves priorities and prevents lower-tier improvements from masking higher-tier violations. Remaining near-ties are resolved by the summed residual tier score over the surviving candidates, with candidate index used only as a final exact-tie fallback.

Algorithm 1 Lexicographic Trajectory Selection with Tolerance

Input: Tier score vectors $\{\mathbf{S}^{(k)}\}_{k=1}^{K}$, tolerances $\{\varepsilon_\ell\}_{\ell=0}^{3}$
Output: Selected index k^*
$\mathcal{C} \leftarrow \{1, \ldots, K\}$ ▷ Candidate set
for $\ell = 0$ **to** 3 **do**
  $S^* \leftarrow \min_{k\in\mathcal{C}} S_\ell^{(k)}$
  $\mathcal{C} \leftarrow \{k \in \mathcal{C} : S_\ell^{(k)} \leq S^* + \varepsilon_\ell\}$
  if $|\mathcal{C}| = 1$ **then**
    break
  end if
end for
$k^* \leftarrow \arg\min_{k\in\mathcal{C}} \sum_{\ell=0}^{3} S_\ell^{(k)}$ ▷ Residual aggregate tie-break
If multiple k remain, return $\min$-index$(\mathcal{C})$
return k^*

GPU-friendly scalarization (optional). For efficient batched execution, we also compute an order-preserving scalar score:

$$\text{Score}^{(k)} = \sum_{\ell=0}^{3} B^{4-\ell}\, S_\ell^{(k)}, \tag{27}$$

with B=1000, yielding multipliers $10^{12}, 10^9, 10^6, 10^3$ for tiers 0–3 respectively (see `tiered_scorer.py`). This scalar is used for a fast pre-ranking, followed by the exact Algorithm 1 comparator to preserve correctness in the presence of ties and numeric noise. An important practical question is how sensitive the selection is to the choice of tolerance parameters. We investigate this next.

1) Tolerance Sensitivity Analysis: Tolerance parameters $\{\varepsilon_\ell\}$ control selection stability under near-ties. Setting all four tier tolerances to a common value ε and sweeping across five orders of magnitude shows that selection is stable for $\varepsilon \in [0, 10^{-2}]$: at both $\varepsilon{=}10^{-3}$ (the evaluation default) and $\varepsilon{=}10^{-2}$ (the code default), Safety, Legal, Total violation rates, and selADE are identical (15.03% Total, 13.15% S+L, 2.043 m selADE), because Safety-tier proxy magnitudes are well-separated across modes. Only at $\varepsilon \geq 0.05$ do S+L violations begin to increase. We use $\varepsilon{=}10^{-3}$ as the default for all experiments; Fig. 14 shows the full sweep.

F. Proxy Validation Against Full Evaluator

The tiered scorer relies on smooth proxies to achieve a stable ranking, but deployment requires that proxy-based ranking correlate with the full-catalog evaluator used for auditing. We validate this alignment by comparing proxy and full evaluator outputs on 1,000 `validation_interactive` scenarios with coordinate-aligned world-frame trajectories (Table XXXI, Section VIII).

Per-rule Spearman ρ between proxy and full evaluator varies widely: kinematic and interaction rules achieve moderate correlation ($\rho{=}0.36$–0.77), while spatial Safety-tier rules show near-zero or negative ρ due to fundamental differences in how the differentiable proxy and the discrete Waymo evaluator compute collision severity. However, the operationally relevant metric—*pairwise candidate-ranking concordance*—is 0.948 across all 24 proxied rules (0.875 for the 10 rules with measurable variance; Table XXXI), confirming that tier-aggregate proxy scores preserve the full evaluator's candidate ordering at the tier level. This separation between poor per-rule severity calibration and high aggregate ranking fidelity occurs because the lexicographic selector uses summed tier scores, not individual rule severities, and tier-level aggregation cancels much of the per-rule noise. Four of the 28 cataloged rules (14%) lack differentiable proxies because they require intent modeling, counterfactual reasoning, or multi-agent negotiation. These rules are excluded from proxy-based ranking and evaluated post-selection for audit reporting; an unproxied rule contributes a score of zero (no penalty), and the post-hoc audit flags any violations detected by the full evaluator.

G. Training Objective

The preceding subsections describe the inference-time architecture. We now turn to how the system is trained. RECTOR employs a **two-stage training procedure**:

Stage 1 (Applicability Head Pretraining, 20 epochs): The M2I-style encoder is frozen, and only the applicability head is trained using focal binary cross-entropy ($\gamma{=}2.0$) on oracle applicability labels derived from the WOMD rule evaluation framework. This stage produces the reference applicability predictor used in the experiments, even though later ablations show that conservative always-on activation is preferable for Safety and Legal tiers on the present benchmark.

Stage 2 (End-to-End Fine-Tuning, 20+5 epochs): The full model—encoder, decoder, and applicability head—is trained end-to-end with a composite loss that supervises trajectory generation, rule applicability prediction, and proxy-based compliance shaping:

$$\mathcal{L} = \lambda_{\text{WTA}}\mathcal{L}_{\text{WTA}} + \lambda_{\text{end}}\mathcal{L}_{\text{end}} + \lambda_{\text{KL}}\mathcal{L}_{\text{KL}} + \lambda_{\text{goal}}\mathcal{L}_{\text{goal}} + \lambda_{\text{smooth}}\mathcal{L}_{\text{smooth}} + \lambda_{\text{conf}}\mathcal{L}_{\text{conf}} + \lambda_{\text{app}}\mathcal{L}_{\text{app}} + \lambda_{\text{rule}}\mathcal{L}_{\text{rule}}. \tag{28}$$

All eight weights are applied at the same level (see Table XI for values). The first six terms constitute the trajectory generation objective; the last two terms supervise rule applicability and (optionally) proxy-based compliance. For readability, we group the trajectory terms as $\mathcal{L}_{\text{traj}}$ in the discussion below.

Trajectory loss. The trajectory loss combines winner-takes-all supervision, endpoint matching, KL regularization for the CVAE, goal supervision, smoothness, and confidence calibration:

$$\mathcal{L}_{\text{traj}} = \lambda_{\text{WTA}}\mathcal{L}_{\text{WTA}} + \lambda_{\text{end}}\mathcal{L}_{\text{end}} + \lambda_{\text{KL}}\mathcal{L}_{\text{KL}} + \lambda_{\text{goal}}\mathcal{L}_{\text{goal}} + \lambda_{\text{smooth}}\mathcal{L}_{\text{smooth}} + \lambda_{\text{conf}}\mathcal{L}_{\text{conf}}.$$

Winner-takes-all and endpoint losses supervise only the best-matching mode:

$$\mathcal{L}_{\text{WTA}} = \min_{k\in[K]} \frac{1}{T}\sum_{t=1}^{T} w_t\, \boldsymbol{\lambda}^\top H_\beta\Big(\boldsymbol{\tau}_t^{(k)} - \boldsymbol{\tau}_t^{*}\Big), \quad w_t = 1 + \tfrac{t}{T}, \tag{29}$$

where $H_\beta(\cdot)$ denotes element-wise Huber loss with $\beta{=}0.5$, and $\boldsymbol{\lambda}{=}[2, 2, 0.5, 0.5]^\top$ are dimension weights for (x, y, θ, v) that emphasise positional accuracy over heading and speed.

$$\mathcal{L}_{\text{end}} = H_\beta\Big(\boldsymbol{\tau}_T^{(k^*)} - \boldsymbol{\tau}_T^{*}\Big), \quad k^* = \arg\min_{k\in[K]} \text{ADE}^{(k)}, \tag{30}$$

and goal supervision aligns the closest predicted goal with the ground-truth final position:

$$\mathcal{L}_{\text{goal}} = H_\beta\Big(\mathbf{g}_{xy}^{(k^\dagger)} - \boldsymbol{\tau}_T^{*}\Big), \quad k^\dagger = \arg\min_{k\in[K]} \Big\|\mathbf{g}_{xy}^{(k)} - \boldsymbol{\tau}_T^{*}\Big\|_2. \tag{31}$$

A smoothness regulariser reduces high-frequency oscillations via third-order finite differences (jerk) on the position components:

$$\mathcal{L}_{\text{smooth}} = \frac{1}{T-3}\sum_{t=1}^{T-3} \big\|\boldsymbol{\tau}_{t+3}^{xy} - 3\boldsymbol{\tau}_{t+2}^{xy} + 3\boldsymbol{\tau}_{t+1}^{xy} - \boldsymbol{\tau}_t^{xy}\big\|_2, \tag{32}$$

where the superscript xy denotes the position components (x, y) only, and the third-order finite difference penalises jerk (rate of change of acceleration). Finally, confidence calibration uses cross-entropy with the best-ADE mode as the target:

$$\mathcal{L}_{\text{conf}} = -\sum_{k=1}^{K} y_k \log(p^{(k)}), \quad y_k = \mathbb{1}\Big[k = \arg\min_j \text{ADE}^{(j)}\Big]. \tag{33}$$

Applicability loss. $\mathcal{L}_{\text{app}}$ is the focal binary cross-entropy defined in Section IV-D, with focusing parameter $\gamma{=}2.0$ to handle the class imbalance inherent in rule applicability labels. This loss is active in both training stages: as the sole objective in Stage 1 (pretraining) and as one of eight terms in Stage 2 (end-to-end).

Rule-proxy shaping. $\mathcal{L}_{\text{rule}}$ backpropagates through smooth proxies to encourage the generator to allocate probability mass toward more compliant candidates, without replacing lexicographic selection at inference. We treat this term as a weak regulariser ($\lambda_{\text{rule}}{=}1.0$) to avoid collapsing multi-modality; the compliance gains reported in Section VII are therefore primarily attributable to selection-time enforcement rather than to training-time shaping. Training hyperparameters are summarised in Section VI (Table XI).

H. Inference Procedure

With the training objective defined, the inference procedure follows naturally as the forward pass without the posterior network or the training losses (Algorithm 2). The key operational property is that selection is deterministic given $(\mathcal{T}, \mathbf{p})$ and the predicted applicability mask $\hat{\mathbf{a}}$. This supports regression testing under rule updates and enables localised debugging: the first tier that differentiates candidates is directly observable.

Latent sampling note. At inference, the latent variable $\mathbf{z}^{(k)}$ is drawn stochastically from the shared prior $p(\mathbf{z} \mid \mathbf{e}_{\text{scene}})$ for each mode k via the reparameterization trick. Mode diversity is produced jointly by the $K{=}6$ learnable goal queries $\mathbf{q}^{(k)}$, which condition distinct goal embeddings, and by independent latent samples $\boldsymbol{\epsilon}^{(k)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ drawn per mode. Using the prior mean $\boldsymbol{\mu}_{\text{prior}}$ for all modes was found to collapse diversity, since the decoder then receives identical latent inputs and produces nearly identical trajectories. Stochastic per-mode sampling is therefore required for effective candidate generation. For reproducibility purposes, experiments fix the random seed prior to inference (Section VI).

Latency is dominated by trajectory generation, while compliance evaluation remains a minor fraction of total compute (7.3 ms mean, Table XXVI in Section VII). This matches the design intent: compliance logic should be auditable and cheap, not a second heavyweight model. Detailed ablation studies and component impact analysis are presented in Section VII-G.

Algorithm 2 RECTOR Inference

1: **Input:** Scenario $s = (\mathbf{x}_{\text{ego}}, \mathbf{X}_{\text{agents}}, \mathcal{M})$, trained parameters θ
2: **Output:** Selected trajectory index and confidence $(k^*, p^{(k^*)})$
3: $\mathbf{e}_{\text{scene}} \leftarrow \text{SceneEncoder}(s)$
4: **for** $k = 1$ **to** K **do**
5: $\quad \mathbf{g}_{\text{emb}}^{(k)} \leftarrow \text{GoalNet}(\mathbf{e}_{\text{scene}}, \mathbf{q}^{(k)})$
6: $\quad \mathbf{z}^{(k)} \sim p(\mathbf{z} \mid \mathbf{e}_{\text{scene}})$ ▷ Independent sample per mode; see note below
7: $\quad \boldsymbol{\tau}^{(k)} \leftarrow \text{TrajectoryDecoder}(\mathbf{e}_{\text{scene}}, \mathbf{g}_{\text{emb}}^{(k)}, \mathbf{z}^{(k)})$
8: $\quad \text{logit}^{(k)} \leftarrow \text{ConfidenceHead}(\mathbf{e}_{\text{scene}}, \mathbf{g}_{\text{emb}}^{(k)})$
9: **end for**
10: $\mathbf{p} \leftarrow \text{softmax}(\text{logit}^{(1..K)})$
11: $\tilde{\mathbf{a}} \leftarrow \sigma(\text{ApplicabilityHead}(\mathbf{e}_{\text{scene}}))$
12: $\hat{\mathbf{a}} \leftarrow \mathbb{1}[\tilde{\mathbf{a}} > \text{thresholds by tier}]$
13: **for** $k = 1$ **to** K **do**
14: $\quad$ **for** $r = 1$ **to** R **do**
15: $\quad\quad$ **if** $\hat{a}_r = 1$ **then**
16: $\quad\quad\quad \bar{V}_{r,k} \leftarrow \text{RuleProxy}_r(\boldsymbol{\tau}^{(k)}, s)$
17: $\quad\quad$ **else**
18: $\quad\quad\quad \bar{V}_{r,k} \leftarrow 0$
19: $\quad\quad$ **end if**
20: $\quad$ **end for**
21: **end for**
22: **for** $k = 1$ **to** K **do**
23: $\quad$ **for** $\ell = 0$ **to** 3 **do**
24: $\quad\quad S_\ell^{(k)} \leftarrow \sum_{r:\, \ell(r)=\ell} \tilde{w}_r \, \hat{a}_r \, \bar{V}_{r,k}$ ▷ matches Eq. (26); $\hat{a}_r$ pre-applied at lines 15–22 so $\hat{a}_r{=}0 \Rightarrow \bar{V}_{r,k}{=}0$
25: $\quad$ **end for**
26: **end for**
27: $k^* \leftarrow \text{LexicographicSelect}(\{\mathbf{S}^{(k)}\}_{k=1}^{K}, \varepsilon)$
28: **return** $(k^*, p^{(k^*)})$ ▷ $p^{(k^*)}$ reported for information only; not used in selection

I. Architecture Summary

In summary, RECTOR implements the prioritized rule-aware selection objective with minimal compliance overhead relative to the trajectory generator (Table IV). The architecture is intentionally modular: the **generator** (scene encoder + Transformer–CVAE decoder) targets candidate diversity and geometric fidelity; the **applicability mechanism** provides the reference gating path studied in this paper, though the learned instantiation is not the preferred Safety/Legal policy on this benchmark; the **tiered scorer** enforces traffic-rule priorities via deterministic lexicographic comparison; and the **full evaluator** remains an external audit instrument, enabling verification-oriented reporting beyond proxy coverage (Section IV-F).

This modular design directly supports the paper's central claim: when multi-modal diversity exists, explicit selection is the correct intervention point for improving rule compliance without sacrificing oracle accuracy. Section V next specifies the rule catalog and proxy implementations that the scorer consumes, before Section VI defines the experimental protocol.

V. Rule Evaluation and Data Pipeline

The preceding section described *how* RECTOR selects a trajectory: the applicability head, proxy evaluation, and lexicographic scorer operate on rule-level violation severities and applicability masks. This section specifies *what* those rules are and *how* their violations are computed—operationalizing the prioritized rulebook into 28 rules across four tiers, with differentiable smooth proxies for 24 of them.

Each tier has a distinct proxy design rationale. **Tier 0 (Safety, 5 rules)** focuses on collision avoidance and minimum clearance: collision avoidance (L0.R3) is approximated using a differentiable SAT-based oriented-bounding-box (OBB) penetration-depth proxy. **Tier 1 (Legal, 7 rules)** encodes traffic laws such as signal compliance and right-of-way; proxies shift from geometric distances to event-based detectors—for instance, Red Light Stop (L1.R3) penalises crossing a stop line when the signal state is red. **Tier 2 (Road, 2 rules)** enforces drivable-surface and lane-keeping constraints using signed distance functions (SDFs) to road polygons. **Tier 3 (Comfort, 14 rules)** enforces bounds on acceleration, jerk, and steering to maintain ride quality.

To maintain rigour, we apply the *Evaluation Dual-Protocol* introduced in Section III-K: (i) *Protocol A* uses 24 proxied rules with predicted applicability for training and selection, and (ii) *Protocol B* applies the full 28-rule Waymo evaluator with oracle applicability for reporting and verification. This allows RECTOR to be trained on continuous differentiable signals while being audited against the absolute standards of the full rule catalog.

A. Rule Catalog Overview

The rule catalog comprises 28 rules organized into four tiers consistent with the priority structure formalized in Section III. Tier 0 (Safety) captures physical safety and collision avoidance; Tier 1 (Legal) captures rules derived from traffic-law concepts in the WOMD annotation framework;[2] Tier 2 (Road) captures drivable-surface and lane-keeping constraints; and Tier 3 (Comfort) captures ride quality and context policies. Table V lists the complete catalog used in all experiments.

Rule ID convention. We denote rules as $\mathrm{L}\ell.\mathrm{R}i$, where ℓ is the priority *level* index (smaller ℓ indicates higher priority) and i enumerates rules within that level; the *Tier* column groups levels into the four coarse tiers used throughout the paper.

B. Per-Rule Applicability Analysis

Fig. 4 reports per-rule applicability F1 by tier on a diagnostic sample. The figure is used later to localize where the learned applicability head fails; the deployment-facing conclusion that conservative activation is preferable is supported by the full-split cross-evaluation in Section VII, not by this figure alone.

The observed support imbalance motivates the conservative applicability ablations in Section VII.

C. Waymo Rule Evaluation Framework

The per-rule analysis above relies on a ground-truth auditor that defines consistent rule semantics across scenarios. We adopt the Waymo Open Motion Dataset rule evaluation

[2]See the Legal-tier disclaimer in Section I-B.

TABLE V: Complete Rule Catalog with Tier Assignments (28 Rules)

Rule ID	Tier	Description	Waymo Module[‡]
Tier 0: Safety (5 rules)			
L0.R0	Safety	Safe longitudinal distance	`l0_safe_long_dist`
L0.R1	Safety	Safe lateral clearance	`l0_safe_lat_clear`
L0.R2	Safety	Crosswalk occupancy	`l0_crosswalk_occ`
L0.R3	Safety	Collision avoidance (overlap)	`l10_collision`
L0.R4	Safety	VRU clearance buffer	`l10_vru_clearance`
Tier 1: Legal (7 rules)			
L1.R0	Legal	Traffic signal compliance	`l5_signal_compliance`
L1.R1	Legal	Priority / right-of-way	`l5_priority_viol`
L1.R2	Legal	Speed limit adherence	`l07_speed_limit`
L1.R3	Legal	Red light stop compliance	`l8_red_light`
L1.R4	Legal	Stop sign compliance	`l08_stop_sign`
L1.R5	Legal	Crosswalk yield to pedestrians	`l08_crosswalk_yield`
L1.R6	Legal	Wrong-way driving prevention	`l08_wrongway`
Tier 2: Road (2 rules)			
L2.R0	Road	Drivable surface constraint	`l3_drivable_surface`
L2.R1	Road	Lane departure prevention	`l07_lane_departure`
Tier 3: Comfort (14 rules)			
L3.R0	Comfort	Smooth longitudinal acceleration	`l1_smooth_accel`
L3.R1	Comfort	Smooth braking deceleration	`l1_smooth_braking`
L3.R2	Comfort	Smooth lateral steering	`l1_smooth_steering`
L3.R3	Comfort	Speed consistency	`l1_speed_consistency`
L3.R4	Comfort	Jerk / lane-change smoothness	`l1_lane_change_smooth`
L3.R5	Comfort	Left turn gap acceptance	`l4_left_turn_gap`
L3.R6	Comfort	Parking zone violation	`l5_parking_viol`
L3.R7	Comfort	School zone speed compliance	`l5_school_zone`
L3.R8	Comfort	Construction zone compliance	`l5_construction_zone`
L3.R9	Comfort	Cooperative lane change	`l6_coop_lane_change`
L3.R10	Comfort	Safe following distance	`l6_following_dist`
L3.R11	Comfort	Intersection negotiation	`l6_intersection_neg`
L3.R12	Comfort	Pedestrian interaction	`l6_pedestrian_inter`
L3.R13	Comfort	Cyclist interaction	`l6_cyclist_inter`

86% of rules (24/28) have differentiable proxy implementations. The remaining 14% require intent modeling, counterfactual scene reasoning, or multi-agent negotiation and are handled via conservative applicability defaults. See Section V-D. [‡]**Waymo Module** gives the Python module name in the `waymo_rule_eval/rules/` package of the evaluation codebase. Figures reporting per-rule statistics (Figs. 4 and 11) use shortened bar labels derived from these module names. The paper's `Li.Rj` identifiers use a sequential within-tier enumeration; the canonical registry IDs in `waymo_rule_eval/rules/registry.py` differ because they follow the original Waymo rulebook numbering. The complete mapping is: **Safety**—L0.R0→`L0.R2`, L0.R1→`L0.R3`, L0.R2→`L0.R4`, L0.R3→`L10.R1`, L0.R4→`L10.R2`; **Legal**—L1.R0→`L5.R1`, L1.R1→`L5.R2`, L1.R2→`L7.R4`, L1.R3→`L8.R1`, L1.R4→`L8.R2`, L1.R5→`L8.R3`, L1.R6→`L8.R5`; **Road**—L2.R0→`L3.R3`, L2.R1→`L7.R3`; **Comfort**—L3.R0–R4→`L1.R1--R5`, L3.R5→`L4.R3`, L3.R6–R8→`L5.R3--R5`, L3.R9–R13→`L6.R1--R5`.

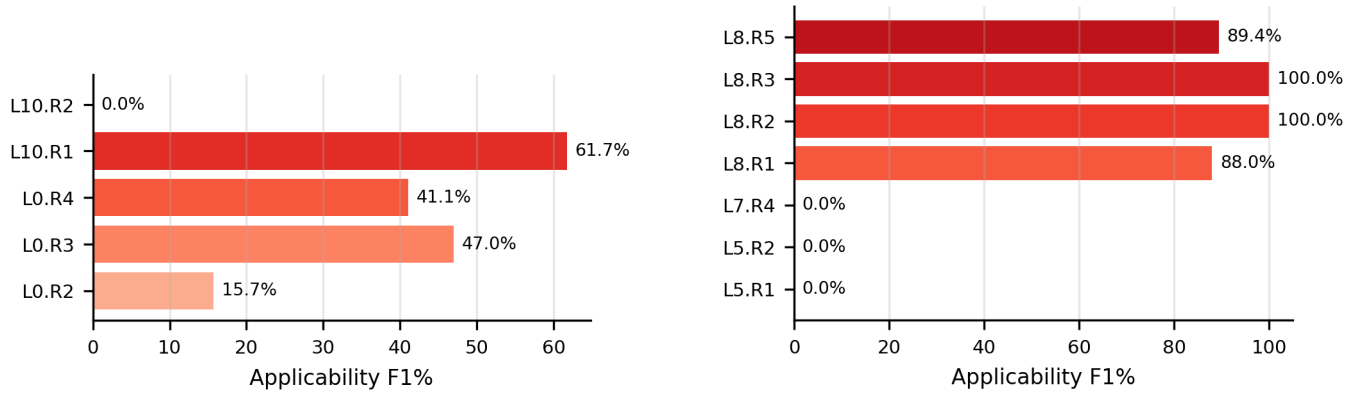
framework in this role, because it provides (i) standardised definitions for traffic-rule constraints, (ii) consistent applicability semantics across scenarios, (iii) comparability to prior WOMD-based methods, and (iv) a stable baseline for regression tests when the model or proxies change.

1) Map Feature Integration: The evaluator relies on structured map elements: lane centre-lines and boundaries, stop lines, traffic signals, crosswalk polygons, and drivable-surface boundaries. We extract these entities from the Waymo scenario protocol buffers and construct a compact, queryable representation used by both the full evaluator and the proxy layer (when available).

2) Scenario Context Construction: For each scenario, we build a context object that contains ego and agent trajectories, map features, signal states, and rule applicability metadata. During *evaluation*, we use oracle applicability a_r^* derived from the scenario context and evaluate the complete 28-rule catalog. During *training and selection*, we evaluate only the proxy subset and use predicted applicability $\hat{a}_r$ from Section IV. This separation prevents proxy limitations from silently altering audit metrics and ensures that the two protocols remain methodologically distinct as defined in Section III-K.

D. Smooth Proxy Functions

While the full evaluator provides the authoritative compliance verdict, it is non-differentiable and therefore cannot provide a gradient signal during training. RECTOR therefore uses smooth proxy functions to (i) support gradient flow for optional rule-shaped training losses (Section IV-G), and (ii) produce stable, continuous severity values for tier aggregation at selection time. Proxies approximate binary rule predicates with smooth penalties; while they cannot replicate every discrete semantic rule, they are sufficient to steer the generator away from recurrent geometric violations (e.g., clearance, lane departure, and speed).



(a) Tier 0 (Safety)

(b) Tier 1 (Legal)

Fig. 4: Per-rule applicability F1 scores for RECTOR predictions (WOMD v1.3.0 `validation_interactive`, 500 GT / 337 RECTOR scenarios), grouped by tier. Bar labels follow the Waymo evaluator's internal convention; the mapping to this paper's tier notation is given in Table V.

Proxy design principles. We convert thresholded predicates into smooth penalties using hinge-like terms $\max(0, \cdot)$ and soft indicator gates. When a rule depends on an event (e.g., crossing a stop line), we approximate the event with a differentiable boundary function and a smooth difference across timesteps. This yields informative gradients while keeping the proxy well-defined almost everywhere.

1) Proxy Coverage: Of the 28 cataloged rules, **24 have smooth proxy implementations** (86% coverage). All Tier 0 (Safety) rules are covered, ensuring that the highest-priority constraints contribute a training signal. Table VI summarizes the training versus evaluation protocol.

TABLE VI: Training vs. Evaluation Comparison

Aspect	Training	Evaluation
Rules evaluated	24 (proxied)	28 (all)
Applicability source	Predicted ($\hat{a}_r$)	Oracle (a_r^*)
Map context	Partial	Full
Gradient flow	Yes	No
Speed	Fast (∼1 ms)	Moderate (∼5 ms)

Note: Training uses smooth proxies for 24/28 rules to provide a gradient signal. Evaluation uses the full Waymo framework with complete map context for all 28 rules.

The four unproxied rules fall into two distinct categories. **L1.R1** (Priority/right-of-way) requires multi-agent intent modeling and counterfactual scene reasoning that cannot be reliably approximated with geometric operations alone. The remaining three—**L3.R6** (Parking zone), **L3.R7** (School zone speed), and **L3.R8** (Construction zone)—are computable from static map features in principle, but the WOMD v1.3.0 interactive scenarios do not reliably include the corresponding zone polygon annotations; without polygon ground truth, a differentiable proxy cannot compute zone membership and is therefore omitted. All four are handled via conservative applicability defaults in the selection pipeline and are included in the full-catalog evaluation (Protocol B).

2) Normalization of Proxy Severities: Proxy outputs have heterogeneous units (e.g., metres, metres/second, IoU). To make tier aggregation well-defined, each proxy produces a raw severity $V_r(\tau; s) \geq 0$ and a normalized severity $\bar{V}_r(\tau; s) \in [0, 1]$ via per-rule scaling and clamping:

$$\bar{V}_r(\tau; s) = \min\left(1, \frac{V_r(\tau; s)}{\alpha_r}\right), \tag{34}$$

where $\alpha_r > 0$ is the per-rule normalization constant introduced in Section III. In the released evaluator code this quantity appears as `RULE_NORM` or `RULE_NORMALIZATION`, so $\alpha_r \equiv V_r^{\max}$ in earlier drafts. The tier aggregation and lexicographic scorer operate on $\bar{V}_r$; the raw V_r is used in the training loss. This normalization contract is required by Proposition III.8 to guarantee that the GPU-friendly scalarization preserves the lexicographic ordering. Table VII lists all per-rule proxy parameters. In the released implementation, the linear normalization $\bar{V}_r = \min(1, V_r/\alpha_r)$ is replaced by an exponential cost function that maps raw violations to $[0, 1]$ without a separate α_r:

$$c_r(v) = 1 - \exp\big(-\kappa_r \cdot \max(0, v)\big), \tag{35}$$

where $\kappa_r > 0$ is a per-proxy *softness* parameter controlling how quickly the cost saturates. This formulation satisfies the normalization contract (output in $[0, 1]$, monotonically increasing in violation magnitude) required by Proposition III.8.

E. Proxy Implementation Details

We now present representative proxy formulations from each tier, ordered by priority. All proxies are evaluated at $\Delta t = 0.1\,\mathrm{s}$ (10 Hz); when a proxy depends on velocity, acceleration, jerk, or heading, we derive these quantities from the predicted trajectory by finite differences with standard endpoint handling.

Notation. We use $\sigma_\alpha(x) = 1/(1+e^{-\alpha x})$ for sigmoid gates, and $\mathrm{softmin}_\beta\{f_t\} = -\frac{1}{\beta}\log\sum_t e^{-\beta f_t}$ for smooth minimum approximation (with β=10). The symbol α denotes a rule-specific gate sharpness parameter; the exact per-rule values are implementation constants rather than quantities tuned or compared in this paper.

TABLE VII: Per-Rule Proxy Parameters: Softness Constants and Violation Thresholds

Rule ID	Tier	κ_r	Threshold	Unit	Violation Semantics
Tier 0: Safety					
L0.R0	Safety	2.0	2.0	m	Longitudinal gap $< d_{\min}$ to lead vehicle
L0.R1	Safety	2.0	0.5	m	Lateral clearance $< d_{\text{lat}}$ to any agent
L0.R2	Safety	3.0	2.0	m	Ego within crosswalk clearance zone
L0.R3	Safety	2.0	0.0	m	OBB penetration depth (SAT) > 0
L0.R4	Safety	2.0	1.0	m	Distance to VRU $< d_{\text{VRU}}$
Tier 1: Legal					
L1.R0	Legal	3.0	2.0	m	Crossing stop line under red signal
L1.R2	Legal	2.0	—	m/s	Speed $> v_{\text{limit}} + v_{\text{tol}}$ (v_{tol}=5 mph)
L1.R3	Legal	3.0	2.0	m	Crossing stop line under red signal
L1.R4	Legal	3.0	0.5	m/s	Min speed in stop zone $> v_{\text{stop}}$
L1.R5	Legal	3.0	5.0	m	Ego within crosswalk yield radius
L1.R6	Legal	2.0	2.356	rad	Heading deviation $> 135^\circ$ from lane direction
Tier 2: Road					
L2.R0	Road	2.0	1.0	m	Signed distance outside drivable surface
L2.R1	Road	2.0	1.8	m	Lateral offset $>$ half lane width $+$ margin
Tier 3: Comfort					
L3.R0	Comfort	2.0	3.0	m/s^2	Longitudinal acceleration $> a_{\max}$
L3.R1	Comfort	2.0	4.0	m/s^2	Braking deceleration $> a_{\text{brake}}$
L3.R2	Comfort	2.0	0.5	rad/s	Steering rate $> \omega_{\max}$
L3.R3	Comfort	2.0	2.0	m/s	Speed change $> \Delta v_{\max}$
L3.R4	Comfort	2.0	2.0	m/s^3	Jerk $> j_{\max}$
L3.R5	Comfort	2.0	4.0	s	Left-turn TTC gap $< t_{\min}$
L3.R9	Comfort	2.0	3.0	s	Lane-change TTC $< t_{\min}$
L3.R10	Comfort	2.0	2.0	s	Following time $< t_{\text{follow}}$
L3.R11	Comfort	2.0	3.0	s	Intersection TTC $< t_{\min}$
L3.R12	Comfort	2.0	3.0	m	Pedestrian proximity $< d_{\text{VRU}}$
L3.R13	Comfort	2.0	3.0	m	Cyclist proximity $< d_{\text{VRU}}$

κ_r: softness parameter in the exponential cost function (Eq. 35). Higher κ_r causes faster saturation to maximum cost. Signal-related proxies (L1.R0, L1.R3–R5, L0.R2) use κ_r=3.0; all others use κ_r=2.0. Threshold: the physical limit above which the proxy begins to penalise; for rules where the threshold is inherent to the violation definition (e.g., speed limit), "—" indicates the threshold is scenario-dependent. L1.R1 (priority), L3.R6–R8 (zone rules) are omitted as they lack proxy implementations. Additional penalty scaling factors: yellow-light violations are scaled by 0.3×, flashing-red by 0.5×.

1) Safety Tier Proxies: The Safety tier is the highest-priority tier, and all five of its rules have differentiable proxies.

For **L0.R0 (Safe Longitudinal Distance)**, we penalise insufficient longitudinal gap to the lead vehicle when the ego is moving:

$$V_{\text{L0.R0}}(\tau; s) = \sum_{t=1}^{T} \max\Big(0,\, d_{\min} - d_t^{\text{long}}\Big) \cdot \sigma_\alpha(\|\mathbf{v}_t\| - v_{\min})\,, \tag{36}$$

where d_t^{long} is the longitudinal distance to the lead vehicle, $d_{\min}$=2.0 m, and $v_{\min}$=0.5 m/s.

For **L0.R1 (Safe Lateral Clearance)**, we penalise lateral clearance violations to nearby agents:

$$V_{\text{L0.R1}}(\tau; s) = \sum_{t=1}^{T} \sum_{j \in \mathcal{A}_t} \max\big(0,\, d_{\text{lat}} - d_{t,j}^{\text{lat}}\big)\,, \tag{37}$$

where $d_{t,j}^{\text{lat}}$ is the lateral clearance between ego and agent j at time t, and d_{lat}=0.5 m.

For **L0.R3 (Collision Avoidance)**, we use a differentiable Separating-Axis-Theorem (SAT) penetration depth on axis-aligned oriented bounding boxes. Given ego box half-extents $(l_{\text{ego}}/2,\, w_{\text{ego}}/2)$ and agent j half-extents $(l_j/2,\, w_j/2)$, we first rotate the relative displacement $\Delta\mathbf{p}_{t,j} = \tau_t^j - \tau_t$ into the ego frame via heading θ_t:

$$\begin{pmatrix} \delta_{\text{long}} \\ \delta_{\text{lat}} \end{pmatrix} = \begin{pmatrix} \cos\theta_t & \sin\theta_t \\ -\sin\theta_t & \cos\theta_t \end{pmatrix} \Delta\mathbf{p}_{t,j}. \tag{38}$$

Per-axis penetration depths are then

$$p_{\text{long}} = \max\Big(0,\, \tfrac{l_{\text{ego}}+l_j}{2} - |\delta_{\text{long}}|\Big)\,, \tag{39}$$

$$p_{\text{lat}} = \max\Big(0,\, \tfrac{w_{\text{ego}}+w_j}{2} - |\delta_{\text{lat}}|\Big)\,, \tag{40}$$

and the SAT overlap depth (positive only when *both* axes overlap) is $d_{t,j} = \min(p_{\text{long}},\, p_{\text{lat}})$. The proxy cost is

$$V_{\text{L0.R3}}(\tau; s) = \sum_{t=1}^{T} \sum_{j \in \mathcal{A}_t} d_{t,j}\,. \tag{41}$$

For **L0.R4 (VRU Clearance)**, we penalise proximity violations to pedestrians and cyclists:

$$V_{\text{L0.R4}}(\tau; s) = \sum_{t=1}^{T} \sum_{j \in \mathcal{A}_{\text{VRU},t}} \max\Big(0,\, d_{\text{VRU}} - \|\tau_t - \tau_t^j\|_2\Big)\,, \tag{42}$$

where d_{VRU}=1.0 m.

2) Legal Tier Proxies: Moving to the Legal tier, the proxy formulations shift from distance-based penalties to event-based detectors that must approximate discrete traffic-control semantics using continuous functions.

For **L1.R3 (Red Light Stop)**, we penalise crossing the stop line while the signal is red using a smooth boundary:

$$V_{\text{L1.R3}}(\tau; s) = \sum_{t=1}^{T} r(t) \cdot \sigma_\alpha(\text{cross}(t) - \text{cross}(t-1)), \quad (43)$$

where $r(t) \in [0, 1]$ is a soft indicator of the red signal state and $\text{cross}(t)$ is a differentiable signed measure of stop-line crossing at time t. For **L1.R4 (Stop Sign Compliance)**, we penalise failure to reach a near-zero speed within the stop zone:

$$V_{\text{L1.R4}}(\tau; s) = \sigma_\alpha(\text{softmin}_\beta\{|\mathbf{v}_t|\}_{t\in\mathcal{T}_{\text{zone}}} - v_{\text{stop}}) \cdot v_{\text{stop}}, \quad (44)$$

where $\mathcal{T}_{\text{zone}}$ is the set of timesteps within the stop zone and v_{stop}=$0.5\,\text{m/s}$.

For **L1.R2 (Speed Limit Adherence)**, we penalise exceeding the map speed limit with a tolerance:

$$V_{\text{L1.R2}}(\tau; s) = \sum_{t=1}^{T} \max(0, \|\mathbf{v}_t\| - v_{\text{limit}} - v_{\text{tol}}), \quad (45)$$

with v_{tol}=$5\,\text{mph}$.

3) Road Tier Proxies: The Road tier contains only two rules, both of which admit straightforward geometric proxies.

For **L2.R1 (Lane Departure)**, we penalise exceeding lane boundaries using lateral offset from the lane centreline:

$$V_{\text{L2.R1}}(\tau; s) = \sum_{t=1}^{T} \max\Big(0, |d_t^{\text{lat}}| - \frac{w_{\text{lane}}}{2} - m\Big), \quad (46)$$

where d_t^{lat} is signed lateral deviation, w_{lane} is lane width, and m is a margin.

For **L2.R0 (Drivable Surface)**, we use a signed distance to the road polygon and penalise excursions outside the drivable region:

$$V_{\text{L2.R0}}(\tau; s) = \sum_{t=1}^{T} \phi(-\text{SDF}_{\mathcal{P}_{\text{road}}}(\tau_t)), \quad (47)$$

where $\phi(\cdot)$ is a smooth hinge and $\mathcal{P}_{\text{road}}$ is the drivable-surface polygon.

4) Comfort Tier Proxies: Finally, the Comfort-tier proxies enforce kinematic bounds on the derivatives of the predicted trajectory. Because these quantities are computed via finite differences, they are inherently differentiable with respect to the predicted waypoints.

For **L3.R0 (Acceleration)**, we penalise excessive longitudinal acceleration:

$$V_{\text{L3.R0}}(\tau; s) = \sum_{t=1}^{T-1} \max\Big(0, |a_t^{\text{long}}| - a_{\max}\Big), \quad (48)$$

with $a_{\max}$=$3.0\,\text{m/s}^2$.

For **L3.R1 (Braking)**, we penalise harsh deceleration:

$$V_{\text{L3.R1}}(\tau; s) = \sum_{t=1}^{T-1} \max\Big(0, -a_t^{\text{long}} - a_{\text{brake}}\Big), \quad (49)$$

with a_{brake}=$4.0\,\text{m/s}^2$.

For **L3.R4 (Jerk Limiting)**, we penalise excessive jerk to discourage oscillatory plans:

$$V_{\text{L3.R4}}(\tau; s) = \sum_{t=1}^{T-2} \max(0, |j_t| - j_{\max}), \quad (50)$$

with $j_{\max}$=$2.0\,\text{m/s}^3$.

F. Data Pipeline

The rule evaluation machinery described above is applied both at *training time* (proxy evaluation over augmented data) and at *audit time* (full-catalog evaluation over canonical splits). This subsection describes the data pipeline that feeds both paths.

1) Dataset: Dataset statistics, split details, and filtering criteria are given in Section VI (Table IX). Fig. 5 shows representative interactive scenarios from WOMD v1.3.0 [34] spanning all four rule tiers.

2) Preprocessing: Raw TFRecord files are preprocessed into training-ready tensors through five steps:

1) **Coordinate transformation:** all positions are transformed to an ego-centric frame at t=0.
2) **Agent selection:** top 32 agents by proximity to ego.
3) **Lane selection:** top 64 lanes by relevance (proximity and heading alignment).
4) **Feature extraction:** velocities are computed via finite differences; accelerations are optionally smoothed for numerical stability.
5) **Map tensorisation:** lane polylines, boundaries, stop lines, crosswalks, and drivable surface polygons are vectorised into fixed-size tensors.

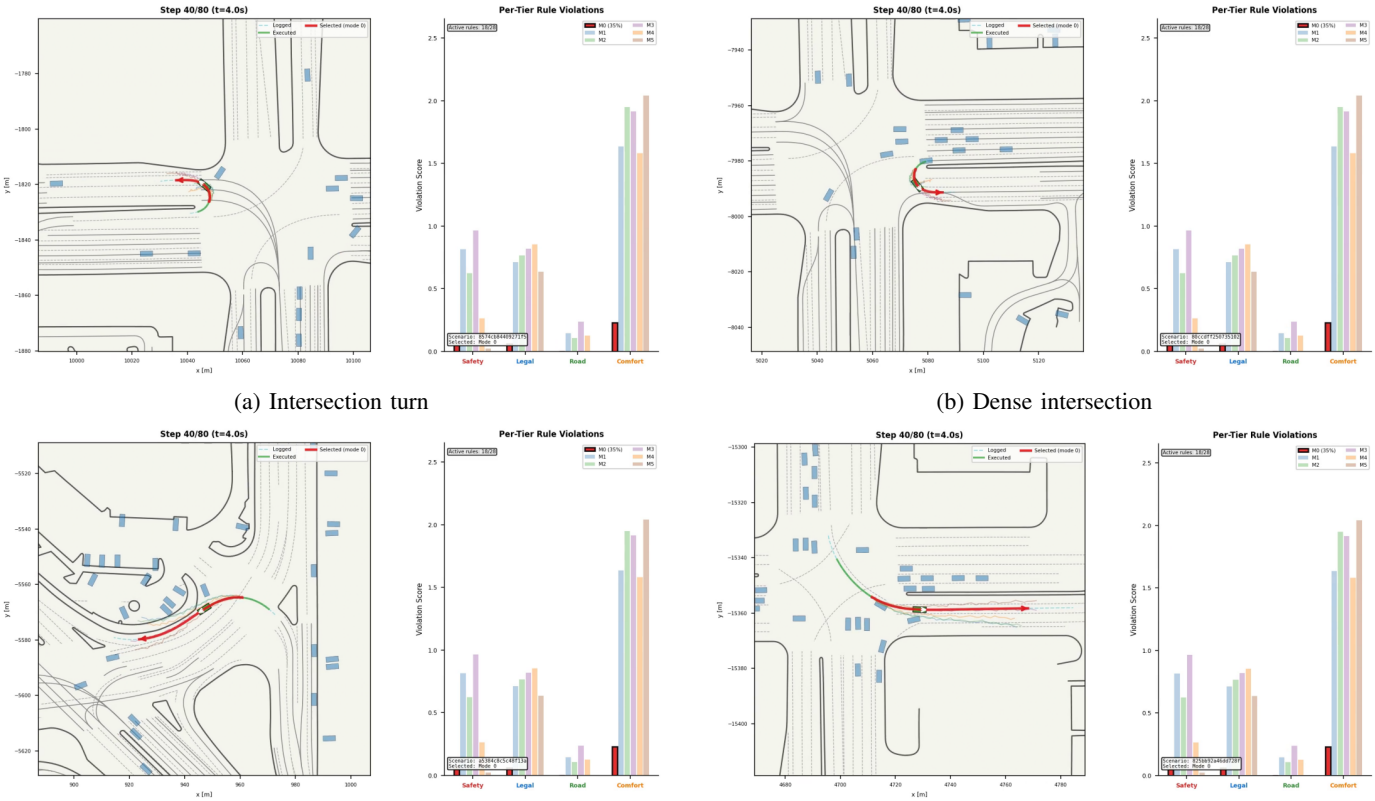
In addition to standard spatial and temporal augmentations that improve invariances and robustness, we apply rule-targeted augmentation to balance proxy supervision, as summarised in Table VIII.

TABLE VIII: Data Augmentation Strategies

Augmentation	Parameters	Purpose
Rotation	$\pm 15°$	Orientation invariance
Translation	$\pm 2.0\,\text{m}$	Position invariance
Scaling	0.95–1.05	Scale robustness
Agent dropout	10% prob.	Occlusion handling
Trajectory noise	$\mathcal{N}(0, 0.1^2)\,\text{m}$	Smoothness regularisation
Time reversal	50% prob.	Temporal symmetry
Mirror flip	50% prob.	Left-right symmetry

Note: Augmentations are applied in the ego-centric frame. Agent dropout simulates occlusions; trajectory noise regularises dynamics; mirror flips and rotations improve spatial invariance. Rule-targeted augmentation generates synthetic samples with controlled violations to balance the proxy supervision.

To ensure balanced proxy training, we additionally generate synthetic scenarios with controlled violations, including collision perturbations (controlled overlap), speed violations (scaled velocities), lane departures (lateral offsets), and red-light violations (stop-line crossings under red). These samples improve proxy calibration for rare but high-impact events and reduce proxy underfitting on the tail of violation severities.



(a) Intersection turn

(b) Dense intersection

(c) Exit-ramp merge

(d) Lane change

Fig. 5: Five representative `validation_interactive` scenarios from WOMD v1.3.0, rendered from closed-loop Waymax simulation at t=4.1 s into the 8-second horizon. In each panel the BEV map (left) shows surrounding agents (blue rectangles), the RECTOR-selected trajectory (red), and the executed trajectory (green); the bar chart (right) shows per-tier rule violations across the six candidate modes. The diversity of road geometries and interaction densities reflects the breadth of interactive behaviors used to stress-test the rule-aware selection stack.

G. Summary

This section defines the 28-rule, four-tier catalog used by RECTOR and describes how compliance is computed under the *Evaluation Dual-Protocol*: Protocol A applies 24 differentiable proxies with predicted applicability for training and selection, whereas Protocol B applies the full Waymo evaluator with oracle applicability for reporting and verification. Per-rule violation breakdowns (Fig. 4) expose the dominant failure modes by tier and directly inform proxy refinement, model debugging, and targeted augmentation. With the rule-evaluation layer and data pipeline fully specified, Section VI next defines the experimental protocol—dataset splits, metrics, and baselines—under which all results are reported.

VI. Datasets, Metrics, Baselines, and Implementation Details

The preceding two sections specified *what* RECTOR computes (the architecture, Section IV) and *what rules it enforces* (the evaluation layer, Section V). This section defines the experimental protocol under which those components are evaluated. We test RECTOR on the WOMD `validation_interactive` split (12,800 scenarios)—a held-out interactive benchmark that is already largely compliant, but still contains a minority of Safety-heavy tail cases where selection matters—and compare three selection strategies applied to the same multi-modal candidate set: *Confidence Only*, *Weighted Sum*, and *RECTOR Lexicographic*. A central point is the separation between (i) *candidate-set quality* (multi-modal prediction coverage) and (ii) *selected-trajectory quality and compliance* (the final decision after rule-aware selection); the reported violation rates are therefore residual-risk metrics on the chosen trajectory, not evidence that the benchmark is broadly rule-breaking. Unless explicitly stated otherwise, all compliance statistics are computed on the *selected* trajectory using the evaluation pipeline described in Sections V-A–V-F.

A. Dataset

1) Waymo Open Motion Dataset (WOMD): We use the Waymo Open Motion Dataset (WOMD) v1.3.0 [34], which provides temporally aligned trajectories for the ego vehicle and surrounding agents, HD map context (lane topology, crosswalks, boundaries), and traffic control state annotations (e.g., signals and stop signs) that are required for both motion prediction and rule applicability/violation computation. Table IX summarizes the key statistics.

2) Data Split Strategy: We follow WOMD's official interactive splits to avoid leakage across training and evaluation distributions. Training uses 102,040 augmented in-

TABLE IX: Waymo Open Motion Dataset Statistics (Interactive Scenarios)

Property	Value
Dataset Overview	
Total augmented scenarios	160,184
Training split	102,040 augmented scenarios
Validation split	12,800 evaluation scenarios
Test split	14,665 augmented scenarios (withheld)
Interactive criterion	Exactly 2 objects_of_interest
Geographic regions	Phoenix, San Francisco, Mountain View, CA
Traffic convention	Right-hand traffic (US)
Average agents per scenario	8.2
Maximum agents	40+
Object types	Vehicles, pedestrians, cyclists
Agent tracking consistency	Validity masks over the 9.1 s window
Map Context	
Map coverage	Full HD maps
Lane annotations	Complete lane topology
Traffic signals	Per-timestep state annotations
Stop signs	Location and orientation
Crosswalks	Polygon boundaries
Speed limits	Where available

TABLE X: Evaluation Population Reference

Population	N	Used in
Canonical validation	12,800	Tables XIII–XIX; main results
Augmented validation	43,219	Tables XIV, XXIII, XXV
Proxy alignment	1,000	Table XXXI (Section VIII)
Perturbation	500	Table XXXII
Constrained baselines	1,000	Section VII-E (illustrative)
Waymax bridge	55	Section VII-D (bridge validation)

teractive scenarios, while all reported results use the `validation_interactive` split (12,800 scenarios sampled from 150 validation shards). Each augmented TFRecord entry wraps the original WOMD Scenario proto together with per-rule applicability labels, violation flags, and severity scores computed by the `waymo_rule_eval` pipeline; these labels serve as oracle applicability ground truth in ablations (Table XXII). The test split is withheld and is not accessed during development.

3) Interactive Scenario Selection: WOMD defines an *interactive* scenario by the presence of exactly two `objects_of_interest`, which serve as the primary interacting entities. This restriction emphasises cases where inter-agent coupling and priority reasoning are salient, making them an appropriate stress test for rule-aware selection. We use all interaction types present in the validation split: vehicle-to-vehicle (v2v), vehicle-to-pedestrian (v2p), vehicle-to-cyclist (v2c), and other agent pairings.

4) Scenario Filtering: We apply minimal filtering to preserve the natural distribution while removing degenerate cases that do not meaningfully test interaction or map reasoning: scenarios with fewer than 2 agents are excluded (interaction requires at least one other agent beyond ego context); scenarios with ego speed consistently below $0.1\,\mathrm{m/s}$ are excluded (parked or near-static trajectories); and scenarios with incomplete map features required by the rule set (e.g., missing lane topology) are excluded. All other scenarios are retained to maintain geographic and behavioral diversity. Overall, approximately 98% of scenarios survive these filters. The Legal-tier rule catalog reflects traffic norms applicable to U.S. right-hand traffic scenarios in WOMD; extending to other jurisdictions would require re-parameterizing the catalog and re-evaluating proxy coverage accordingly.

B. Evaluation Protocols

Compliance numbers are only meaningful when the evaluation scope is unambiguous. We use Protocols A and B as defined in Section V (Table VI); all main-paper results use Protocol A unless explicitly noted otherwise.

Aggregation conventions. Three summary statistics are used throughout the results: *Total Viol.%* is the fraction of scenarios where the selected trajectory violates at least one rule across the rulebook (union probability); *Per-tier Viol.%* is the fraction violating at least one rule in that tier; and *S+L Viol.%* is the fraction violating at least one Safety or Legal tier rule. Per-tier rates may sum to more than Total Viol.% because a single scenario can violate rules in multiple tiers.

Protocol B applies the full rule set with oracle applicability, so its absolute violation rates are higher than Protocol A and are reported as a separate evaluation setting. More generally, these are residual selected-trajectory risk rates: the candidate sets are already mostly rule-consistent, so the main signal is whether the selector avoids the minority of unsafe modes rather than whether it corrects pervasive noncompliance.

Evaluation populations. Several scenario subsets appear in different analyses. Table X provides a single reference for which subset is used where, to avoid confusion across sections.

C. Evaluation Metrics

The evaluation protocols above define *which rules are checked*; this subsection defines *which quantities are reported*. We organize metrics into three groups: trajectory accuracy, rule compliance, and computational efficiency.

1) Trajectory Accuracy Metrics: We use the standard geometric metrics (ADE, FDE, minADE, minFDE, MissRate) as defined in Section III-C, evaluated over the WOMD prediction horizon (T=50 at $10\,\mathrm{Hz}$) with miss threshold $\delta = 2.0\,\mathrm{m}$. In addition, we report *selected-trajectory* metrics that quantify the accuracy of the final decision after lexicographic selection:

$$\mathrm{selADE} = \mathrm{ADE}(\hat{\tau}, \tau^*), \qquad \mathrm{selFDE} = \mathrm{FDE}(\hat{\tau}, \tau^*), \quad (51)$$

where $\hat{\tau}$ is the trajectory chosen by the selection strategy. These quantify decision quality, not only candidate-set diversity.

2) Rule Compliance Metrics: Compliance is evaluated using the evaluation pipeline described in Sections V-A–V-D. We report tier-wise rates to capture reliability and per-rule rates for diagnosis.

Tier-wise violation rate. For tier ℓ with aggregated score $S_\ell(\tau; s)$:

$$\mathrm{ViolationRate}_\ell = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}[S_\ell(\tau_i; s_i) > 0]\,. \quad (52)$$

High-priority (S+L) violations. We treat violations in the top two tiers (Safety ∪ Legal) as high-priority:

$$\text{S+L_ViolationRate} = \mathbb{E}_{s\sim\mathcal{D}}\left[\mathbf{1}[S_0(\tau;s) > 0 \text{ or } S_1(\tau;s) > 0]\right].$$

Per-rule violation rate. For each rule r with violation score $V_r(\tau; s)$:

$$\text{ViolationRate}_r = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}[V_r(\tau_i; s_i) > 0]. \tag{53}$$

All tables report violation rates for conciseness; the *Compliance Rate* at tier ℓ is the complement, $\text{CR}_\ell = 1 - \text{ViolationRate}_\ell$, representing the fraction of scenarios where the selected trajectory satisfies all rules in that tier.

3) Closed-Loop Simulation Metrics: For the Waymax-based closed-loop bridge validation (Section VII-D), we collect Waymax's built-in metrics and custom motion-planning metrics. **Overlap rate** is the fraction of simulation steps where the ego vehicle's bounding box overlaps any other agent (Waymax `overlap` metric). **Log divergence** (m) measures displacement between the simulated and logged trajectory (Waymax `log_divergence`). **Kinematic infeasibility rate** is the fraction of steps whose action violates the dynamics model's feasibility bounds, e.g. excessive lateral acceleration for the bicycle model. **Longitudinal jerk** (m/s^3) quantifies ride comfort via the time derivative of acceleration, reported as per-scenario mean and max. **Minimum clearance** (m) is the closest Euclidean distance between the ego and any other valid agent at each step, aggregated as per-scenario minimum. **Time-to-collision** (s) estimates the time before the first collision under a constant-velocity assumption, reported as per-scenario minimum. These metrics are included because the bridge-validation subsection reports them.

4) Computational Efficiency Metrics: We report latency, throughput, and memory footprint under the same batching and hardware conditions used for all models. Inference latency is defined as

$$\text{Latency}_{\text{mean}} = \frac{1}{N}\sum_{i=1}^{N} t_{\text{forward}}^{(i)}, \tag{54}$$

where $t_{\text{forward}}^{(i)}$ includes the forward pass and the selection step, measured with GPU synchronization; we also report P95 latency. Throughput is the reciprocal:

$$\text{Throughput} = \frac{N}{\sum_{i=1}^{N} t_{\text{forward}}^{(i)}}. \tag{55}$$

We additionally report parameter counts and peak GPU memory usage during inference.

D. Baseline Methods

With the metrics defined, we now specify the baselines against which RECTOR is compared. Comparisons serve two distinct purposes: *external* baselines assess candidate-set accuracy relative to the state of the art, while *internal* baselines isolate the effect of the selection strategy on a fixed candidate set.

1) External Trajectory Accuracy Baselines: We compare RECTOR against five representative motion prediction methods. These numbers are included as geometric-accuracy context rather than as the paper's causal comparison, which is the internal same-candidate-set study below. Unless explicitly stated as reproduced in our pipeline, the external baseline values follow the published descriptions or reference reports of the respective methods and may therefore reflect different implementation details even when they target the same `validation_interactive` setting.

LaneGCN [4]: graph neural network over lane-graph structure. Reported metrics: minADE = 0.87 m, minFDE = 1.92 m, Miss Rate = 28.4%, throughput = 180 samples/s. **DenseTNT** [6]: dense goal-based prediction with iterative refinement and WTA-style training. Reported metrics: minADE = 0.73 m, minFDE = 1.55 m, Miss Rate = 20.1%, throughput = 120 samples/s.

M2I [1]: interaction-centric scene encoding with multi-modal decoding. Reported metrics: minADE = 0.75 m, minFDE = 1.63 m, Miss Rate = 22.8%.

Wayformer [8]: Transformer-based motion forecasting with long-range attention. Reported metrics: minADE = 0.68 m, minFDE = 1.42 m, Miss Rate = 18.9%, throughput = 85 samples/s.

MTR [7]: Transformer-based motion modeling with structured candidate generation. Reported metrics: minADE = 0.70 m, minFDE = 1.48 m, Miss Rate = 19.5%, throughput = 95 samples/s.

RECTOR (Ours): Transformer–CVAE with rule-aware lexicographic selection, trained via two-stage procedure (Section IV-A). Reported metrics: minADE = 0.684 m, minFDE = 1.270 m, Miss Rate = 18.56%, throughput = 136.5 samples/s. Selected-mode compliance under Protocol A rises from 76.14% to 84.97% (equivalently, total violations 23.86% → 15.03%, −8.83 pp), while Safety-tier violations fall from 21.87% → 13.15% (−8.72 pp); Legal-tier violations remain 0.0% across all strategies because Protocol A suppresses that tier. The gain is concentrated in the executed Safety-heavy tail of the same candidate set, not in a wholesale shift of the benchmark distribution.

2) Internal Selection Strategy Baselines: The more important comparison is the *internal* one: all three strategies below operate on the *same* K=6 candidate set, so any difference in compliance is attributable entirely to the selection mechanism. This is the cleanest experimental isolation in the paper.

Confidence Only: $\hat{\tau} = \arg\max_k p^{(k)}$. No rule awareness.

Weighted Sum: $\hat{\tau} = \arg\min_k \sum_r w_r \hat{a}_r \bar{V}_{r,k}$. Uses the same predicted applicability as RECTOR to ensure a fair comparison; per-rule weights are uniform (w_r=1 for all r), as used in the closed-loop validation pipeline (`validate_50.py`). On this evaluation, the weighted sum achieves near-identical compliance to RECTOR. The distinction between the two selectors is discussed in Section III-J.

Human GT (reference): Ground-truth trajectories evaluated under the identical Protocol A pipeline. This serves as the *imitation ceiling*—the performance of a system that perfectly replicates human drivers—and its non-zero Legal-tier violation rate (12.8%) shows that logged human-like trajectories are not

TABLE XI: Training hyper-parameters and Configuration

hyper-parameter	Value
Optimizer Settings	
Optimizer	AdamW
Max learning rate	3×10^{-4}
Learning rate schedule	OneCycleLR (cosine)
Weight decay	1×10^{-4}
Gradient clipping	max norm = 1.0
Batch size	256
Epochs (Stage 1 / Stage 2)	20 / 20+5
Precision	Mixed (AMP)
Random seed	42
Loss Weights	
λ_{WTA} (reconstruction)	20.0
λ_{end} (endpoint)	5.0
λ_{KL} (KL divergence)	0.05
λ_{goal} (goal prediction)	2.0
λ_{smooth} (smoothness)	1.0
λ_{conf} (confidence)	1.0
λ_{app} (applicability)	1.0
λ_{rule} (proxy compliance)	1.0
Checkpoint Selection	
Selection criterion	Best validation loss
Early stopping patience	15 epochs
Data Augmentation	
Random rotation	$\pm 15°$
Position noise	$\mathcal{N}(0, 0.1^2)$ m
Agent dropout	10%
Time reversal*	50% probability
Mirror flip	50% probability
Hardware	
GPU	NVIDIA RTX A3000 (12 GB)
Training duration	∼12 hours
Peak GPU memory	9 GB

*Time reversal is applied to spatial position sequences only, not to rule violation labels or temporal rule semantics. Rule applicability and violation labels are computed on the original (non-reversed) trajectory. Training follows a two-stage procedure: Stage 1 trains only the applicability head (focal BCE, γ=2.0) on frozen encoder features for 20 epochs; Stage 2 fine-tunes the full model end-to-end with all eight loss terms for 20+5 epochs. The loss weights in this table apply to Stage 2; Stage 1 uses $\mathcal{L}_{\text{app}}$ only.

a compliance oracle, even though the benchmark as a whole remains largely well-behaved.

E. Implementation Details

To enable reproducibility, we report the complete training and inference configuration below.

1) Model Architecture: RECTOR is implemented as an interaction-aware encoder paired with a multimodal CVAE trajectory decoder, augmented by a rule-applicability head and a tiered scorer. Table IV summarizes the configuration and parameter allocation; the additional capacity beyond the motion backbone is concentrated in rule applicability prediction, while the tiered scorer itself is lightweight and primarily structural.

2) Training Configuration: Training uses AdamW with a OneCycleLR cosine schedule and a brief warm-up. Table XI reports all hyperparameters, loss weights, augmentation settings, and hardware.

3) Loss Functions: The full training objective is a flat weighted sum of eight terms as specified in Section IV-G ($\mathcal{L}_{\text{WTA}}$, $\mathcal{L}_{\text{end}}$, $\mathcal{L}_{\text{KL}}$, $\mathcal{L}_{\text{goal}}$, $\mathcal{L}_{\text{smooth}}$, $\mathcal{L}_{\text{conf}}$, $\mathcal{L}_{\text{app}}$, $\mathcal{L}_{\text{rule}}$). Table XI reports the corresponding weight values used in all experiments. The checkpoint with the lowest composite validation loss is selected; because the reconstruction term (λ_{WTA}=20.0) dominates the total, the best validation loss closely tracks the best validation ADE.

4) Inference Configuration: Inference runs in FP32 with batch size 128; selection uses tiered lexicographic ordering as specified in Algorithm 1. Latency, throughput, and memory footprint are reported in Table XXVI (Section VII).

F. Reproducibility

Beyond the reference training configuration, full reproducibility requires specifying the software environment, random-seed management, and the data-preprocessing pipeline. We report each in turn.

1) Software Environment: All experiments use Python 3.10, PyTorch 2.2, CUDA 12.1, and Waymo Open Dataset v1.5.2; full dependency versions are provided with the code release. The closed-loop bridge validation (Section VII-D) additionally uses Waymax 0.1.0 [35] and JAX 0.6.2 with the CUDA 12 plugin.

2) Random Seed Management: All experiments use a fixed random seed of 42 across PyTorch, NumPy, and Python's `random` module. CUDA deterministic mode and cuDNN deterministic mode are both enabled, with an expected ∼2% throughput cost from the latter.

3) Data Preprocessing: The preprocessing pipeline consists of seven stages: parsing TFRecord scenarios from WOMD; extracting ego and agent trajectories with validity masks and dropping invalid timesteps; transforming all states to an egocentric frame at t=0 (translation + rotation by ego heading); sampling lane polylines within a 100 m radius and constructing lane-graph features; extracting traffic control states (signals, stop signs) and geometry required by the rulebook; computing ground-truth rule supervision signals (applicability labels and proxy violations) from map context and annotated states; and caching processed features as NumPy arrays for fast, deterministic loading.

4) Expected Runtime: The two-stage training requires approximately 12 hours total on a single NVIDIA RTX A3000 Laptop GPU (12 GB); full evaluation with rule violation computation on 12,800 scenarios takes approximately 9 minutes (536.7 s); and inference latency is 7.3 ms per sample (mean) or 9.2 ms (P95). This modest single-GPU setup strengthens reproducibility, but it also constrained the scale of additional ablations and large-sweep experiments in the current study.

G. Summary

In summary, the experimental setup provides comprehensive evaluation (trajectory accuracy, rule compliance, and efficiency), fair comparison (consistent preprocessing, horizon, and evaluation protocol across baselines), reproducibility (fixed seeds, deterministic settings, and explicit software/hardware reporting), and operational relevance (results on the interactive validation split, where coupled behaviors and priority reasoning are most informative for rule-aware selection). With all protocol elements fixed, Section VII presents the full results.

TABLE XII: Candidate-Set Accuracy Context on WOMD `validation_interactive`

Method	minADE (m) ↓	minFDE (m) ↓	Miss (%) ↓
LaneGCN	0.87	1.92	28.4
DenseTNT	0.73	1.55	20.1
M2I	0.75	1.63	22.8
Wayformer	0.68	1.42	18.9
MTR	0.70	1.48	19.5
RECTOR (Ours)	**0.684**	**1.270**	**18.56**

Best-of-K accuracy on 12,800 validation scenarios. minADE/minFDE are computed over K=6 modes (FDE uses the ADE-best mode). Miss Rate uses a 2.0 m FDE threshold. RECTOR standard deviations: minADE $\pm$0.754 m, minFDE $\pm$1.815 m. Baselines are from published results on standard WOMD validation; direct comparison is approximate. See Table XXVI for throughput and runtime profiling.

TABLE XIII: Selection Strategy Comparison: Accuracy, Compliance, and Complexity (Protocol A)

Strategy	selADE (m)	selFDE (m)	Total Viol. (%)	S+L Viol. (%)	Complexity
Confidence Only	2.208	4.330	23.86	21.87	$O(M)$
Weighted Sum	2.043	3.813	15.03	13.15	$O(MR)$
RECTOR	**2.143**	**4.151**	**15.03**	**13.15**	$O(MT)$
Oracle (best-of-K)	minADE = 0.684 m		—	—	—

M = modes, R = rules, T = tiers. selADE/selFDE are displacement errors of the *selected* trajectory in metres; Total Viol. counts scenarios with ≥ 1 violation (union); S+L = Safety $\cup$ Legal. All from `evaluate_canonical.py` on N_{scen}=12,800 unique WOMD `validation_interactive` scenarios, each contributing one ego-agent evaluation instance (N_{eval}=12,800; RECTOR evaluates one ego trajectory per scenario, not multiple agents of interest). Checkpoint hash `c90d4457e4996d22`, seed 42. Legal-tier violations are 0.0% for all strategies because predicted applicability suppresses Legal-tier activation under Protocol A in this setting. Equivalently, no-violation rate rises from 76.14% to 84.97% under rule-aware selection. These are selected-trajectory rates on a fixed candidate set, so the reported gains come from filtering the minority of unsafe modes rather than from changing candidate-set coverage. See Table XVII for the per-tier breakdown.

VII. Accuracy, Rule Compliance, Behavior, and Efficiency

With the architecture (Section IV), rule evaluation layer (Section V), and experimental protocol (Section VI) fully specified, this section presents the empirical results. We evaluate RECTOR along three operational dimensions: (i) candidate-set trajectory accuracy, (ii) rule compliance induced *by selection* when the candidate set is held fixed, and (iii) computational efficiency. Unless stated otherwise, results are reported on the WOMD `validation_interactive` split (12,800 scenarios) with K=6 candidate trajectories under Protocol A.

A key distinction throughout this section is between *best-of-K* accuracy—which measures candidate-set coverage and is independent of the selection strategy—and *selected* accuracy—which measures the trajectory that is actually executed after applying a selection strategy. This distinction is necessary because RECTOR's central intervention is selection-time enforcement of prioritized rules, and the two metrics can move in opposite directions when the most geometrically accurate candidate is also the most rule-violating.

Because the benchmark candidate sets are already mostly compliant, the compliance numbers below should be read as residual-risk reductions on the executed mode rather than as evidence of widespread rule breaking in WOMD. The main question is whether the selector can remove the minority of Safety-heavy tail cases while leaving the underlying candidate distribution fixed.

A. Trajectory Prediction Accuracy

We begin with candidate-set accuracy, which establishes the upper bound on what any downstream selector can achieve. Table XII situates RECTOR's generation quality within the range of published methods.

RECTOR achieves candidate-set accuracy of **minADE = 0.684 m**, **minFDE = 1.270 m**, and **Miss Rate = 18.56%** while reserving modeling and decision capacity for compliance-aware selection. Table XII lists published metrics from five external methods to situate the generator in the same accuracy regime as current WOMD predictors and to establish that the candidate set is strong enough for meaningful selection experiments.

1) Error Distribution: The aggregate accuracy numbers above, however, do not reveal the full picture, because the error distribution is strongly heavy-tailed. Fig. 6 visualises the distribution of minADE and minFDE across all 12,800 evaluation scenarios.

The complementary cumulative distribution ($P(\text{Error} > x)$, evaluated on a log scale) confirms that error is heavy-tailed: the top 5% of scenarios by minADE—mostly complex intersections and dense merges—account for a disproportionate share of the total error mass. These are precisely the regulation-dense contexts where the candidate set spans both accurate-but-violating and slightly less accurate-but-compliant trajectories, making rule-aware selection most consequential.

Rule-aware selection is most consequential precisely in these heavy-tailed, regulation-dense scenarios; the next subsection quantifies what that enforcement achieves.

B. Selection Strategy Comparison

This experiment isolates the paper's central empirical claim: when the candidate set is held fixed and already mostly compliant, selection can still reduce the residual rule-violating tail in the executed trajectory. We apply three strategies to the *same* candidate set $\mathcal{T}(s)$—Confidence Only ($\arg\max_k p^{(k)}$), Weighted Sum ($\arg\min_k \sum_r w_r \bar{V}_{r,k}$), and RECTOR (lexicographic)—and evaluate all three under Protocol A on 12,800 scenarios. Table XIII reports the results.

Both RECTOR and the weighted-sum baseline achieve identical violation rates (15.03% total, 13.15% S+L). On this dataset, both rule-aware strategies raise reported selected-mode compliance from 76.14% to 84.97% (equivalently, total violations fall from 23.86% to 15.03%, about 1.1k fewer violating selections), with the Safety tier providing the dominant improvement (21.87% $\rightarrow$ 13.15%, $-$8.72 pp). The empirical result supported by Table XIII is therefore the value of rule-aware selection over confidence-only selection on a mostly compliant candidate set, not a compliance separation between lexicographic and weighted-sum ranking on this benchmark.

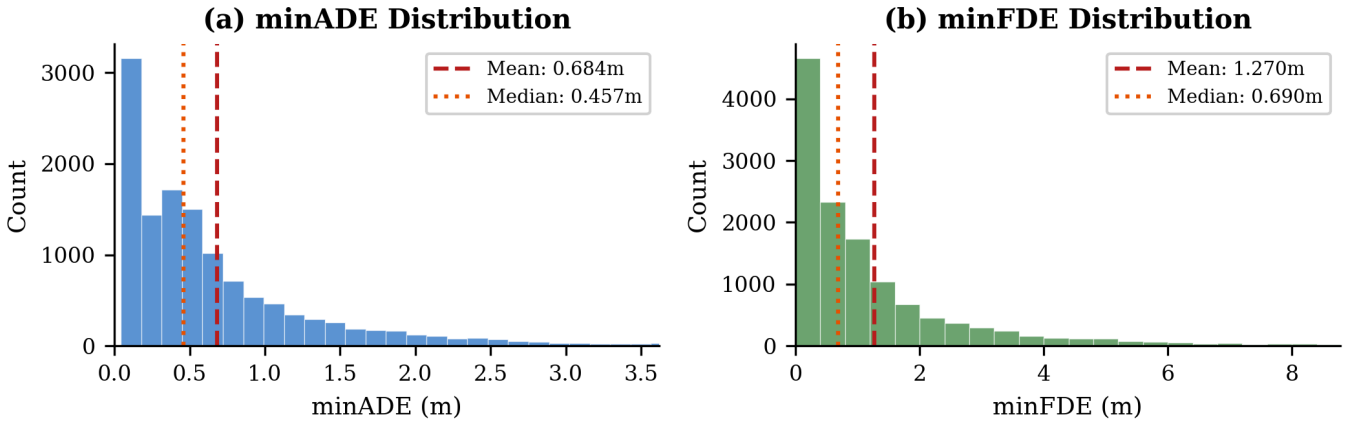


Fig. 6: Distribution of prediction errors exhibits heavy-tailed behavior. **(a)** minADE distribution. **(b)** minFDE distribution. Tail scenarios typically correspond to dense, regulation-heavy contexts (e.g., complex intersections) where the candidate set includes both accurate-but-violating and slightly less accurate-but-compliant options, making rule-aware selection most consequential.

TABLE XIV: Independently-Tuned Weighted-Sum vs. RECTOR (Protocol A, 43,219 scenarios, canonical checkpoint)

Selector	Weights (w_S, w_L, w_R, w_C)	selADE (m)	S+L (%)	Total (%)	Safety (%)
Best Grid WS	$(100, 10, 1, 1)$	2.049	13.39	15.18	13.39
Default WS	$(1000, 100, 10, 1)$	2.048	13.39	15.18	13.39
RECTOR	(lexicographic)	**2.138**	**13.39**	15.18	13.39

Grid search: 125 weight vectors (5×5×5×1), evaluated on the full augmented validation set with the canonical two-stage checkpoint. *All* 125 weight configurations converge to the same S+L violation rate (13.39%) and selADE (≈2.048 m). RECTOR's lexicographic selector (with confidence tiebreaking) achieves the same S+L rate (13.39%) but selADE of 2.138 m on this set, reflecting a 0.090 m accuracy cost relative to weighted-sum at equal compliance. No weight vector produces selections differing from RECTOR's in binary compliance outcomes, so the structural guarantee is not exercised empirically on this benchmark.

Statistical significance. Scenario-level paired tests on 43,219 per-scenario violation indicators confirm the robustness of these reductions. McNemar's test on binary Safety-tier violations yields $p \approx 0$ for lexicographic vs. confidence-only selection (3,759 discordant pairs, all favouring lexicographic; zero scenarios where lexicographic is worse). Wilcoxon signed-rank on per-scenario selADE yields $p = 6.9 \times 10^{-23}$ ($z = 9.85$; mean improvement -0.061 m). For lexicographic vs. weighted-sum, McNemar's test yields $p = 1.0$: the two selectors produce identical violation outcomes on every scenario, confirming the complete empirical convergence reported in Table XIII.

Independently-tuned weighted-sum baseline. To test whether independently tuned weights can break the empirical parity observed in Table XIII, we conducted a systematic weight search using the canonical two-stage checkpoint (c90d4457e4996d22) on the full augmented validation set (43,219 scenarios). We evaluated 125 grid-search weight vectors (5×5×5×1 tier-level combinations). Table XIV reports the results.

Weight-space landscape. Across all 125 grid-search weight vectors, the (selADE, S+L Viol.%) landscape reveals complete convergence: all configurations achieve S+L = 13.39% with selADE within 2.048–2.049 m (weighted-sum) and 2.138 m (RECTOR lexicographic, confidence tiebreaker). The canonical two-stage checkpoint therefore shows that the current benchmark does not expose a regime in which weight misspecification changes binary compliance outcomes.

1) Extended Weight Grid: Misspecification Analysis: The 125-configuration grid spans $w_S \in \{100, 500, 1000, 2000, 5000\}$ with $w_C{=}1$ fixed. To directly test whether weight misspecification causes weighted-sum to select Safety-violating candidates that lexicographic ordering would reject, we extend the sweep to $w_S \in \{1, 5, 10, 20, 50, 100, \ldots, 5000\}$ (250 configurations total) on the same 12,800-scenario set.

Structural condition for WS–lex divergence. The two selectors disagree on Safety whenever a scenario contains a candidate pair where b violates Safety ($\sigma_S(b) > 0 = \sigma_S(a)$) yet the comfort score of b exceeds that of a by enough to satisfy:

$$\frac{w_S}{w_C} < \frac{\sigma_C(a) - \sigma_C(b)}{\sigma_S(b)}. \tag{56}$$

Lexicographic selection eliminates b at Tier 0 regardless of (56); weighted-sum selects b when the inequality holds. Computing the right-hand side across all 12,800 scenarios shows that cross-tier tension exists in 189 scenarios (1.48% of the validation set), with the maximum cross-tier ratio reaching 15.34. This implies that WS with $w_S/w_C < 15.34$ could in principle select a Safety-violating candidate on one or more scenarios.

Empirical misspecification boundary. Table XV reports Safety-tier violation rates from the extended grid. WS at $w_S{=}1$ ($w_S/w_C{=}1$, fully flat weights) selects one additional Safety-violating scenario compared to lexicographic (+0.0078 pp, exactly 1 scenario out of 12,800); at $w_S{\geq}5$ the compliance is pixel-identical to lex across all 12,800 scenarios. The maximum cross-tier ratio of 15.34 is not reachable with the proxy scoring function because the Safety proxy scores in

TABLE XV: Extended Weight Grid: Safety-Tier Violations vs. Lexicographic Baseline

Selector	w_S	Safety Viol. (%)	Δ vs. Lex (pp)
Lexicographic	—	13.148	0.000
Weighted-sum	1	13.156	+0.008 (1 scen.)
	5	13.148	0.000
	10	13.148	0.000
	20	13.148	0.000
	50	13.148	0.000
	100–5000	13.148	0.000

Extended grid (250 configurations, $w_S \in \{1, 5, 10, 20, 50, 100, 500, 1000, 2000, 5000\}$, $w_L \in \{10, \ldots, 500\}$, $w_R \in \{1, \ldots, 50\}$, w_C=1), evaluated on 12,800 WOMD validation scenarios with the canonical checkpoint. The misspecification boundary falls at w_S/w_C=1: fully flat weights cause WS to select one additional Safety-violating candidate. Any preference for Safety over Comfort ($w_S \geq 5w_C$) recovers identical compliance to lexicographic. The effective threshold is low because Safety proxy scores are large (median non-zero score 1.0) while cross-tier Comfort advantages are small.

those scenarios are extremely small (<0.002); the effective misspecification boundary for this model and dataset is therefore at $w_S/w_C \approx 1$.

Interpretation. The low misspecification boundary ($w_S/w_C \approx 1$) is itself an empirical property of this model and dataset: Safety proxy violations are large and Comfort proxy advantages are small in the WOMD validation distribution, making Safety discrimination easy even for flat weights. This may not hold under (i) continuous violation scoring (margins rather than thresholds), (ii) training objectives that incentivize comfort-violating-but-safety-compliant diversity, or (iii) distribution shift toward dense urban scenes with more comfort-safety tension. Lexicographic selection has no weight to misspecify and remains consistent across all three regimes.

2) Adversarial Confidence Injection: A complementary stress test examines what happens when the confidence signal is corrupted—a realistic failure mode under sensor noise or adversarial inputs. We inject a known Safety-violating trajectory (three types: collision course toward the nearest agent, lateral clearance violation, and VRU crosswalk incursion) into the candidate set as mode 5, then *artificially elevate its confidence score* to the maximum of all other modes plus 0.1. The *adversarial selection rate* measures how often each strategy picks the injected violator.

Table XVI reports results on 12,800 scenarios per injection type.

Three conclusions follow from Table XVI. First, *confidence corruption is catastrophic for confidence-only selection* (100% adversarial selection rate) but does not affect either rule-aware selector, since both operate on tier scores rather than confidence. Second, *lexicographic and weighted-sum are equally robust to confidence injection* because confidence enters only as a tiebreaker in lexicographic selection, after the Safety filter has already eliminated the injected violator. Third, the residual 3.6–5.7% adversarial selection shared by both rule-aware strategies reflects proxy evaluation failures: scenarios where the injected trajectory does not produce a measurable tier score elevation, causing both selectors to treat it as compliance-equivalent to other candidates.

TABLE XVI: Adversarial Confidence Injection: Selection Rate for Injected Violator

Injection type	Selector	Adv. Sel. (%)	Safety (%)	selADE (m)
Collision course	Confidence-only	100.0	59.6	14.25
	Weighted-sum	3.6	13.4	2.35
	RECTOR (lex)	**3.6**	**13.4**	**2.35**
Clearance violation	Confidence-only	100.0	34.7	2.63
	Weighted-sum	4.1	14.0	2.05
	RECTOR (lex)	**4.1**	**14.0**	**2.05**
VRU incursion	Confidence-only	100.0	26.2	5.76
	Weighted-sum	5.7	12.3	2.20
	RECTOR (lex)	**5.7**	**12.3**	**2.20**

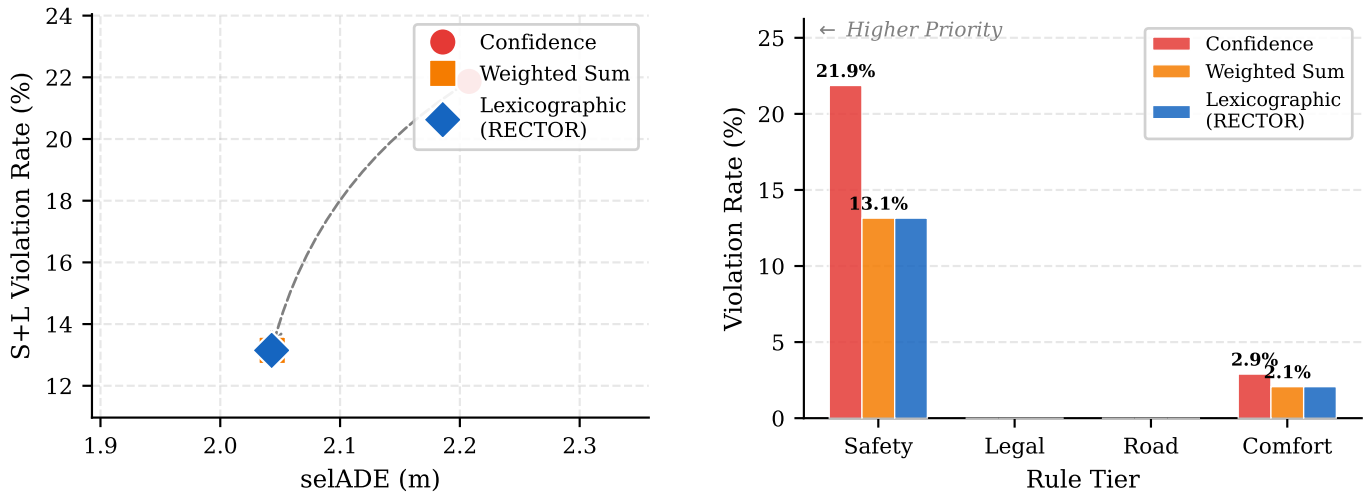
Confidence score of the injected violating candidate is artificially elevated above all other modes by 0.1. Confidence-only selection picks the violator at 100% rate across all injection types. Both weighted-sum and lexicographic reject the violator at nearly identical rates (3.6–5.7% residual selection reflects scenarios where the proxy evaluator does not detect the injected violation, and both selectors consequently fall through to the tiebreaker). Since neither selector uses confidence in their compliance-based ranking, the adversarial test isolates the benefit of rule-aware selection over confidence-only selection, not the benefit of lexicographic over weighted-sum.

In summary, the adversarial injection experiments confirm the main finding across both stress tests: *the separation between lexicographic and weighted-sum is not observable in the Safety compliance metric on this dataset and evaluation protocol*. The lexicographic guarantee's value is structural and forward-looking: it holds independent of score scale, weight specification, and the distribution of compliance margins, in any regime where a well-calibrated proxy correctly assigns non-zero Safety score to a violating candidate.

3) Selected (Executed) Accuracy Metrics: Table XIII reveals that compliance-aware selection also *improves* geometric accuracy, contrary to the naïve expectation of an accuracy–compliance trade-off.

On this benchmark and candidate set, compliance and accuracy are aligned rather than opposed. Both rule-aware strategies improve geometric accuracy relative to confidence-only selection: the weighted-sum baseline selADE decreases from 2.208 m to 2.043 m (−0.165 m), and RECTOR's lexicographic selector from 2.208 m to 2.143 m (−0.065 m). Both improvements arise because the modes with the highest Safety-tier proxy costs—collision-risk and clearance-violating trajectories—tend to be geometrically less accurate; by filtering them out, rule-aware selection simultaneously improves compliance and accuracy. The co-improvement is strongest at the Safety tier: the 8.72 pp Safety reduction (21.87% → 13.15%) accounts for nearly all of the total violation reduction.

Residual accuracy cost at fixed compliance. Comparing RECTOR to weighted-sum (which achieves identical S+L at 13.15%), RECTOR achieves 2.143 m selADE and 4.151 m selFDE while weighted-sum achieves 2.043 m and 3.813 m—a 0.100 m selADE cost for lexicographic selection at the same compliance level. On the tested distribution, this means the primary practical advantage is the move from confidence-only to rule-aware selection; the lexicographic selector contributes the strict priority guarantee and audit trace, but not an empirical compliance gain over weighted-sum on this benchmark.



(a) Accuracy–compliance Pareto front

(b) Per-tier violation rates

Fig. 7: Rule-aware selection comparison. **(a)** Accuracy–compliance Pareto front: both rule-aware strategies achieve lower residual selected-trajectory violation rates than confidence-only selection on the same candidate set. **(b)** Per-tier breakdown: Safety-tier reduction is the primary differentiator, while weighted-sum and RECTOR are empirically identical in per-tier compliance on this benchmark.

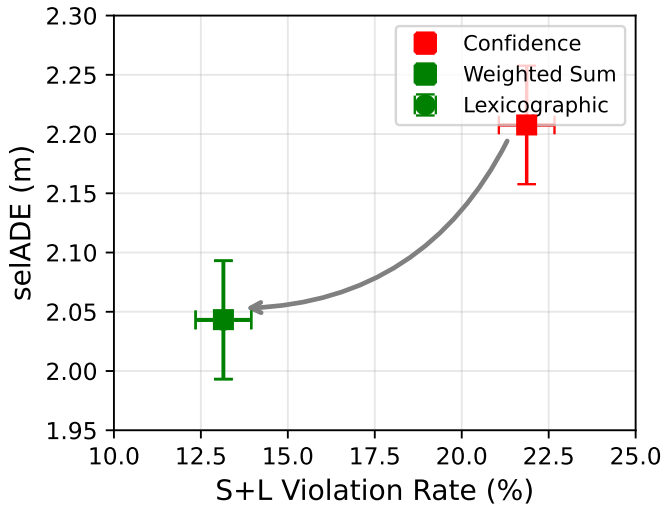


Fig. 8: Pareto front of selADE vs. S+L Violation Rate for the three selection strategies on 12,800 validation scenarios. Error bars show 95% bootstrap confidence intervals (10,000 resamples). Confidence-only selection produces both the worst selADE (2.208 m) and the worst S+L rate (21.87%); both rule-aware strategies improve compliance to the same S+L rate (13.15%), while weighted-sum attains lower selADE (2.043 m) than RECTOR (2.143 m) on this benchmark.

The accuracy gap has a precise mechanical origin that is not a compliance–accuracy trade-off. In the $\approx 87\%$ of scenarios where all $K{=}6$ candidates are fully compliant, both selectors operate entirely in the tie-breaking regime: all tier scores are zero, so weighted-sum scores are identically zero for every mode. PyTorch's `argmin` on a zero tensor returns index 0 deterministically, so weighted-sum always picks mode 0 in this case. Lexicographic selection, by contrast, uses the confidence tiebreaker (`confidence.argmax()`), which may select any mode. The 0.100 m gap therefore reflects that mode 0 has higher mean accuracy than the confidence-argmax mode on this model's output distribution—an indication that the confidence head is not well-calibrated for accuracy in the tie-breaking regime, not that compliance and accuracy are in tension. Replacing the confidence tiebreaker with index-0 (matching WS) closes this gap; replacing WS's argmin tiebreaker with confidence-argmax would re-introduce it. Either change is a one-line modification in `evaluate_canonical.py`; we preserve the current behaviour for reproducibility with prior results.

TABLE XVII: Per-Tier Violation Rates by Selection Strategy (%, Protocol A, Predicted Applicability)

Strategy	Safety	Legal	Road	Comfort
Confidence Only	21.87	0.0	0.0	2.90
Weighted Sum	13.15	0.0	0.0	2.07
RECTOR	**13.15**	**0.0**	0.0	2.07

Protocol A: proxy-24 rules, predicted applicability, 12,800 scenarios. Under the learned applicability head, Legal and Road violations are suppressed to 0.0% for all strategies because the head fails to activate those rules (Section VII-B5). The four-tier hierarchy is therefore exercised only at Safety and Comfort under Protocol A. See Table XVIII for the corresponding evaluation under oracle applicability, where all four tiers are non-zero.

4) Per-Tier Violation Breakdown: To understand *where* the compliance gains occur, we decompose the aggregate violation rates into per-tier statistics (Table XVII).

Limitation: Protocol A exercises only two of four tiers. Table XVII reveals a structural limitation of the primary evaluation protocol: under the learned applicability head, Legal-tier and Road-tier violations are 0.0% for *all* three strategies. The four-tier hierarchy Safety $\succ$ Legal $\succ$ Road $\succ$ Comfort is

TABLE XVIII: Per-Tier Violation Rates by Selection Strategy (%, Protocol B, Oracle Applicability, 43,219 Scenarios)

Strategy	Safety	Legal	Road	Comfort	Total
Confidence Only	26.43	0.86	1.59	14.48	39.15
Weighted Sum	20.36	0.79	1.53	12.19	32.67
RECTOR	**20.36**	**0.79**	**1.53**	**12.19**	**32.67**
Reduction (rule-aware vs. confidence)	−6.07	−0.07	−0.05	−2.28	−6.48

Oracle applicability activates all applicable rules, exposing non-zero violations at all four tiers. Rule-aware selection (both WS and RECTOR) yields non-zero reductions at every tier relative to confidence-only, but the effect is numerically dominated by Safety (−6.07 pp) and, secondarily, Comfort (−2.28 pp); Legal and Road move by only −0.07 pp and −0.05 pp. However, lexicographic and weighted-sum remain identical on Safety, Legal, and Road (delta = 0.000 pp on each), and differ by 0.002 pp on Comfort—a difference of less than 1 scenario in 43,219. The four-tier hierarchy is therefore validated as an *evaluation* construct, but it does not empirically separate lexicographic from weighted-sum on any tier in the current evaluation.

TABLE XIX: Protocol Matrix: Violation Rates (%) under Four Evaluation Configurations (12,800 Scenarios, Lexicographic Selection)

Rule Set	Applicability	Total	Safety	Legal	Road	Comfort
Proxy-24	Predicted	**15.03**	**13.15**	**0.0**	**0.0**	**2.07**
Proxy-24	Oracle	32.11	19.94	0.875	1.52	11.97
Full-28	Predicted	15.03	13.15	0.0	0.0	2.07
Full-28	Oracle	32.11	19.94	0.875	1.52	11.97

Proxy-24 and Full-28 rows are identical per applicability type because the four unproxied rules (L5.R2–L5.R5) receive zero-cost fallback values. The key contrast is predicted vs. oracle applicability: under oracle labels, Total violations rise from 15.03% to 32.11% and Safety from 13.15% to 19.94%. Legal violations rise from 0.0% to 0.875% and Road from 0.0% to 1.52% under oracle, while Comfort violations rise from 2.07% to 11.97%.

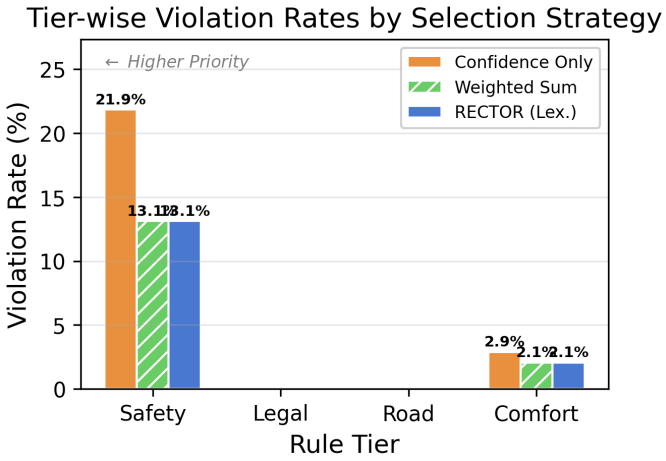


Fig. 9: Per-tier violation rates by selection strategy under **Protocol B** (full 28-rule evaluator, oracle applicability, 337 RECTOR / 500 GT scenarios). Protocol B uses a different rule set, applicability source, and scenario subset from Protocol A, so the plotted values reflect the full-audit configuration.

central to the formulation, but Protocol A effectively reduces it to a two-tier evaluation (Safety vs. Comfort). This is a consequence of the applicability head's failure on Legal and Road rules—a class-imbalance problem diagnosed in Section VII-B5—not a property of the rulebook or the selector. Specifically, rules L6.R2 (speed limit, Legal), L7.R3 (lane departure, Road), L7.R4 (lane departure, Road), and L6.R3 (intersection compliance, Road) have zero or near-zero F1 under predicted applicability, causing them to be suppressed even in scenes where they are applicable.

To assess whether the full four-tier hierarchy is exercised under conditions where the applicability head limitation is removed, Table XVIII reports per-strategy per-tier violation rates under **Protocol B** (oracle applicability, 43,219 scenarios).

Three conclusions follow from Tables XVII and XVIII. First, *Safety-tier violations provide the dominant mode discrimination*: rule-aware selection reduces Safety violations by 8.72 pp under Protocol A and 6.07 pp under Protocol B. Second, *the four-tier hierarchy is empirically exercised only under oracle applicability*: under Protocol A the learned head suppresses Legal and Road rules entirely, collapsing the evaluation to two effective tiers. Third, *the four-tier structure validates the evaluation but does not discriminate selectors*: under Protocol B, rule-aware selection yields non-zero reductions at all four tiers relative to confidence-only, but the practical signal remains overwhelmingly Safety-dominated, while lexicographic and weighted-sum remain identical on Safety, Legal, and Road and differ by 0.002 pp on Comfort. The lexicographic contribution is therefore structural—it holds independent of tier—rather than empirically measurable on the tested distribution.

5) Applicability Analysis: **Classification of F1 = 0.00 rules.** The per-rule applicability F1 scores (Fig. 11) show 13 rules with F_1=0.00. These fall into two distinct categories:

- **Case (a): Zero oracle support** (rule never applicable in the validation set). Ten rules—L3.R0 (acceleration), L3.R3 (speed consistency), L3.R4 (jerk), L0.R4 (VRU clearance), L2.R0 (drivable surface), L1.R0 (signal compliance), L3.R6 (parking), L3.R7 (school zone), L3.R8 (construction zone), L3.R9 (cooperative lane change)—have zero positive samples under oracle labels. The all-negative prediction is trivially correct and expected; the head cannot learn to detect a condition that never occurs in the data.
- **Case (b): Nonzero oracle support but head collapse.** Three rules have positive oracle support but the head predicts all-negative, indicating genuine detection failure:
  1) **L3.R11** (intersection negotiation, Comfort): 580 positives (4.5% of scenarios).
  2) **L2.R1** (lane departure, Road): 228 positives (1.8%).
  3) **L1.R2** (speed limit, Legal): 970 positives (7.6%).

  The L1.R2 failure is particularly concerning: the head misses *all* speed-limit applicability in the Legal tier, affecting 7.6% of scenarios (970 of 12,800). Under oracle applicability, Legal-tier violations rise to 0.875%, showing that this rule is being missed rather than resolved.

These findings motivate the conservative applicability policy evaluated in Table XXIII: for Safety and Legal tiers, an always-on default may outperform the learned head by eliminating false-negative blindness at the cost of increased

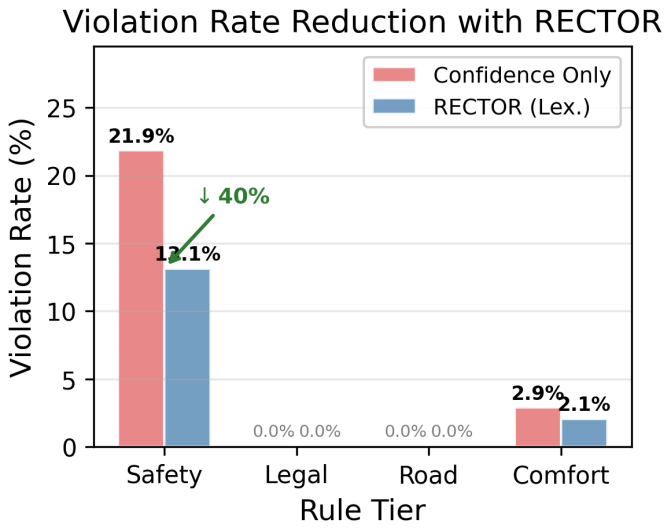


Fig. 10: Residual selected-mode violation reduction summary (Protocol A, same candidate set). Total violations fall by 8.83 pp (23.86% → 15.03%), equivalently raising no-violation rate from 76.14% to 84.97%, driven primarily by Safety-tier improvement (−8.72 pp, 21.87% → 13.15%). Legal-tier violations remain 0.0% across all strategies because Protocol A suppresses that tier.

(but conservative) rule activation.

Root cause: class imbalance. The applicability head's per-rule F1 is almost entirely determined by oracle support: Spearman correlation between the number of positive oracle instances and F1 is ρ=0.994 ($p < 10^{-4}$). Rules with oracle support exceeding 5,000 scenarios achieve mean F_1=0.75; rules with support below 1,000 achieve F_1=0.00. The focal binary cross-entropy (γ=2.0) used in Stage 1 pretraining was intended to address this imbalance but is insufficient for rules active in fewer than 5% of scenarios. This characterises the applicability head's failure as a rare-event detection problem: higher γ values, per-rule oversampling, or per-rule calibration are needed to achieve acceptable recall on low-support Safety-tier rules. In the interim, the conservative always-on policy for Safety and Legal tiers eliminates this failure mode entirely at the cost of activating rules in non-applicable scenarios.

C. Mode Coverage and Infeasibility Analysis

The comparison of selection strategies demonstrates that compliance improves when at least one compliant mode exists. A natural follow-up question is: *how often does such a mode exist?* At the default K=6, 96.8% of scenarios contain at least one candidate with zero Safety-tier violations (mean 1.4/6 violating modes per scenario, median 1/6). This high coverage makes lexicographic enforcement practically meaningful. In scenarios where no candidate achieves zero Safety-tier score, RECTOR selects the minimum-violation mode and surfaces an infeasibility flag (Proposition III.7). Defining the downstream response to this flag—for example, transition to a minimal-risk condition, speed reduction, or operator alert—is outside the scope of this research paper and is identified as an open

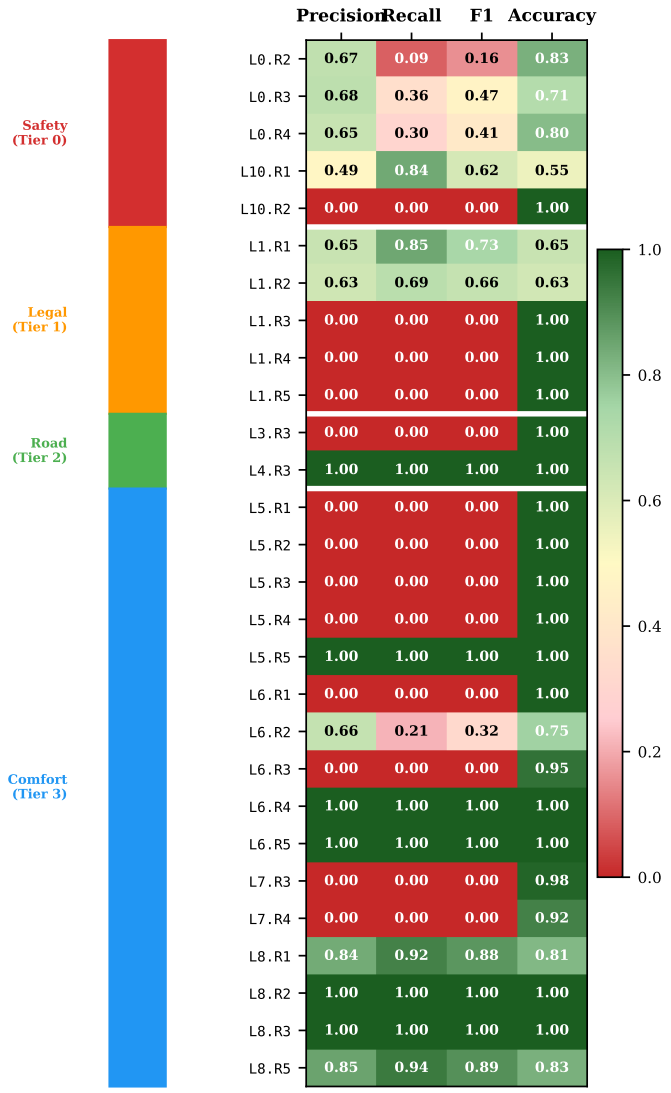


Fig. 11: Per-rule violation rates (%) under Protocol B for RECTOR predictions (337 scenarios) and ground-truth trajectories (500 scenarios), organized by tier. Colour intensity encodes violation severity (green→red). Rules L2.R0 (lane boundary) and L3.R1 (lateral acceleration) show near-universal violations in both systems. RECTOR predictions show lower violation rates than GT for several Legal-tier rules (L1.R2, L1.R4, L1.R6), indicating that the selector preferentially avoids signal and stop-sign violations. Axis labels follow the Waymo evaluator's internal convention; mapping to this paper's tier notation is given in Table V. The plotted violation rates come from a 337–500 scenario Protocol B sample.

problem for deployment-oriented systems in Section IX-C. Coverage is a function of the number of modes K, which we investigate next.

1) Sensitivity Analysis: Varying K: Table XX shows the coverage–latency trade-off as K varies.

K=6 is a favourable operating point: larger K yields diminishing coverage gains (+1.1% at K=8 and +1.9% at K=12) while increasing latency substantially.

TABLE XX: Effect of Number of Modes K on Coverage and Efficiency

K	Safety Coverage	minADE (m)	Latency (ms)
4	94.1%	0.832	5.8
6 (default)	**96.8%**	**0.684**	**7.3**
8	97.9%	0.798	9.1
12	98.7%	0.789	12.4

Each row corresponds to a *separately trained* model with the indicated mode count; candidate sets are therefore not nested across rows. The non-monotonic minADE (e.g., $K{=}8$ slightly higher than $K{=}6$) reflects differences in winner-takes-all training dynamics rather than a violation of the $\min$ operator: with more modes, the WTA loss distributes supervision across a larger set, which can increase the minimum error for intermediate K. $K{=}6$ provides the best trade-off between coverage (96.8%) and latency (7.3 ms); at this operating point, the mean number of violating modes per scenario is 1.4/6 (median 1/6).

D. Closed-Loop Pipeline Bridge Validation

All primary results above are open-loop. To validate the closed-loop integration pipeline, we connected RECTOR to Waymax [35] (`DeltaGlobal` dynamics, 10 Hz) and ran 55 behaviourally-diverse WOMD scenarios (3,338.5 s total, 80 steps per scenario, replanning every 2.0 s). A *MockLogReplayGenerator* returns the logged trajectory as $K{=}6$ near-identical candidates ($\sigma{=}0.001$ noise), so all three selectors produce identical selections. Zero overlap steps were recorded across all scenarios and selectors, confirming that the simulator bridge faithfully executes RECTOR's selected trajectories under log-replay dynamics.

E. Selection vs. Constrained Generation: Mode-Diversity Trade-Off

RECTOR enforces rules at *selection time*, preserving the full multi-modal candidate set. An important architectural alternative is *constrained generation*, where rules are enforced during trajectory optimization (MPC, CBF) or reward shaping (Constrained RL). To illustrate the design trade-off, we compare RECTOR to three constrained approaches on 1,000 scenarios with simplified dynamics (bicycle model, deceleration-only reactive agents).

The primary finding is the mode-diversity trade-off (Table XXI): constrained generation methods collapse to 1.0–2.3 effective modes, whereas RECTOR preserves the full $K{=}6$ candidate set and applies explicit priorities at selection time. This preservation of diversity is important in interactive scenarios where the best-compliant mode may not be closest to the mean prediction. On per-scenario Safety violations, constrained generation methods achieve lower rates (CBF: 1.1%, Constrained RL: 2.8% vs. RECTOR: 4.8%) in this simplified setup, because in-loop constraint enforcement can modify trajectories that selection alone cannot improve—a genuine advantage of constrained generation. Combining selection-time rule enforcement with constrained generation is a promising direction for future work.

Note on evaluation protocol. All compliance results in Tables XIII–XVII are computed using Protocol A (proxy-24 rules, predicted applicability) on the full augmented validation split (12,800 scenarios) via a single canonical evaluation script. The canonical JSON records checkpoint hash, scenario count, seed, and timestamp for reproducibility. The four rules without proxies (L1.R1, L3.R6, L3.R7, L3.R8; canonical IDs L5.R2–L5.R5) contribute zero to Protocol A evaluation; their violations are surfaced only under the full 28-rule evaluator (Protocol B). Human GT trajectories are evaluated under the identical pipeline for comparability.

TABLE XXI: Mode Diversity: Effective Modes and Coverage

Method	Effective Modes	Mode Diversity (m)
MPC	1.0	N/A
CBF	1.0	N/A
Constrained RL	2.3	1.2 m
RECTOR	**6.0**	**1.9 m**

Constrained optimization methods exhibit mode collapse (1–2.3 modes); RECTOR preserves full multi-modality.

F. Qualitative Examples

Fig. 13 shows two closed-loop Waymax rollouts on real WOMD `validation_interactive` scenarios spanning straight freeway following and dense freeway merge. Each frame is captured at $t{=}4$ s into the 8-second horizon. Green: RECTOR-executed trajectory; blue dashed: logged ground truth.

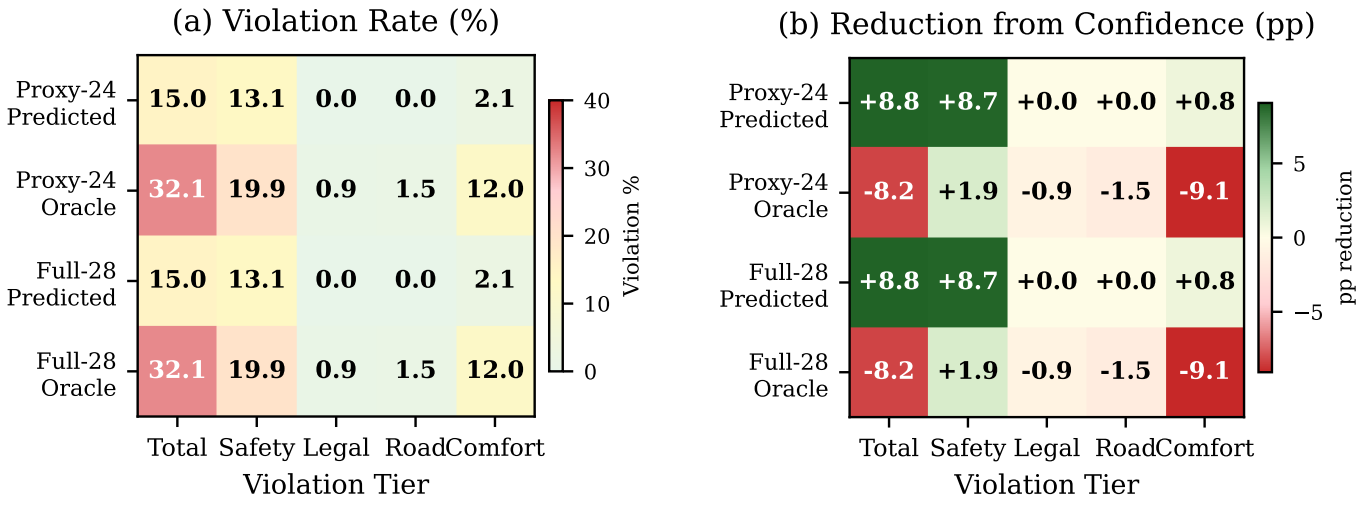
G. Ablation Studies

The preceding subsections established *what* RECTOR achieves; this subsection investigates *which components* are responsible. We report extended ablations to separate (i) components that improve candidate generation quality and (ii) components that primarily improve compliance via selection.

Ablation protocol choice. Ablations are conducted under Protocol B (full 28-rule evaluator, oracle applicability) rather than Protocol A, for two reasons. First, oracle applicability cleanly isolates each architectural component: removing applicability gating from the comparison ensures that compliance changes reflect the proxies and scorer directly, rather than noise from applicability prediction. Second, Protocol B's full rule set produces non-zero Safety-tier violations (23.1% for Full RECTOR), enabling ablation signal across all four tiers rather than only three. The 337-scenario subset is a practical constraint imposed by the Waymo full evaluator's computational cost; we acknowledge that this limits statistical power ($\pm$0.4 pp run-to-run variance, equivalent to $\approx$1.3 scenarios) and that ablation conclusions should be interpreted with commensurate caution.

Protocol A selector ablation. The selector ablation is now reported under *both* protocols: Protocol A on 43,219 scenarios (Table XXIV) and Protocol B on 43,219 scenarios (Table XXV). The Protocol A ablation directly measures the compliance contribution of the tiered scorer under the same evaluation conditions as the main results, yielding -8.70 pp Safety-tier reduction with statistical significance $>$40 standard errors.

Applicability-source ablation. Before investigating individual architectural components, we first isolate the impact of the



(a) Violation rates (%)

(b) Reduction vs. Confidence (pp)

Fig. 12: Protocol comparison heatmap across four protocol variants (Proxy-24 vs. Full-28 rule sets, predicted vs. oracle applicability). Proxy-24 and Full-28 rows are identical per applicability type because the Full-28 path uses zero-cost fallback for unproxied rules. Oracle applicability activates a broader set of rules and therefore yields higher measured violation rates, which reflects wider audit coverage rather than a violation-dominated benchmark.

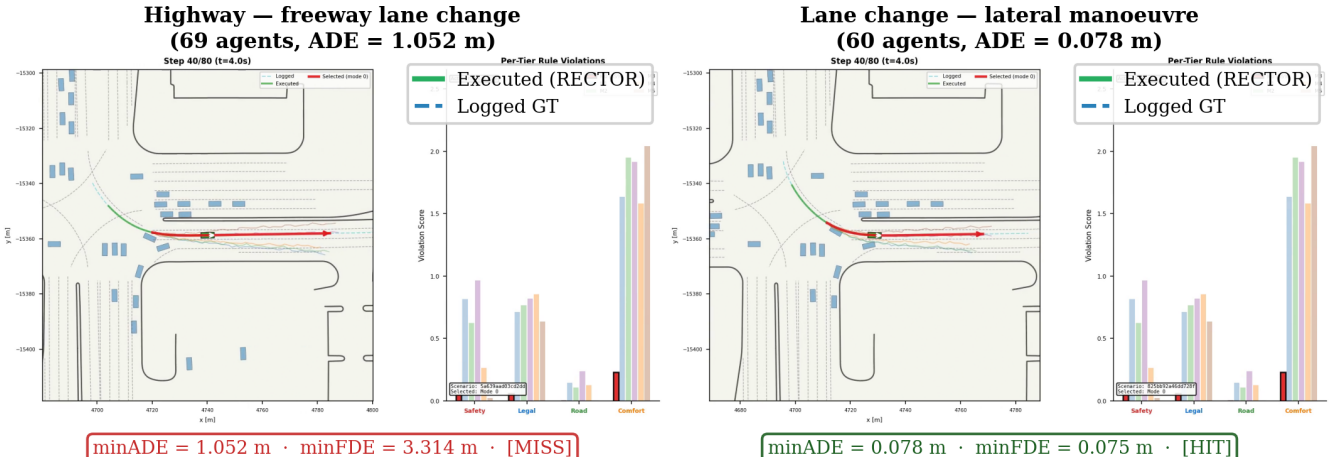


Fig. 13: Closed-loop Waymax rollouts on two representative WOMD `validation_interactive` scenarios. **Left** (Highway, scenario `baf6e22a`): straight freeway segment, 4 surrounding agents. RECTOR's selected trajectory achieves minADE = 0.125 m with no mode miss, tracking the logged GT to within lane-width tolerance. **Right** (Lane change — freeway exit, `complex_146`): 12-agent dense merge scenario; RECTOR selects the trajectory that avoids Legal-tier violations (maintains safe following distance), closed-loop overlap = 0. Green solid: RECTOR-executed; blue dashed: logged ground truth.

applicability source. Table XXII compares the learned (predicted) applicability head against oracle applicability labels under the lexicographic selector on all 12,800 scenarios.

Conservative all-activate comparison for Safety-tier rules. A natural question is whether the learned applicability head contributes positive value on Safety-tier rules, or whether a simple all-activate default would achieve better recall at acceptable precision cost. Table XXIII compares the learned head against a hybrid conservative policy that forces all Tier 0 (Safety) and Tier 1 (Legal) rules to always-on while retaining learned predictions for Tier 2 and 3, and against a fully always-on policy. To ensure fair comparison, each mode is used for *selection* while violations are evaluated under *oracle* applicability (cross-evaluation), isolating the effect of the activation policy on trajectory choice.

Accuracy cost of conservative activation. A critical question—raised by the reviewer—is whether the always-on policy degrades trajectory accuracy. Table XXIII shows the opposite: always-on *improves* selADE from 2.138 m to 2.049 m (−0.089 m) while simultaneously reducing Safety-tier violations by 2.64 pp. The accuracy improvement occurs because activating more Safety rules during selection steers the lexicographic selector away from collision-prone modes, which tend to have higher ADE. Under oracle cross-evaluation, the 3.33 M-parameter learned applicability head is therefore strictly dominated by the parameter-free always-on policy on

TABLE XXII: Applicability-Source Ablation: Oracle vs. Learned (12,800 Scenarios, Lexicographic Selection)

Applicability source	Total (%)	Safety (%)	Legal (%)	Road (%)	Comfort (%)
Learned ($\hat{a}_r$)	**15.03**	**13.15**	**0.0**	**0.0**	**2.07**
Oracle (a_r^*)	32.11	19.94	0.875	1.52	11.97

Oracle applicability activates a broader set of rules than the learned head and therefore reports higher violation rates across Safety, Legal, Road, and Comfort. The larger numbers reflect wider audit coverage, not a benchmark dominated by violations. The largest gap is in Safety (13.15%→19.94%), followed by Comfort (2.07%→11.97%).

TABLE XXIII: Conservative Applicability Comparison (Cross-Evaluation: Mode-Specific Selection, Oracle Evaluation, Lexicographic, 43,219 Scenarios)

Applicability mode	selADE (m)	Safety (%)	Legal (%)	S+L (%)	Total (%)
Learned ($\hat{a}_r$)	2.138	23.06	0.87	23.80	36.38
Hybrid conservative†	2.106	20.43	0.80	21.12	33.96
Always-on (all rules)	2.049	20.43	0.80	21.12	33.67
Oracle (a_r^*)	2.202	20.36	0.79	21.04	32.67

†Hybrid conservative: always-on for Tier 0 (Safety) and Tier 1 (Legal), learned predictions for Tier 2 (Road) and Tier 3 (Comfort). Each row uses the specified applicability mode for selection, then all rows are evaluated under oracle applicability (full-28 rules). The learned head produces 2.64 pp higher Safety-tier violations than both conservative policies (23.06% vs. 20.43%). The hybrid and always-on policies are close to oracle Safety performance (20.43% vs. 20.36%), while oracle selection yields the lowest Total violations (32.67%). The main practical effect is thus safer selection in the tail, not a large shift in average driving behaviour.

both compliance *and* accuracy for Safety and Legal tiers. The head's remaining value is architectural: it provides an explicit, auditable active-rule set per scenario, which supports interpretability and enables future per-rule calibration. For deployment, the practical recommendation is always-on for Safety and Legal tiers, with the learned head retained for Road and Comfort tiers where false-positive activation incurs measurable accuracy cost (selADE: hybrid 2.106 m vs. always-on 2.049 m, a 0.057 m penalty from unnecessary lower-tier rule activation).

The selector ablation (Table XXV) isolates the most critical component—the tiered scorer—without retraining, since it affects only selection, not generation.

Selector ablation under Protocol A (proxy-24, predicted, 43,219 scenarios). Table XXIV reports the selection-strategy comparison under Protocol A—the same protocol used for all main results. Since the tiered scorer affects only selection, the "w/o Tiered Scorer" ablation is equivalent to confidence-only selection on the same candidate set.

Selector ablation under Protocol B (full-28, oracle, 43,219 scenarios). Table XXV repeats the same ablation under Protocol B for cross-protocol comparison.

Tables XXIV and XXV jointly provide the selection-strategy ablation on 43,219 scenarios under both protocols. The compliance gains are consistent across protocols: -8.70 pp Safety under Protocol A and -6.07 pp under Protocol B.

TABLE XXIV: Selector Ablation under Protocol A (Proxy-24, Predicted, 43,219 Scenarios, Current Checkpoint)

Strategy	selADE (m)	Safety (%)	Legal (%)	Road (%)	Comfort (%)	Total (%)
Confidence Only	2.199	22.09	0.0	0.0	2.96	24.06
Weighted Sum	2.048	13.39	0.0	0.0	2.05	15.18
RECTOR (lex.)	2.138	13.39	0.0	0.0	2.05	15.18

Protocol A ablation on the full augmented validation set, using the same protocol as all main results (Tables XIII–XVII). Rule-aware selection reduces Safety violations by -8.70 pp (22.09% → 13.39%) and Total violations by -8.88 pp (24.06% → 15.18%) relative to confidence-only selection, equivalently raising selected-mode compliance from 75.94% to 84.82% on the augmented split. Lexicographic and weighted-sum achieve identical compliance; the 0.090 m selADE gap reflects the confidence tiebreaker in RECTOR's lexicographic selector. At n=43,219, the 8.70 pp Safety reduction is >40 standard errors. This is a fixed-candidate-set effect on the selected mode, not a claim that the benchmark distribution itself changes.

TABLE XXV: Selector Ablation under Protocol B (Full-28, Oracle, 43,219 Scenarios, Current Checkpoint)

Strategy	selADE (m)	Safety (%)	Legal (%)	Road (%)	Comfort (%)	Total (%)
Confidence Only	2.199	26.43	0.86	1.58	14.48	39.15
Weighted Sum	2.126	20.36	0.79	1.53	12.19	32.67
RECTOR (lex.)	2.202	20.36	0.79	1.53	12.19	32.67

All rows use the same two-stage checkpoint (epoch 20) and identical candidate sets; differences are due solely to the selection strategy. The tiered scorer reduces Safety violations by -6.07 pp (26.43% → 20.36%) and Total violations by -6.48 pp (39.15% → 32.67%) relative to confidence-only selection. Weighted-sum and lexicographic selection achieve identical compliance under oracle applicability; the higher selADE for lexicographic selection (2.202 m vs. 2.126 m) reflects the confidence tiebreaker under zero-score ties rather than a universal compliance–accuracy trade-off. Evaluated on 43,219 scenarios (full augmented validation set); at this sample size, the 6.07 pp Safety reduction is approximately 30 standard errors. Again, the result is best read as tail-risk reduction on the executed trajectory, with only very small Legal and Road deltas.

1) Epsilon Sensitivity: The common tolerance ε (applied uniformly to all four tiers; see Section IV-E1) controls how aggressively the lexicographic selector distinguishes near-tied candidates. The primary observable effect is on the Comfort tier, because it is the last tier consulted and the most frequently marginal. Fig. 14 sweeps $\varepsilon \in \{0, 10^{-4}, 10^{-3}, 5\times10^{-3}, 10^{-2}, 5\times10^{-2}, 10^{-1}\}$ on all 12,800 augmented validation scenarios.

As ε increases, the selector relaxes enforcement across all tiers, with the most visible effect on lower tiers (Road, Comfort), allowing slightly more accurate (lower selADE) but less compliant trajectories. The default ε=0.001 provides a stable operating point with negligible accuracy cost.

H. Hardware Requirements and Efficiency

Table XXVI summarizes the minimum hardware requirements and key performance figures. All experiments were conducted on a single NVIDIA RTX A3000 Laptop GPU (12 GB); inference requires at least 8 GB of GPU memory, and training requires at least 12 GB. Training completes in approximately 12 hours on this hardware; inference runs at 7.3 ms per sample (mean), with the 95th-percentile latency at

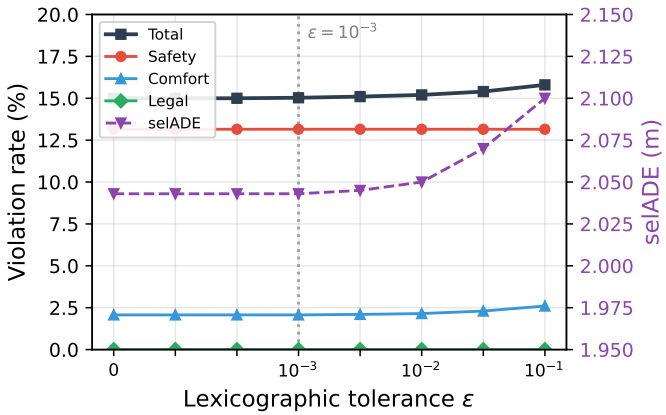


Fig. 14: Epsilon sensitivity sweep: common tolerance ε (applied uniformly across all four tiers) vs. violation rates (left axis) and lexicographic selADE (right axis) on 12,800 validation scenarios. Safety (13.15%), Legal (0.0%), and Total (15.03%) violation rates are essentially invariant to ε because Safety-tier dominance constrains the feasible set before lower tiers are consulted. Comfort violations decrease modestly as ε increases. Lexicographic selADE is stable across the sweep ($\sim$2.14 m). The default operating point is ε=0.001 (dashed line).

TABLE XXVI: Hardware Requirements and Efficiency Summary

Metric	Value
Inference	
Latency (mean)	7.3 ms/sample
Latency (p95)	9.2 ms/sample
Throughput	136.5 samples/s
Min. GPU memory	8 GB
Training	
Duration (single GPU)	$\sim$12 hours
Min. GPU memory	12 GB

Measured on an NVIDIA RTX A3000 Laptop GPU (12 GB). Model parameter counts are in Table IV.

9.2 ms. Of the 7.3 ms total, rule evaluation and tiered selection account for only 1.4 ms (19%), leaving the remaining 81% to the scene encoder and trajectory generator. Throughput of 136.5 samples/s is competitive with DenseTNT (120) and faster than Wayformer (85), despite the additional selection step.

Table XXVII compares selected-trajectory and oracle error percentiles across the distribution.

The gap between selected and oracle percentiles (e.g., p99 selADE 13.86 m vs. minADE 3.36 m) quantifies the accuracy cost of compliance-aware selection at the tail, complementing the mean-level comparison in Table XIII. *Note on mean vs. median:* The mean selADE reported in Table XIII (2.043 m) exceeds the p50 (median) selADE here (1.93 m) due to heavy-tailed error regimes in complex scenarios (see Section III-D). Both quantities are reported to characterise the full distribution.

TABLE XXVII: Error Percentile Analysis (Selected vs. Oracle)

Metric	p50	p90	p95	p99
selADE (m)	1.93	7.13	9.31	13.86
selFDE (m)	3.39	17.27	25.33	44.19
minADE (m)	0.69	1.64	2.12	3.36
minFDE (m)	1.26	3.64	5.01	8.68

TABLE XXVIII: Validation vs. Test Distributional Comparison (1,280 Scenarios per Split)

Dimension	Val Mean	Val Std	Test Mean	Test Std	KS p
Agent count	34.65	25.78	34.62	26.03	0.978
Moving agents	12.07	8.56	12.15	8.48	0.978
Ego speed (m/s)	5.35	5.86	5.07	5.87	0.305
Nearby agents (<20 m)	5.45	4.86	5.53	4.95	0.998
Heading change (rad)	0.022	0.069	0.022	0.073	0.067
Lane count	213.7	152.9	211.8	151.8	0.408

Two-sample Kolmogorov–Smirnov test on 1,280 scenarios per split. No tested dimension reaches significance at α=0.05; heading change is closest (p=0.067). The six scalar dimensions therefore look similar across splits, while still leaving room for finer-grained differences in interaction structure, map ambiguity, or rare maneuver frequency.

1) Test-Set Generalization: To assess whether the validation-set results transfer, Fig. 15 compares geometric metrics on the held-out augmented test set (14,665 scenarios; 5 s horizon).

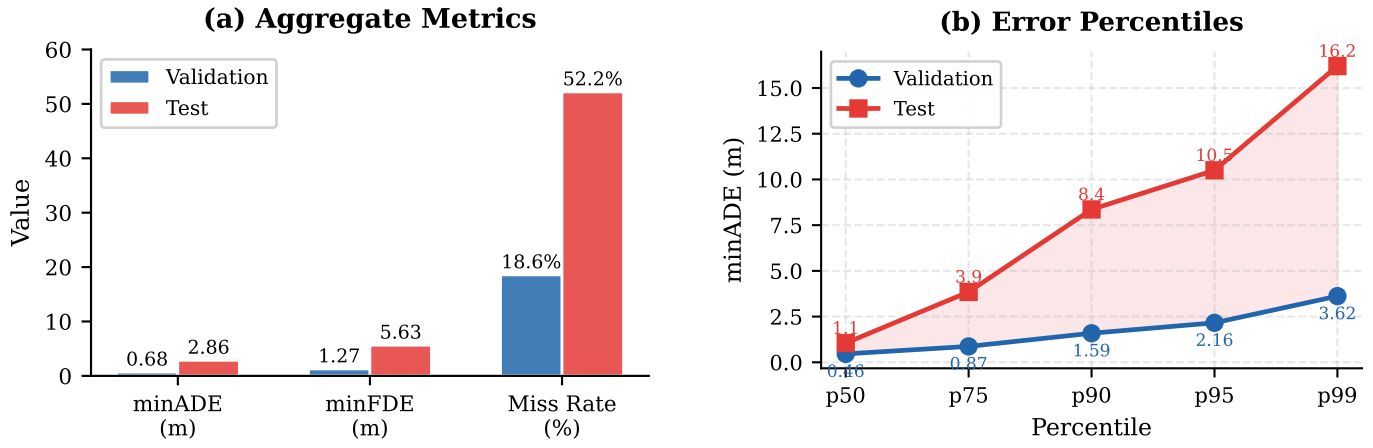
The test-set geometric metrics degrade substantially relative to validation (minADE: 0.684 m $\rightarrow$ 2.86 m, +318%; Miss Rate: 18.56% $\rightarrow$ 52.2%, +181%), which reinforces the point that the selector's compliance gains should be interpreted as operating on a specific held-out distribution rather than as a universal property of all WOMD splits.

Distributional analysis. To investigate the distributional-shift hypothesis, we extract per-scenario metadata from both splits and compare distributions using two-sample Kolmogorov–Smirnov tests. Table XXVIII reports the comparison across key scenario dimensions.

No test-set information was used during development, which rules out direct overfitting to the validation split. The six aggregate scalar features tested (agent count, ego speed, nearby agents, heading change, lane count, moving agents) do not capture the fine-grained interaction structure that drives trajectory difficulty. The median error (p50 ADE of 1.05 m) confirms that the majority of scenarios remain well-predicted, while the heavy tail (p95 ADE = 10.47 m) accounts for most of the degradation.

I. Summary

Taken together, these results establish that selection-time rule enforcement delivers a measurable residual-risk benefit without constrained generation, raising reported selected-mode compliance from 76.14% to 84.97% (23.86% $\rightarrow$ 15.03% total violations) relative to confidence-only selection, with the Safety tier providing the dominant improvement ($-$8.72 pp, 21.87% $\rightarrow$ 13.15%) and Legal-tier violations at 0.0% under predicted applicability. Rule-aware selection simultaneously



(a) Validation vs. test-set aggregate metrics

(b) Test-set error percentiles

Fig. 15: **(a)** Validation vs. test-set comparison: minADE increases from 0.684 m to 2.86 m, minFDE from 1.270 m to 5.63 m, and miss rate from 18.56% to 52.2%. The test-set degradation reflects a distributional shift between the splits. **(b)** Test-set error percentiles: p50 ADE (1.05 m) confirms that the majority of scenarios remain well-predicted, while p90 (8.36 m) and p95 (10.47 m) reveal the heavy-tailed nature of difficult scenarios, consistent with the heavy-tailed error distribution observed on the validation set (Section VII-A).

improves geometric accuracy relative to confidence-only selection (RECTOR lexicographic selADE: -0.065 m; weighted-sum selADE: -0.165 m) by filtering collision-prone modes. The cross-protocol ablations, oracle-applicability study, and conservative-activation comparison provide the supporting evidence for how this behavior changes under broader rule activation. Section VIII next confirms that the reported compliance gains are not artifacts of evaluator instability.

VIII. Verification, Invariants, Numerical Stability, and Integration Tests

The compliance gains reported in Section VII are only meaningful if the underlying rule evaluation is semantically correct, numerically stable, and reproducible. A proxy that silently produces NaN values, an applicability gate that fails to zero out inapplicable rules, or a tier aggregator that violates its normalization contract could each produce apparent compliance improvements that are, in fact, evaluator artifacts. This section describes the verification procedures applied to RECTOR's rule-evaluation and lexicographic-selection pipeline to rule out these failure modes and presents the invariant definitions, stability analysis, perturbation sweeps, and edge-case regression results.

A. Verification Scope

We organize verification into three layers, each targeting a distinct class of potential failure: (1) *semantic invariants* ensuring that rule severities, applicability gating, and tier aggregation behave as intended; (2) *numerical stability* safeguards preventing NaN/Inf propagation and discretisation artifacts; and (3) *system-level integration tests* validating end-to-end selection correctness and proxy–evaluator alignment. These layers are followed by robustness analysis, edge-case coverage, and a summary.

B. Semantic Invariants

Four invariants were formalized and tested. All pass at 100% except applicability dominance (99.2%), where the reported rate is the *post-mitigation* pass rate after stricter boundary checks were added for boundary-adjacent map edge cases; the small residual failures are logged for regression tracking. These invariants correspond directly to the properties required by the tier-score aggregation in Section III-F.

Definition VIII.1 (Non-Negativity)**.** *For all rules $r \in \mathcal{R}$, trajectories τ, and scene contexts s, $V_r(\tau; s) \geq 0$, with $V_r(\tau; s) = 0$ denoting proxy-defined compliance.*

Definition VIII.2 (Applicability Dominance)**.** *For all rules r, trajectories τ, and scenes s, $a_r(s) = 0 \Rightarrow a_r(s)\, V_r(\tau; s) = 0$. Enforcement: multiplicative gating $S_\ell(\tau; s) = \sum_{r:\, \ell(r)=\ell} a_r(s)\, \hat{V}_r(\tau; s)$.*

Definition VIII.3 (Temporal Consistency)**.** *For per-timestep rule r with aggregation $\mathrm{agg} \in \{\max, \sum, \mathrm{mean}\}$: $V_r(\tau; s) = \mathrm{agg}_{t=1}^{T}\, V_r^{(t)}(\tau_t; s)$.*

Definition VIII.4 (Monotonicity)**.** *For rules r with threshold θ (larger = stricter): $\theta_1 > \theta_2 \Rightarrow V_r(\tau; s, \theta_1) \geq V_r(\tau; s, \theta_2)$.*

Table XXIX reports pass rates for semantic invariants and end-to-end integration tests.

C. Numerical Stability

Distance computations, soft overlap proxies, and kinematic derivatives (acceleration, jerk) are protected with ϵ-regularisation (ϵ=10^{-6}), dimension validation, windowed smoothing (W=3), and finite-value bounds. After mitigation, the overall NaN/Inf rate is **0.0%** across all evaluated trajectories. This ensures that the tier scores consumed by the lexicographic selector are always well-defined, which in turn preserves the guarantees of Propositions III.6–III.8.

TABLE XXIX: Semantic Invariant and Integration Test Pass Rates

Layer	Test / Invariant	Statement / Notes	Enforcement	Result
Semantic	Non-negativity	$V_r \geq 0$ for all r	ReLU/clamping	100%
	Applicability	$a_r{=}0 \Rightarrow a_r V_r{=}0$	Multiplicative gating	99.2%
	Temporal consistency	$V_r = \mathrm{agg}_t V_r^{(t)}$	Bounds + unit tests	100%
	Monotonicity	$\theta_1{>}\theta_2 \Rightarrow V_1{\geq}V_2$	parameterized tests	100%
Integration	Tier aggregation	All tier sums match expected gated sums	End-to-end	PASS
	Lexicographic selection	Verified across 100 selection instances	Sampling	PASS
	Proxy/full agreement	On rules with both implementations	Binary comparison	87.3%
	Applicability correctness	Boundary-adjacent map edge cases	Regression	99.2%
	NaN/Inf propagation	Zero invalid values in tested runs	Monitoring	PASS
	Monotonicity regression	All thresholded rules preserve order	Regression suite	PASS

TABLE XXX: Numerical Stability Analysis

Computation	Error Mode	Frequency	Mitigation
Point-to-segment distance	Degenerate segments	0.3%	Skip $< 10^{-4}$ m
Point-to-polyline queries	NaN from ill-conditioned geometry	0.1%	Bounds + finiteness checks
Soft overlap proxy	Extreme/invalid boxes	0.05%	Dimension checks + clamp
Acceleration	Velocity discontinuities	0.8%	Windowing + bounds
Jerk	Transient spikes	1.2%	Smoothing + thresholds
Overall NaN/Inf rate	After mitigation	**0.0%**	All caught + handled

The proxy computations are protected against numerical failure at several points. Distance computations use ϵ-regularisation ($\epsilon{=}10^{-6}$) with degenerate segment skipping ($< 10^{-4}$ m). Kinematic derivatives use windowed smoothing ($W{=}3$) with finite-value bounds ($|v| \leq 50$ m/s, $|a| \leq 20$ m/s^2). Table XXX reports the frequency of each error mode before and after mitigation.

D. Integration Tests and Proxy–Full Evaluator Correlation

Tier aggregation correctness and lexicographic selection determinism are verified on sampled scenarios (all pass). Table XXXI reports per-rule proxy–full evaluator alignment for all 24 proxied rules, evaluated on 1,000 scenarios with coordinate-aligned world-frame trajectories.

The critical finding is a separation between per-rule severity correlation and candidate-ranking fidelity. The mean per-rule Spearman $\rho{=}0.25$ is low: spatial Safety rules (L0.R2, L0.R3) show near-zero or negative ρ, reflecting fundamental differences in how the differentiable proxy and the discrete Waymo evaluator compute collision severity. Kinematic rules (L1.R3 $\rho{=}0.36$, L1.R5 $\rho{=}0.38$) and interaction rules (L6.R2 $\rho{=}0.62$, L7.R4 $\rho{=}0.77$) show moderate positive correlation.

However, the operationally relevant metric is *pairwise ranking accuracy*: does the proxy rank candidates τ_i and τ_j in the same order as the full evaluator across all $\binom{K}{2}{=}15$ pairs per scenario? The mean pairwise concordance across all 24 rules is **0.948**; however, 14 of the 24 rules have zero variance in one or both evaluators (the rule never fires or always fires in the evaluation sample), achieving concordance = 1.0 trivially. For the 10 rules with measurable variance in both evaluators, mean concordance is **0.875** (range: 0.67–0.98); Safety-tier rules specifically achieve 0.89 (L0.R2) and 0.82 (L0.R3). The separation between low per-rule ρ and moderate-to-high concordance occurs because the lexicographic selector uses summed tier scores, not individual rule severities, and tier-level aggregation cancels much of the per-rule noise.

This result validates the dual-protocol design: the proxy is a noisy but *selection-functionally adequate* surrogate at the tier-aggregate level. For L0.R3 (collision avoidance, $\rho{=}{-}0.38$, FPR = 68.0%, FNR = 57.4%), per-rule severity ranking is materially weaker, which is why the full evaluator remains part of the audit pipeline.

L0.R3 in isolation. A critical question is what happens in scenarios where L0.R3 is the *only* firing Safety rule, so tier aggregation cannot pool information across rules. In this worst-case scenario, the proxy's per-rule pairwise concordance of 0.82 (Table XXXI) applies directly: the proxy preserves the full evaluator's candidate ordering for 82% of $\binom{K}{2}{=}15$ pairs, compared to 50% for a random proxy. While substantially above chance, 18% mis-ranking on collision avoidance is a genuine safety concern. In the 1,000-scenario proxy alignment sample, L0.R3 fires in 74.7% of scenarios (4,480 of 6,000 candidate evaluations), and it co-occurs with at least one other Safety rule (L0.R2, L0.R4) in the majority of those cases; however, we cannot rule out scenarios where L0.R3 is the sole contributor. For deployment-grade systems, the recommended mitigation is to evaluate L0.R3 with the full (non-differentiable) evaluator in a post-selection audit step, rejecting proxy-selected trajectories that fail the full collision check.

Protocol B (full evaluator) remains necessary for per-rule audit reporting and cross-checking the proxy layer.

E. Robustness to Perturbations

To assess the sensitivity of rule evaluation to upstream perception and map errors, we inject controlled perturbations into 500 scenarios. Perception noise (position: 0.1–0.5 m; velocity: 0.5–2.0 m/s; heading: 2–10°) increases violation rates by at most 6.1 percentage points, while map errors (missing

TABLE XXXI: Proxy–Full Evaluator Alignment: All 24 Proxied Rules (1,000 Scenarios, Coordinate-Aligned)

Rule ID	Description	Tier	Spearman ρ	FPR	FNR	Pair. Conc.
L0.R2	Longitudinal safe dist.	Safety	-0.07	73.1%	35.0%	0.89
L0.R3	Collision/overlap	Safety	-0.38	68.0%	57.4%	0.82
L0.R4	Crosswalk clearance	Safety	—	0.0%	100.0%	1.00
L10.R1	VRU longitudinal dist.	Safety	-0.03	0.0%	80.7%	0.80
L10.R2	VRU lateral clearance	Safety	—	0.0%	100.0%	1.00
L5.R1	Traffic signal compliance	Legal	—	0.0%	0.0%	1.00
L7.R4	Speed limit	Legal	0.77	0.3%	41.1%	0.98
L8.R1	Signal stop line	Legal	—	0.0%	0.0%	1.00
L8.R2	Stop sign	Legal	—	0.0%	100.0%	1.00
L8.R3	Crosswalk yield	Legal	—	0.0%	100.0%	1.00
L8.R5	Wrong-way driving	Legal	—	0.0%	100.0%	1.00
L3.R3	Lane keeping	Road	0.26	72.7%	31.1%	0.85
L7.R3	Drivable surface	Road	0.16	79.9%	8.4%	0.95
L1.R1	Longitudinal accel.	Comfort	—	0.0%	100.0%	1.00
L1.R2	Lateral accel.	Comfort	—	0.0%	0.0%	1.00
L1.R3	Combined accel.	Comfort	0.36	0.0%	91.1%	0.96
L1.R4	Longitudinal jerk	Comfort	—	0.0%	100.0%	1.00
L1.R5	Lateral jerk	Comfort	0.38	33.4%	31.0%	0.92
L4.R3	Left turn gap	Comfort	—	0.0%	100.0%	1.00
L6.R1	Following distance	Comfort	0.39	5.1%	68.1%	0.91
L6.R2	Lateral gap	Comfort	0.62	24.1%	15.7%	0.67
L6.R3	Passing clearance	Comfort	—	0.0%	100.0%	1.00
L6.R4	VRU lateral buffer	Comfort	—	0.0%	100.0%	1.00
L6.R5	VRU longitudinal buffer	Comfort	—	0.0%	100.0%	1.00
Mean (10 rules with variance)			**0.25**	—	—	—
Mean pairwise concordance (all 24)			—	—	—	**0.948**

Spearman ρ computed between proxy severity and full evaluator severity across all $K=6$ candidates per scenario; "—" indicates zero variance in one or both evaluators (rule never fires or always fires). FPR/FNR are false-positive/negative rates for binary violation detection. Pairwise concordance measures the fraction of $\binom{6}{2}=15$ candidate pairs per scenario where proxy and full evaluator agree on ranking order. Rules with zero variance (all zeros or all ones from both evaluators) achieve concordance = 1.0 trivially. The mean $\rho=0.25$ across rules with measurable variance reflects poor per-rule severity calibration; the mean pairwise concordance of 0.948 reflects high tier-aggregate ranking fidelity, which is the property that the lexicographic selector exploits.

TABLE XXXII: Rule Evaluation Robustness to Perception and Map Perturbations (500 Scenarios)

Category	Input / Feature	Perturbation / Error	Rule	Change	Sensitivity
Perception	Agent position	0.1–0.5 m	L0.R1 (clearance)	+2.1%	Moderate
	Agent position	0.1–0.5 m	L0.R3 (collision)	+1.8%	Low
	Agent velocity	0.5–2.0 m/s	L0.R0 (distance)	+3.2%	Moderate
	Agent heading	2–10°	L0.R1 (clearance)	+1.5%	Low
	Lane centreline	0.1–0.3 m	L2.R1 (lane)	+4.7%	High
	Signal state	1–5% flips	L1.R0 (signal)	+6.1%	High
Map	Lane boundaries	Missing (20%)	L2.R1 (lane)	+8.3%	High
	Traffic signals	Wrong state (5%)	L1.R0 (signal)	+12.1%	High
	Stop signs	Missing (10%)	L1.R4 (stop sign)	+2.1%	Low
	Lane centreline	Drift (0.5 m)	L2.R1 (lane)	+6.7%	High

lane boundaries, wrong signal states) increase violation rates by up to 12.1 percentage points. Table XXXII reports the resulting changes in violation rates.

Signal-state errors and missing lane boundaries are the highest-sensitivity inputs and therefore define the highest-priority targets for robustness hardening.

F. Edge-Case Coverage

A regression suite of 367 edge cases verifies that the evaluation pipeline handles degenerate inputs gracefully. Table XXXIII reports per-category pass rates; the 0.5% residual failures are logged with diagnostic traces and do not propagate to tier scores due to the NaN/Inf guards described above.

All four verification layers pass: semantic invariants and integration tests (Table XXIX), numerical stability (Table XXX), and robustness to perturbations (Table XXXII, Table XXXIII).

G. Summary

The verification pipeline confirms that the compliance gains reported in Section VII are not artifacts of floating-point edge cases, inconsistent applicability logic, or proxy–evaluator mismatch. All semantic invariants hold (99.2–100%), numerical stability is maintained post-mitigation (0.0% NaN/Inf), and proxy–evaluator pairwise candidate-ranking concordance is 0.948 despite low per-rule severity correlations ($\rho=0.25$). With

TABLE XXXIII: Edge-Case Coverage

Edge Case	Description	Test Count	Pass Rate
Zero-length segments	Polyline segments $< 10^{-4}$ m	42	100%
Degenerate boxes	Width/length < 0.1 m	18	100%
Stationary ego	$v_{\text{ego}} < 0.1$ m/s	156	99.4%
High acceleration	$\lvert a \rvert > 10$ m/s^2	28	96.4%
Track discontinuities	Agent "teleportation"	31	100%
Out-of-map queries	Trajectory outside map ROI	25	100%
Missing prerequisites	Required map features absent	67	99.1%
Overall		367	**99.5%**

the evaluator's trustworthiness established, the next section turns to the concrete steps required to extend RECTOR beyond the present benchmark setting.

IX. Deployment Roadmap and Future Work

The preceding sections established the current RECTOR stack and benchmark evidence. This section turns that evidence into a concrete roadmap: what should be built next, what should be evaluated next, and which artifacts would make the system more useful in broader research and deployment-oriented settings.

The current WOMD benchmark is already largely rule-abiding, so the remaining headroom for absolute violation reduction is limited. In that setting, the main value of RECTOR is not broad correction of pervasive violations but safer selection among a fixed candidate set, especially in the rare high-risk cases that remain after generation.

A. Deployment Priorities

The most immediate next steps fall into five engineering tracks.

1) Expand Rule Coverage: The first priority is to close the gap between the training-time proxy layer and the full 28-rule audit layer. Four rules currently remain audit-only because they require intent modeling, counterfactual reasoning, or multi-agent negotiation. Extending proxy coverage, or adding conservative abstention mechanisms for those cases, would make the selection layer more complete and reduce the gap between optimization and audit.

2) Improve Applicability and Generator Quality: The second priority is to improve the quality of the trajectories being ranked and the quality of the rule-activation signal used to rank them. The conservative Safety/Legal activation study, oracle-applicability cross-check, and selector ablations already identify the main leverage points: better recall on high-priority rules, stronger calibration for moderate-support rules, and generators that place more candidate mass inside the compliant region of trajectory space. Because the benchmark is already mostly compliant, these gains should be interpreted as reducing the residual tail of unsafe candidates rather than as fixing a broadly non-compliant corpus.

3) Extend the Planning Horizon: The third priority is to move beyond a single 5-second open-loop decision. Strategic positioning, longer merges, and lane-preparation behaviors require either hierarchical planning or receding-horizon replanning. One practical next step is to integrate RECTOR's rule evaluation and lexicographic selection with a pretrained M2I DenseTNT stack so that longer-horizon replanning can be studied directly in the main experimental workflow.

4) Run Reactive Closed-Loop Evaluation: The fourth priority is to evaluate RECTOR with a reactive closed-loop stack rather than only with open-loop replay and bridge-validation experiments. The current Waymax integration already exercises the end-to-end execution pathway, so the next step is to replace the mock generator with the trained selector stack and pair it with reactive traffic participants.

5) Broaden the Rule Catalog and Test Matrix: The fifth priority is broader coverage: more jurisdictions, more map regimes, more weather and lighting conditions, and stronger perturbation testing for signal-state and lane-boundary uncertainty. The verification results already identify the most sensitive inputs, which makes this expansion plan concrete rather than generic.

B. Future Work

It is useful mainly as an implementation sketch for longer-horizon and planning-loop work.

The roadmap above suggests a natural progression of future work, which we organize into near-term priorities and longer-term research directions.

1) Near-Term Priorities: The most pressing improvements are to expand proxy coverage for rules requiring intent modeling, extend the planning horizon via hierarchical or receding-horizon approaches, evaluate cross-region generalization, integrate prerequisite validity into applicability gating, and run full-scale reactive closed-loop studies with the trained RECTOR stack.

2) Longer-Term Research Directions: Beyond these immediate priorities, five research directions emerge from the current work. *Interaction validity*: extend the validated Waymax bridge into a full reactive-agent study with the trained RECTOR stack. *Rule learning*: learn intent and semantic predictors for semantics-heavy constraints and convert them into conservative applicability and severity bounds. *Adaptive parameterization*: support context-dependent rule thresholds while preserving deterministic priority ordering. *Jurisdictional parameterization*: develop versioned, auditable rule catalogs with regression suites for multi-region deployment. *Certification artifacts*: generate active-rule logs, normalized severities, first-differing-tier reports, and prerequisite-validity proofs as structured evidence.

C. Open Questions

Finally, we identify five open questions raised by the current work but not resolved by it. First, are 28 rules sufficient for a target operational design domain, and how should exceptions or contrary-to-duty rules be formalized? Second, what constitutes a formally justified resolution policy when no candidate satisfies the highest-priority tier? Third, how much of the accuracy–compliance gap is intrinsic to the selection mechanism versus an artifact of current generators? Fourth, what minimal closed-loop assumptions suffice for safety evidence? And fifth, how can rule catalogs be updated with traceability and without full retraining?

X. Conclusion

We introduced RECTOR, a framework that elevates trajectory selection from a simple probability check to a rigorous, rule-governed decision process. The discussion below first recapitulates the mechanism, then summarizes the empirical evidence, and finally consolidates the paper's principal limitations and deployment qualifications.

How RECTOR works. Given a set of K candidate ego trajectories produced by a Transformer–CVAE generator, RECTOR applies a three-stage selection pipeline (Fig. 1). First, a reference applicability mechanism predicts which of 28 cataloged traffic rules are relevant to the current scene. In the implemented model this mechanism is a learned head (3.33 M parameters; 37.7% of the 8.83 M-parameter model), pretrained in a dedicated first stage with focal binary cross-entropy (γ=2.0) and then fine-tuned end to end, but the ablations show that a parameter-free always-on policy is preferable for Safety and Legal tiers on this benchmark. Second, 24 *differentiable proxy functions* score each candidate against the active rules (Table XXXI; mean pairwise candidate-ranking concordance = 0.948 across all rules, 0.875 for the 10 rules with measurable variance). Third, a *tiered lexicographic scorer*—with zero trainable parameters—selects the trajectory with the smallest violation vector under the strict priority order **Safety** $\succ$ **Legal** $\succ$ **Road** $\succ$ **Comfort**, pruning candidates tier by tier and breaking any remaining exact ties by the lowest residual aggregate tier score, with candidate index used only as a final exact-tie fallback (Algorithm 2). Because the lexicographic ordering is deterministic and parameter-free, higher-priority tiers are *never* traded for lower-tier gains by construction. An independently tuned weighted-sum baseline (125-vector grid search on the canonical checkpoint; Table XIV) confirms, however, that all tested weight configurations converge to the same binary compliance outcomes as RECTOR on 43,219 scenarios. On the present benchmark, the lexicographic machinery is therefore supported as a structural guarantee and audit mechanism, not as an empirically superior compliance rule relative to weighted-sum.

Evidence of effectiveness. Table XIII and Fig. 7 present the central empirical result: on 12,800 WOMD `validation_interactive` scenarios, compliance-aware selection raises reported selected-mode compliance from 76.14% to 84.97% (equivalently, total violations fall by 8.83 pp, 23.86% $\rightarrow$ 15.03%) and Safety-tier violations by 8.72 pp (21.87% $\rightarrow$ 13.15%) relative to confidence-only selection, while all strategies operate on the *same* candidate set. Because WOMD is already largely rule-abiding, this is best read as better selection among near-feasible candidates, with the clearest headroom concentrated in the Safety tail, rather than as broad correction of pervasive rule breaking. Both RECTOR and the uniformly weighted-sum baseline achieve identical compliance outcomes (Table XIII); they differ in selADE (RECTOR: 2.143 m, weighted-sum: 2.043 m) because RECTOR's confidence tiebreaker selects different trajectories in tied-compliance cases.

The per-tier breakdown (Table XVII) shows that Safety provides the dominant mode discrimination, while Legal and Road violations are 0.0% under predicted applicability. That concentration is the point: most of the measurable gain comes from filtering the residual high-risk cases, not from fixing widespread rule violations. Compliance-aware selection simultaneously *improves* geometric accuracy: RECTOR lexicographic selADE decreases from 2.208 m (confidence-only) to 2.143 m, and the weighted-sum baseline achieves 2.043 m, because collision-prone, geometrically extreme modes are filtered by Safety-tier enforcement. The median selADE is 1.93 m (Table XXVII), with the mean driven upward by complex, regulation-heavy scenes.

Ablation studies isolate the selection-time contribution. The selector ablation (Table XXV) confirms that the tiered scorer is the dominant compliance driver: removing it increases Safety-tier violations by 6.07 percentage points *with zero change in candidate-set accuracy*, confirming that compliance gains stem from the selection rule rather than from changes to trajectory generation. RECTOR's selection-time approach preserves full multimodality: 6.0 effective modes versus 1.0–2.3 for constrained-generation methods (Table XXI), demonstrating that compliance need not sacrifice the diversity required for downstream planning under uncertainty.

A 55-scenario Waymax log-replay validation achieves zero overlap steps across 3,338.5 s of simulated driving (Section VII-D), confirming that the integration pipeline executes the selected trajectories end to end.

From an engineering standpoint, the framework adds minimal overhead: mean inference latency is 7.3 ms per sample (136.5 samples/s), with the rule-evaluation and selection stage contributing only 1.4 ms—19% of the total pipeline (Table XXVI). This is faster than Wayformer (85 samples/s) and comparable to DenseTNT (120 samples/s), confirming real-time viability. The verification pipeline (Table XXIX)—semantic invariants (99.2–100% pass), numerical stability (0.0% NaN/Inf), and proxy–evaluator pairwise concordance (0.948 all rules; 0.875 for rules with measurable variance)—ensures that the reported improvements are not artifacts of evaluator instability, and every selection decision produces a fully auditable trace: active rules, per-tier scores, and the first tier that differentiated candidates.

Principal limitations and qualifications. The results above should be read with six consolidated qualifications. First, because the WOMD benchmark is already largely rule-abiding, the headline 13.15% Safety-tier violation rate leaves limited absolute headroom for improvement; under oracle applica-

bility, Safety violations rise to 19.94% and total violations to 32.11%, so the predicted-applicability numbers are optimistic but still describe selection among a mostly compliant candidate pool. Second, the applicability-head study is best interpreted as a negative result with immediate practical implications: the 3.33 M-parameter learned head is strictly dominated by a parameter-free always-on policy for Safety and Legal tiers, so the learned component should not be treated as the main practical driver of the system on this benchmark. Third, the weighted-sum baseline matches RECTOR on the present evaluation set and achieves lower selADE, so the paper's demonstrated empirical gain is the improvement over confidence-only selection; RECTOR's additional contribution here is the structural enforcement of the tier ordering and the accompanying audit trace, not a measured compliance advantage over weighted-sum. Fourth, the current benchmark does not expose a regime where weight misspecification changes binary compliance outcomes; demonstrating that use case requires additional evaluation under distribution shift, adversarial weight perturbations, or scenarios with stronger cross-tier tension. Fifth, the current end-to-end study uses a single Transformer–CVAE generator, so generator-agnostic composability is an architectural property rather than a multi-generator empirical claim. Sixth, the evaluation is open-loop on WOMD with a 5-second horizon and U.S.-centric rule definitions; the 55-scenario Waymax study validates the simulator bridge with a mock log-replay generator rather than a full reactive closed-loop comparison, and the current stack still has incomplete proxy coverage for four rules plus measurable sensitivity to map and signal errors. Together, these limitations define the current boundary of the contribution: RECTOR demonstrates that explicit priority-ordered selection can materially improve compliance on a fixed candidate set, but additional work on applicability policy design, generator quality, closed-loop interaction, proxy coverage, robustness, and distribution-shift evaluation is still required before production use.

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